The cost of precision-fermented dairy, egg and serum proteins

A Humbird-style techno-economic analysis of β-lactoglobulin, ovalbumin, ovomucoid, β-casein, lactoferrin, serum albumin and α-lactalbumin, from stoichiometry to the physical limit, with notes on rennet-cheese casein, heme, transferrin and binders, and the animals' feed bill

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Version 4.1 (seven proteins; v2.1 audited and AI pre-reviewed) for human review · numbers generated 2026-10-06 11:00 · repository `pf-tea`

Status. No human expert has reviewed this analysis yet. The author built the model, the evidence base and this text with an AI assistant (Claude, Anthropic). Independent AI reviewers then audited each version, and each reviewer had to back every finding with a computation, a quote or a line of code. `docs/AUDIT.md` and `docs/REVIEW.md` log every finding, every decision and every fix, version by version. The model computes every number in this report, and the report reads them from `results/analyses.json` when it compiles. Every input has a source and an evidence grade in `data/` (Appendix A). Inputs graded E are guesses; Section 6 lists them.

Summary

Precision fermentation (PF) uses an engineered microbe to make one animal protein and release it into the broth. This report asks two questions about seven such proteins. What does each one cost to make? And how well must the microbe perform before the protein matches the price of the one it replaces? The seven proteins are:

- β-lactoglobulin (BLG), the main whey protein;
- α-lactalbumin (α-lac), the second whey protein, sold enriched for infant formula;
- ovalbumin (OVA), the main egg-white protein;
- ovomucoid, another egg-white protein (about 11% of it), which EVERY sells as a heat-stable ingredient;
- β-casein, one of the two main proteins of cheese curd;
- lactoferrin, a scarce, high-value milk protein;
- serum albumin, the carrier protein in the serum-free media that grow cultivated meat. It is the only medium protein the industry would need by the thousand tonnes.

We follow Humbird's (2021) method. It balances every element and every electron in the fermentation, it limits the fermenter by how fast it can transfer oxygen, and it prices capital with his factors and fixed costs. With these, his baker's-yeast workbook lands on the cost the market implies for baker's yeast. We add a secreted product, a methanol-fed production phase and three purification routes: membranes, isoelectric precipitation and ion exchange. We then check the model against three published plant designs.

The base case. A secreting **Komagataella phaffii** strain reaches 30 g/L after a 72 h production phase, the design point of Intelligen's (2025) OVA process. The plant runs 200 m³ industrial fermenters, makes 10 kt of protein a year and sits in the US Midwest. All costs are in 2025 dollars. Three proteins depart from this base:

- Lactoferrin uses Intelligen's own lactoferrin design: 10 g/L, purified by cation exchange, at 250 t/yr. That plant sits near the low end of the 0.3–1.6 kt/yr world market. Intelligen's strain needs methanol; we cost it on glucose.
- Albumin uses the two columns of the industrial recombinant-albumin process (cation-exchange capture, then an anion-exchange polish) at 1 kt/yr. That is over a quarter of GFI's lowest 2030 demand scenario and about the size of today's whole human-albumin market.
- α-Lac uses BLG's membrane route, purified to isolate grade, at 1 kt/yr, about 5% of the α-lac market.

Small plants use smaller fermenters.

What they cost. We charge 14.6% of the investment each year as the cost of capital. With that charge, the minimum selling price per kg of target protein is **$21.2 (BLG), $21.3 (OVA), $21.3 (ovomucoid), $22.0 (β-casein), $168 (lactoferrin), $215 (albumin), $33.8 (α-lac)**. NREL-style design reports use lighter accounting; with it, the price is $16.4 (BLG), $16.5 (OVA), $16.5 (ovomucoid), $17.0 (β-casein), $130 (lactoferrin), $185 (albumin), $25.0 (α-lac). Private companies' cost models assume an average titer of 42 g/L (GFI 2025, a survey average over unnamed proteins, with no fermentation time given). At that titer the price is $15.8 (BLG), $15.9 (OVA), $16.0 (ovomucoid), $16.4 (β-casein). To reach 42 g/L in our 72 h production phase, cells must secrete about 7 mg of protein per g of cells per hour, about three times the best rate shown on glucose (2.1). At the best titers in the published literature, the price is $43.0 (BLG), $227 (OVA), $479 (ovomucoid), $803 (β-casein), $413 (lactoferrin), $424 (albumin), $1,848 (α-lac) (Table 1).

What they replace. Per kg of protein, in 2015–2025 and in 2025 dollars, whey protein traded at $9–$18, dried egg white (which OVA and ovomucoid replace) at $10–$21 and casein at $8–$19. Reports put dairy lactoferrin at $250–$1,585, bulk bovine serum albumin at $40–$70 (our reading of GFI's "high-value albumins" in trade data) and dairy α-lactalbumin at $25–$129 per kg of α-lac. These last three prices are grade C.

Nine findings stand out:

1. **Cheap protein needs fast secretion more than a high titer.** At the base cell density, $20/kg BLG needs 29–32 g/L in 48–72 h. Cells would then have to secrete 5–7 mg of protein per g of cells per hour; the base design secretes 4.7. Suppose cells could secrete 40 mg/g/h, the ceiling we allow, with today's biology and capital. The cheapest design the model then finds costs $7.1/kg. At 10 mg/g/h it costs $12.0. No published **K. phaffii** process comes near either rate. The best rate shown in a public glucose fed-batch is 2.1 mg/g/h (Prielhofer et al. 2018). At that rate the cheapest design costs $36.2/kg, and even the best case of finding 3 costs $24.5/kg, which is above the whey price band. The same holds for every protein here: none has a published secretion rate of its own.
2. **The fermenter's design sets the titer ceiling.** Suppose the plant feeds Intelligen's 500 g/L glucose and each batch must start at least 30% full. A 200 m³ fermenter at 107 g/L of cells then reaches about 70 g/L in a 72 h production phase. A 600 g/L feed raises this to about 105 g/L.
3. **This is a best case, not a floor.** We set every assumed limit at its best at once, with 400 m³ vessels and 100 kt/yr. The cost per kg is then $4.1 (BLG), $4.2 (OVA), $4.3 (ovomucoid), $4.5 (β-casein); lactoferrin, capped at its largest market estimate, costs $56.9. This assumes secretion at 40 mg/g/h. At 10 mg/g/h the BLG case costs $8.1, and at 5 mg/g/h $12.1.
4. **The vessel price and the fixed-cost rule drive capital.** Aseptic cell-culture vessels instead of industrial ones would add $9.8/kg to the base BLG. The lighter design-report accounting would remove $4.7. Most of that gap comes from fixed costs: under 2% of the investment a year instead of Humbird's 9%. For these reasons, published NREL-style designs come out 1.8× cheaper than our model of the same process. Our model comes within −4% of Intelligen's food-PF design.
5. **The market decides more than the process.** With the same fermentation and accounting, the commodity proteins cost 1.0–1.2× the top of their incumbents' price bands at the base titer. Lactoferrin lands below its reported range at its design titer, because dairy lactoferrin is scarce rather than a by-product. For casein, the purification route changes the product itself. Precipitated, as dairy casein is, β-casein reaches 90% protein at $22.0/kg, like the one precipitated PF casein on file. The only public **K. phaffii** casein process uses membranes alone and makes a powder that is about 6% β-casein (Section 4.9).
6. **For cultivated-meat albumin, the columns set the cost, not the strain.** We purify albumin to cell-culture grade with the minimal train of the industrial process: cation-exchange capture, then an anion-exchange polish. It then costs $215/kg at the base titer and $205 at 42 g/L. Bulk bovine albumin sells for $40–$70, and GFI's cultivated-meat cost targets assume $10/kg. Resin and the low recovery of the capture step (56%) carry most of the cost, even at the longest resin life on record. Capture alone would cost $143, and a $100/L polymeric resin on both columns $114. Membranes alone would cost $34.3, but they reach only 77% albumin, short of the 96% grade. Cultivated meat uses 9–47 g of albumin per kg, so at the base cost albumin adds $1.95–$10.23 per kg of meat (Section 4.8).
7. **α-Lactalbumin is a whey protein whose strains lag far behind.** On BLG's membrane route at isolate purity, it costs $33.8/kg at the base titer on a 1 kt/yr plant ($21.2 at BLG's 10 kt/yr). That is inside the reported $25–$129 band per kg of α-lac (grade C), and the isolate grade it matches sells at the top of that

band. The best public titer is 0.6 g/L of human α-lac in the supernatant, or 0.39 g/L of broth (Wang X et al. 2025). In yeast, bovine α-lac has reached only 2 mg/L. At the best public titer, α-lac costs $1,848/kg (Section 4.2).

8. **Notes: casein for cheese, heme, transferrin and binders.**
   - A rennet cheese needs κ-casein at about 19% of its casein. Suppose a separate strain makes κ, and κ precipitates together with α_{s1}. Then κ must reach roughly α_{s1}′s own design titer before the blend costs within 10% of α_{s1}-casein alone ($22.3/kg). Purified on its own column, κ falls short at every titer up to 42 g/L. The best published κ titer is a 0.25 g/L patent example; at that titer the blend costs $567–$573/kg (Section 4.11).
   - Soy leghemoglobin from the best published secreting strain costs $243/kg on a 150 t/yr plant. Impossible uses it at 0.45%, so it adds $1.09 per kg of beef analogue at the factory gate, before transport and margins. That equals 7% of the retail price of US ground beef (Section 4.12).
   - Recombinant transferrin for cultivated-meat media, made at food scale at today's best titer, adds $0.35 per kg of cultivated meat. At catalogue prices it would add $81 (Section 4.13).
   - As a binder in plant-based meat, PF ovalbumin costs about what dried egg white does, but 13 times as much as methylcellulose per kg of analogue (Section 4.14).
9. **Sugar is not PF's problem.** Charge all of an animal's feed to its protein. The base case's glucose then costs no more than chickens', hens' or cows' feed per kg of protein: $4.1/kg BLG (19% of its cost) against $4.7–$6.6 for the animals in the US. PF's glucose floor, 1.8 kg ($0.71) per kg of protein, sits 7–9 times below the animals' feed. The gap lies in the plant (Section 4.10).

The least-tested assumption is secretion without growth. The production phase follows Intelligen's design, in which cells secrete 4.7 mg/g/h while barely growing. We found one study near zero growth: a **K. phaffii** strain that secretes an antibody fragment (Rebnegger et al. 2024). There, secretion fell as growth slowed and settled about five times below the trend at faster growth; the authors link part of the fall to the promoter they used. Across the literature, **K. phaffii** secretes fastest while it grows. A review compiles maximal rates of 0.004–0.69 mg/g/h, with the glucose ones peaking at growth rates of 0.15–0.20/h (Looser et al. 2015). The best glucose fed-batch we found reached 2.1 mg/g/h near 0.1/h (Prielhofer et al. 2018). Growth during production does not rescue the design. That strain makes 21 mg of protein per g of new cells, so 30 g/L would need 1.4 kg of new cells per litre. No public study we found shows secretion at the design rate without growth.

Scenario	BLG	OVA	ovomucoid	β-casein	lactoferrin	albumin	α-lac
Best public academic strain	$43.0 ($32.5)	$227 ($163)	$479 ($385)	$803 ($627)	$413 ($315)	$424 ($345)	$1,848 ($1,427)
Base case (72 h production; lactoferrin 83 h)	$21.2 ($16.4)	$21.3 ($16.5)	$21.3 ($16.5)	$22.0 ($17.0)	$168 ($130)	$215 ($185)	$33.8 ($25.0)
Industry benchmark (42 g/L, titer in private cost models)	$15.8 ($12.3)	$15.9 ($12.4)	$16.0 ($12.4)	$16.4 ($12.6)	$106 ($85.1)	$205 ($177)	$28.2 ($20.8)
Best case at the best demonstrated secretion (2.1 mg/g/h)	$24.5	$24.6	$24.6	$25.4	$83.7	$211	$25.5
Physical-limit best case (secretion 40 mg/g/h)	$4.1 ($3.3)	$4.2 ($3.4)	$4.3 ($3.4)	$4.5 ($3.5)	$56.9 ($48.5)	$172 ($153)	$5.2 ($4.0)
Base case with aseptic cell-culture vessels	$30.9	$31.1	$31.1	$32.1	$207	$231	$43.7
Base case as a pioneer plant (capital × 1.28, RAND)	$24.4	$24.6	$24.6	$25.5	$192	$234	$39.7
Incumbent, per kg protein (2025$)	$8.7–$17.8	$9.7–$21.2	$9.7–$21.2	$7.9–$18.7	$250–$1,585	$40.0–$70.0	$25.0–$129
Plant output (base), t/yr	10,000	10,000	10,000	10,000	250	1,000	1,000

Table 1: Minimum selling price ($/kg target protein, 2025) with a capital charge of 14.6% of total capital investment per year: price-anchored capital, with design-report capital in brackets (Section 2.5). US Midwest; physical-limit cases at 100 kt/yr (lactoferrin at its largest market estimate; for albumin, GFI's high 2030 demand scenario). Best public titers: BLG 13.9 g/L (Yao et al. 2026, ethanol-fed production priced as glucose; on glucose, Li et al. 2026, 5.14 g/L: $107/kg); OVA 2 g/L in **T. reesei**, full-length and clipped summed (1 g/L of intact OVA: $452/kg); ovomucoid 0.93 g/L (Kim et al. 2025; EVERY's 30 g/L patent example: $23.7/kg); β-casein 1.0 g/L made inside **K. phaffii** cells and costed as if secreted (Choi and Jiménez-Flores 1996; a 3 g/L **E. coli** patent example: $198/kg); lactoferrin 2.8 g/L, porcine, the best methanol-free; albumin 8.7 g/L of whole broth (17.47 g/L of supernatant) after 192 h of methanol induction, costed on methanol (Zhu et al. 2021; the best methanol-free, 1.5 g/L in a **S. cerevisiae** patent: $997/kg); α-lac 0.39 g/L of whole broth (0.6 g/L of human α-lac in the supernatant) after 90 h on ethanol, priced as glucose (Wang X et al. 2025). α-lac's plant is 1 kt/yr; its physical limit is capped at the α-lac market, about 19 kt/yr of α-lac. Incumbents: whey protein, dried egg white and casein trade unit values 2015–2025; lactoferrin, reported prices 2013–2025; albumin, bulk bovine serum albumin trade values (2023–2025); α-lac, market-report prices (2025; all three grade C where reported).

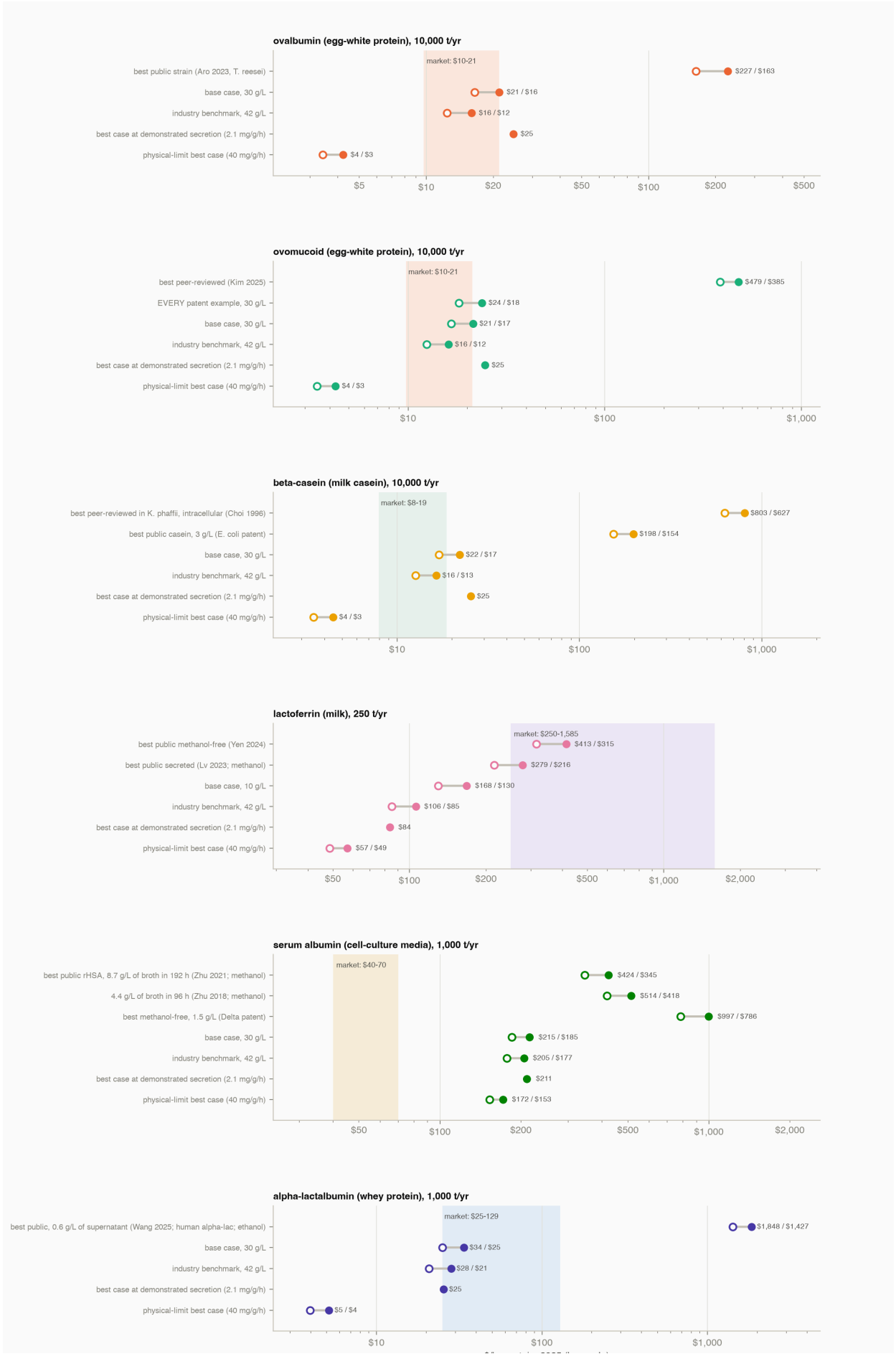


Figure 1: Every case against its market, one panel per protein: filled dots use price-anchored capital (the base), hollow dots design-report capital; the shaded band is the incumbent's price range in 2025 dollars. Model figures are production costs with a return on capital, not selling prices.

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Terms used throughout. *Titer*: grams of product per litre of fermentation broth at harvest. *Secretion rate* (specific productivity): mg of product secreted per g of cells per hour. *Production phase*: the hours after the cells have grown, when they make the product. *MSP* (minimum selling price): the cost per kg including a return on the capital invested (here a 14.6%/yr capital charge), but no margin, sales or transport. *Ex works*: at the factory gate. *DSP* (downstream processing): everything after the fermenter, from cell removal to the dried powder. *UF/DF*: ultrafiltration and diafiltration, membranes that concentrate and wash the protein. *CEX/AEX*: cation- and anion-exchange chromatography columns. k_La: how fast a fermenter transfers oxygen into the liquid. *Nth plant*: a mature plant built with known technology, not the first of its kind. *Grades A–E*: the evidence behind each input, from peer-reviewed measurement (A) to guess (E). *P10–P90*: the range holding 80% of Monte Carlo outcomes.

1 Question and approach

PF companies now sell BLG (Perfect Day, Remilk, Vivici), OVA (EVERY) and lactoferrin (TurtleTree, All G), others are entering (Onego Bio), and the first PF caseins are before regulators (Eden Brew in Australia and New Zealand, Formo in the US). Recombinant serum albumin is already made in yeast and rice for cell-culture media and drugs (Sartorius/Albumedix, InVitria, Healthgen), at catalogue prices of tens of thousands of dollars per kg, and α-lactalbumin is in development (21st.BIO with Novonesis). Whether these proteins can displace animal proteins at scale or stay a premium niche depends on what they cost to make. Published estimates disagree by a factor of several, and company claims are rarely checked against the physics of aerobic fermentation. This study asks:

> What does it cost to make food-grade BLG, α-lactalbumin, OVA, ovomucoid, β-casein and lactoferrin, and serum albumin for cultivated-meat media, by precision fermentation with today's public strain performance, and at the physical limits of aerobic fermentation? What combination of titer, rate, yield, downstream design and scale would bring them to price parity with the whey, egg-white, casein, lactoferrin and bulk albumin they would replace?

#	Sub-question	Answered in
Q1	What do the proteins cost at public (academic) and claimed (industry) performance?	Section 4.2, Figure 3
Q2	What drives the cost, and how good is the evidence for each driver?	Section 4.3, Figure 5, Figure 6, Figure 7
Q3	Which combinations of performance reach price targets across each market ($10–50 per kg protein; lactoferrin $100–400; albumin $10–100; α-lac $25–130)?	Section 4.4, Figure 8
Q4	What is the cost at the physical limits?	Section 4.5, Figure 9
Q5	Are published company claims consistent with the fermenter's limits?	Section 4.6, Table 14
Q6	How much do siting (US vs EU) and scale (1–100 kt/yr) matter?	Section 4.7, Figure 10
Q7	*Added after v1:* what do these costs mean per litre of milk, per egg and (albumin) per kg of cultivated meat?	Section 4.8, Figure 11
Q8	*Added after v1:* how does the downstream route change the product and its cost?	Section 4.9, Figure 12
Notes	*Added in v4 and v4.1:* how does PF's feedstock compare with the animals' feed; what does casein that sets a rennet curd cost; what do heme, transferrin and a protein binder add per kg of food?	Section 4.10, Section 4.11, Section 4.12, Section 4.13, Section 4.14

Table 2: Q1–Q6 were fixed for BLG and OVA before any model output existed (`docs/SCOPE.md`); Q7–Q8 were added at the author's request after the first draft; version 3 extends every question to ovomucoid, β-casein and lactoferrin, and version 4 to serum albumin and α-lactalbumin.

Why these benchmarks. BLG is the main protein of whey, and the PF BLG on the market is sold as a whey-protein ingredient for bars, drinks and dairy analogues (FDA GRN 1056 specifies a concentrate of at least 70% protein). A buyer's alternative is whey protein concentrate or isolate, so their price is the one PF BLG must meet; the same logic sets dried egg white against OVA and ovomucoid, dairy casein and caseinates against β-casein, and dairy lactoferrin and α-lactalbumin against their PF versions. α-Lac is fractionated from whey into enriched ingredients (about 45% of the protein) and isolates (90–96%) for infant formula, which it brings closer

to human milk; it has no trade code, so its price comes from market reports (grade C). For whey it is a hard comparison. Whey is a by-product of cheese, sold for whatever clears the market once cheese has paid for the milk, and it is among the cheapest animal proteins; if PF displaced much of it, its price would fall further. Casein is the opposite case: it is what cheese is made of, and the milk is bought largely for it, so displacing casein would cut milk output far more than displacing whey (Section 4.8). Lactoferrin is scarce (about 0.3–1.6 kt/yr worldwide) and sells for hundreds of dollars per kg. Two consequences follow. Price parity is not the only commercial route: pure, animal-free or functional proteins can sell at a premium. And the products differ in what displacing them would do to animal numbers, which a cost model cannot settle but should state (Section 4.8). Albumin is different again: it is not a food. Cultivated-meat media use it at about 1 g/L (GFI 2023), and recombinant proteins are almost all of a serum-free medium's cost at today's prices. Its incumbent is bovine serum albumin, a slaughter by-product traded in bulk at tens of dollars per kg for technical grades; GFI's cost targets for cultivated meat assume $10/kg. Its leverage on animals is indirect, through making cultivated meat cheaper, and it is shrinking as firms remove albumin from their media (Wildtype's FDA file says it is "no longer used"; rapeseed protein, methylcellulose and cyclodextrins replace it in published media).

Why Humbird's method. Humbird's (2021) cultured-meat analysis builds costs from conservation laws: element and electron balances, oxygen transfer, heat removal. His full cost model reproduces the market-implied cost of one fermented commodity, baker's yeast at about $1.90/kg; capital-related charges are less than half of that cost, so the check constrains his capital factors only loosely (Section 3). We port his baker's-yeast workbook exactly and extend it with a secreted protein, a downstream train, and the specifics of **K. phaffii** and **Trichoderma reesei**.

Rules we followed (written protocol, `PLAN.md`):

- Four scenarios are never merged: academic, industry, required and physical limit.
- Every input carries an evidence grade, A to E.
- The model structure is frozen before the calibration anchors are run; misses are recorded, not tuned away.
- Priors about the results were written down before the model existed; they are scored in Section 7.
- Model-to-model agreement is a consistency check, not validation.
- After the first draft, eight independent reviewers audited it; their findings and every resulting change are logged in `docs/AUDIT.md` (R1–R8, P-1 to P-13). Version 2 was then read by four simulated expert reviewers (AI); their reviews are kept verbatim in `docs/ai-review/` and answered comment by comment in `docs/REVIEW.md`.

2 Methods

2.1 Plant, products and functional unit

The functional unit is 1 kg of target protein in a spray-dried powder that meets the product specification. The base specification is a **concentrate**, following Remilk's GRAS notice for BLG (FDA GRN 1056): at least 70 wt% protein, at least 80% of it BLG, at most 7% moisture. OVA uses the same specification by analogy. The **isolate** variant (99% target protein, dry basis) adds an ammonium-sulfate precipitation, following Intelligen's (2025) OVA design. Ovomucoid follows the cleared product itself, EVERY's GRN 967: a UF/DF concentrate of at least 75% protein, 66–74% of the powder ovomucoid (87.6–92.6% of its protein; the rest is yeast protein). β-casein follows the only precipitated PF casein on file, Formo's α_{s1}-casein (FDA GRN 1312, pending): at least 90% protein, more than 97% of it the casein, at most 5% moisture, sold here as sodium caseinate. Lactoferrin follows the stricter of the recombinant bovine lactoferrin notices, All G's FDA GRN 1284: at least 90% protein, at least 95% of it lactoferrin (TurtleTree's GRN 1219 allows 80% protein). Albumin has no food-grade or media-grade specification; we use the cell-culture grade of recombinant human albumin (InVitria's Cellastim S: at least 96% pure, at most 1 endotoxin unit per mg, the latter not modelled because yeasts make no endotoxin). α-Lactalbumin follows the one high-purity α-lac specification in the GRAS inventory, Agropur's dairy isolate (FDA GRN 763): at least 95% protein, at least 90% of it α-lac, so 85.5% of the powder at the minimum and 90% at its typical 95% α-lac, which we use.

The plant is greenfield, single-product and nth-of-a-kind, built to food (not pharmaceutical) good manufacturing practice. The base is 10 kt/yr of protein in the US Midwest, swept from 1 to 100 kt/yr, with a north-west

Europe (Netherlands) variant. Lactoferrin's base is 250 t/yr, its published design's output (Intelligen), swept from 0.1 to 10 kt/yr; a 10 kt/yr lactoferrin plant would exceed the world market several times over. Albumin's base is 1 kt/yr (a design choice, grade D): about today's whole human-albumin market and over a quarter of the lowest of GFI's 2030 cultivated-meat demand scenarios (3.6–100 kt/yr). α-Lac's base is also 1 kt/yr (grade D): about 5% of the α-lac market, which GFI Europe (2025, C) puts at 31.8 kt/yr of a food-grade ingredient or 3.7 kt/yr of a high-purity one, the same $462 million priced two ways; at about 60% α-lac, the first is about 19 kt/yr of α-lac (our arithmetic). Money is 2025 US dollars. Excluded costs: R&D, strain licences, regulatory approval, sales and distribution, land, first-of-a-kind premiums (shown separately: Section 2.5), carbon pricing, start-up and commissioning, working capital (except in the discounted-cash-flow price), failed batches, per-lot release testing and packaging. A reviewer (AI) put the last four at roughly $1–2/kg together. Figure 2 shows the base-case plant.

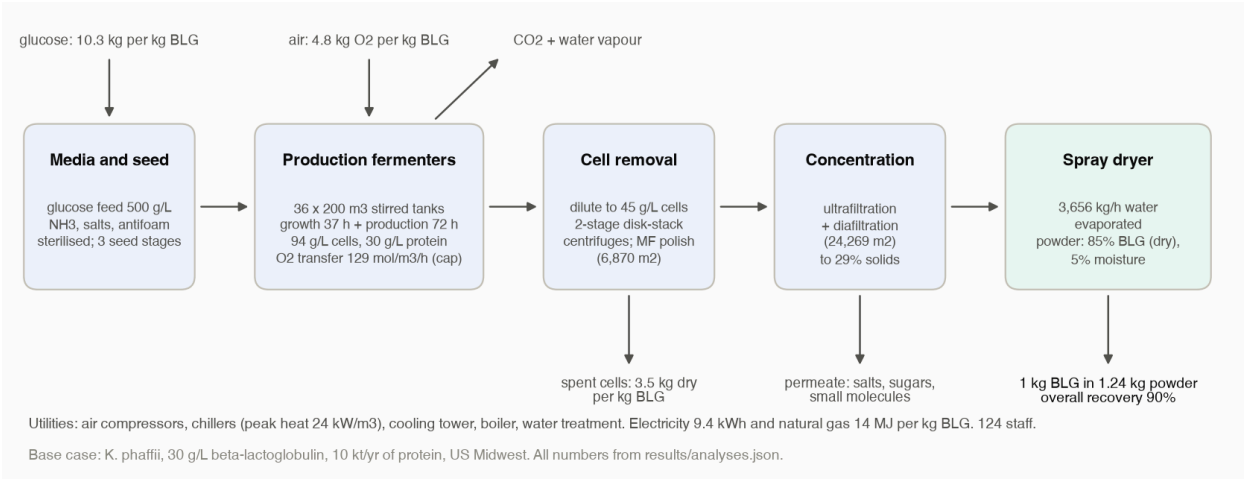


Figure 2: Base-case process: **K. phaffii** secreting BLG at 30 g/L; flows per kg of BLG. α-Lac, OVA and ovomucoid use the same train; β-casein replaces the membrane concentration with isoelectric precipitation, and lactoferrin adds cation exchange before it and albumin cation then anion exchange (Section 2.4, Figure 12).

2.2 Stoichiometry and energetics

Each metabolic process is one element-balanced "black-box" reaction on C, H, O and N: glucose + NH_3 + O_2 ⟶ biomass (or protein) + CO_2 + H_2O. Electrons are tracked through the degree of reduction γ per C-mol (glucose 4, **K. phaffii** biomass 4.06, BLG 4.24, OVA 4.18, ovomucoid 4.00, β-casein 4.25, lactoferrin 4.12, albumin 4.15, α-lac 4.13). This closes the O_2 and CO_2 balances once the carbon split is fixed.

Growth. Biomass yield comes from the Gibbs-energy dissipation correlation of Heijnen and van Dijken (1992), as in Humbird's workbook. The energy dissipated per C-mol of biomass formed is

$$D = 200 + 18(6 - c)^{1.8} + \exp\left\{\left[(3.8 - \gamma_s)^2\right]^{0.16}(3.6 + 0.4c)\right\} \quad \text{kJ/C-mol},$$

where c and γ_s are the carbon number and degree of reduction per C-mol of the substrate (valid for $1 \le c \le 6$). The catabolic reaction (glucose to CO_2) is run just enough to supply D. For glucose and our **K. phaffii** biomass formula (Carnicer et al. 2009) this gives a maximum biomass yield of 0.57 g/g; the measured value is 0.50–0.56 g/g (`data/evidence/hosts.yaml`, grade A). Biomass is counted as ash-free organic matter; ignoring the 7% ash overstates the yield per g of dry cells, but the same convention understates maintenance per g, and the two nearly cancel in glucose per g of cells (audit R1-F3).

Maintenance. In the growth phase maintenance follows Tijhuis et al. (1993), $m_G = 4.5 \exp[-69000/R(1/T - 1/298)]$ kJ per C-mol biomass per hour: 0.0174 g glucose per g cells per hour at 30 °C, close to the 0.0147–0.0161 measured for protein-producing **K. phaffii** in glucose chemostats (Maurer et al. 2006; Rebnegger et al. 2016). In the production phase, where cells barely grow, maintenance falls: Rebnegger et al. (2016) found it threefold lower at near-zero growth. We use the value measured for a protein-secreting strain at low growth, 0.0044 g/g/h (Rebnegger et al. 2024; range 0.0023–0.0161).

Product. The protein's elemental formula comes from its mature UniProt sequence:

- BLG (P02754, 162 residues): $C_{817}H_{1312}N_{206}O_{248}S_9$, 18.3 kDa;
- OVA (P01012, 385 residues): $C_{1901}H_{2997}N_{499}O_{575}S_{22}$, 42.75 kDa;
- hen ovomucoid (P01005, 186 residues, 9 disulfides): $C_{840}H_{1306}N_{240}O_{292}S_{20}$, 20.1 kDa;
- bovine β-casein, A2 variant (P02666, 209 residues): $C_{1080}H_{1692}N_{268}O_{310}S_6$, 23.6 kDa;
- bovine lactoferrin (P24627, 689 residues): $C_{3358}H_{5251}N_{945}O_{1002}S_{38}$, 76.1 kDa;
- human serum albumin (P02768, 585 residues, 17 disulfides, no N-glycan): $C_{2936}H_{4590}N_{786}O_{889}S_{41}$, 66.4 kDa (bovine albumin is within 0.5% of it in composition);
- bovine α-lactalbumin (P00711, 123 residues, 4 disulfides): $C_{626}H_{958}N_{162}O_{196}S_9$, 14.2 kDa (human α-lac, the protein of every **K. phaffii** titer, is within 0.5% in carbon and 3% in nitrogen);
- for the notes, bovine α_{s1}-casein (P02662, 199 residues): $C_{1035}H_{1587}N_{265}O_{317}S_5$, 23.0 kDa; bovine κ-casein (P02668, 169 residues, 1 disulfide): $C_{849}H_{1321}N_{223}O_{262}S_4$, 19.0 kDa; and soy leghemoglobin (P02236, 144 residues, without its heme): $C_{702}H_{1096}N_{176}O_{212}$, 15.4 kDa.

Glycans and phosphates are left out of the formulas: 3–6% of the mass of glycosylated OVA and 5–10% of lactoferrin, which changes the mass balance by less than 1%. Milk β-casein carries five phosphoserines (1.7% of its mass); yeasts and bacteria do not add them without a co-expressed milk kinase (they lack Fam20C, which does): Formo's **E. coli** casein had none (GRN 1312), nor had a **K. phaffii** β-casein without the kinase (NewMilkBuzz).

The product reaction makes protein from glucose and ammonia and respires enough extra glucose to pay for synthesis, so the product-reaction yield Y_{ps} fixes its O_2 demand. **No measured BLG or OVA yield on sugar exists**; the design values are lumped yields that already include maintenance and some growth (Intelligen 0.28 g/g; Behm et al. 2022 0.223 g/g). We therefore derive Y_{ps} so that the model reproduces Intelligen's expression phase (their Eq. 2: 100 g glucose ⟶ 28 g OVA + 8 g cells + 4 g host proteins):

$$\frac{1}{Y_{ps}} = \frac{100}{28} - \frac{8/28}{Y_x} - \frac{m_{s,\mathrm{prod}}}{q_p} = 3.571 - 0.501 - 0.919$$

with our growth yield $Y_x = 0.5705$ g/g and Intelligen's specific productivity $q_p = 0.00475$ g/g/h, giving $Y_{ps} = 0.465$ g/g. The growth rate in production (0.0014/h) and the cell density at its start (107 g/L) follow from the same design and its volume balance. With chemostat-level maintenance Intelligen's Eq. 2 would be impossible: maintenance alone would use all of its glucose. Three cautions:

- **The inversion is sensitive to maintenance.** Maintenance and product enter as a ratio against a total of only 3.571: at the low end of the measured maintenance (0.0023 g/g/h) it gives $Y_{ps} = 0.39$, and above about 0.0146 it has no solution. With the product made at the energetic ceiling (below), Intelligen's expression phase leaves room for at most 0.0062 g/g/h of production-phase maintenance: above the one measurement at near-zero growth (0.0044, for a strain secreting far less) and below every maintenance coefficient compiled for growing cultures on methanol or glycerol, 0.009–0.032 (Looser et al. 2015, which gives none for glucose).
- **It charges Intelligen's host-cell protein to the target.** Read with the 4 g of host protein as product, the same equation gives 0.53 g/g.
- **It assumes secretion without growth.** The base production phase takes up glucose at only 0.016 g per g cells per hour, a few times maintenance, while secreting 4.7 mg/g/h. In the **K. phaffii** chemostat studies on file, secretion rises with growth rate, linearly or up to an optimum near 0.08–0.1/h (Rebnegger et al. 2014; Garrigós-Martínez et al. 2019; Nieto-Taype et al. 2020). The best glucose fed-batch we found, run with a glucose-regulated promoter, peaked at 2.1 mg/g/h near 0.1/h (Prielhofer et al. 2018: 6.4 g/L in 60 h). A review of **K. phaffii** cultivation compiles maximal secretion rates of 0.004–0.69 mg/g/h; its four entries on glucose (0.05–0.49) all peak at growth rates of 0.15–0.20/h, and it calls such secretion growth-associated (Looser et al. 2015). In the one near-zero-growth study we found secretion settled about fivefold below the trend at faster growth (Rebnegger et al. 2024, 0.02 mg/g/h at doubling times of two months, partly attributed to the promoter). Intelligen's design point is not a measurement.

The model's production phase then runs at an effective yield of 0.277 g/g at the base design point, close to Intelligen's 0.28. The effective yield is an output: it rises with productivity, to 0.32 at 42 g/L and 0.52 in the physical-limit case, and neither is anchored to data. The v1 reading (0.28 as the reaction yield, plus maintenance and growth on top) is the low end of the yield range.

Energetic ceiling. Counting ATP gives an upper bound on Y_{ps}. Each residue costs the peptide bond (4 ATP; Martin et al. 2025), the net cost of making the amino acid from glucose and ammonium (0.29 ATP on average: Stouthamer 1973 as recomputed by Martin et al. 2025, whose note is that the common amino acids yield ATP) and, for the protein's sulfur, sulfate assimilation (3 ATP plus 8 electrons per S). That is 4.79 ATP per residue for BLG, paid by respiring glucose at 4 + 12 × P/O ATP per glucose, with P/O = 1.48 for **K. phaffii** (a genome-scale-model value, Chung et al. 2010), i.e. 21.8 ATP per glucose. The ceiling is 0.563 g/g for BLG, 0.570 for OVA, 0.603 for ovomucoid, 0.557 for β-casein, 0.580 for lactoferrin, 0.576 for albumin and 0.580 for α-lac (BLG 0.582 and OVA 0.590 counting peptide bonds alone, as in version 2). Membrane transport, secretion, folding and turnover are not counted, so it stays an upper bound. The base yield is 83% of it. A different estimate, which treats protein synthesis as if it dissipated energy like growth (Heijnen's correlation applied to the protein), gives 0.50 g/g for BLG; it is not a bound, but it suggests the real ceiling sits below the ATP count.

Methanol (version 4). Many published titers come from strains induced with methanol (**AOX1** promoter), among them every recombinant albumin titer above about 1.5 g/L. Version 3 priced those production phases as glucose. The production phase can now run a fraction of its reactions on methanol (`strain.methanol_frac_prod`: 0 in every base case, 1 for the methanol-induced academic cases). Four pieces of physics change:

- **Carbon.** Methanol carries more electrons per carbon (γ = 6) than protein or biomass (γ ≈ 4.1), so making protein from it leaves surplus electrons, not surplus carbon. Every carbon of the product comes from methanol and the extra electrons go to O_2 yeasts do not use them to fix CO_2 (Heijnen and van Dijken 1992; Kleerebezem and van Loosdrecht 2010). That caps the product yield at about 0.70 g per g methanol on carbon alone.
- **Energy.** The first step, alcohol oxidase, hands methanol's first two electrons to O_2 as hydrogen peroxide and conserves no energy; only the formaldehyde and formate dehydrogenases feed respiration, so a methanol yields 2 × P/O ATP, against 3.63 per carbon of glucose. The product yield on methanol is set at the same fraction of its energetic ceiling as the glucose yield (an assumption, grade D): albumin's ceiling on methanol is 0.531 g/g and its yield 0.428 g/g. Without maintenance the biomass yield on methanol comes out at 0.47 g/g, just above the 0.40 g/g measured in methanol chemostats (Tomàs-Gamisans et al. 2018) and the 0.43 of a pilot fed-batch (Gao et al. 2023, via Shemesh and Fishman 2024), which include maintenance; on glucose the model gives 0.57 against a measured 0.50–0.56.
- **Maintenance** is measured, not derived, because the alcohol-oxidase loss is invisible to a Gibbs-energy correlation: 0.016 g methanol per g cells per hour in a methanol fed-batch (Dietzsch et al. 2011), scaled by the same threefold fall at near-zero growth that the glucose value carries (Rebnegger et al. 2016): 0.0050 g/g/h (grade D; the measured 0.016 is the high end). Per unit of Gibbs energy that is about 1.5–1.7× the glucose maintenance, roughly the share of methanol's energy that alcohol oxidase wastes. Methanol chemostats fitted with a genome-scale model give a lower non-growth demand, 0.44 mmol ATP per g cells per hour (Tomàs-Gamisans et al. 2018), i.e. 0.0048 g/g/h at 2.96 ATP per methanol (our arithmetic). That supports a value well below the measured 0.016, though for a different reason than the near-zero-growth scaling: it does not confirm the scaling itself.
- **Feed and heat.** Methanol is fed neat (791 g/L), with no water, at $0.91/kg. Heat and O_2 follow from the formation enthalpies as for glucose; per mol of O_2 they are nearly the same for both substrates, so methanol costs more through its price per electron, its extra O_2 per kg of product and its maintenance.

Element and electron balances close for any methanol fraction (`tests/test_model.py`). Ethanol, which one BLG strain is fed (Yao et al. 2026), is still priced as glucose. The flammable-solvent handling a methanol plant needs is inside the missing-equipment allowance, not costed separately.

Heat. Heat release is computed from formation enthalpies of the dissolved species: about 480–510 kJ per mol O_2 for growth and maintenance and 520–570 for the product reaction. The fermenter uses its own computed heat when it slows growth to the heat-removal limit and when a design's peak heat is checked; the 460 kJ per mol O_2 of Roels (1983, via Almeida Benalcázar et al. 2025; calorimetry gives 455 ± 15, Maskow et al. 2010) only screens the O_2 limit.

> **Check (reader):** The dissipation formula and its units; the conversion of m_G to 0.0174 g/g/h; the Y_{ps} derivation (`tests/test_model.py::test_y_ps_reproduces_intelligen_eq2`); the ATP ceiling (`src/pftea/plant.py`, `atp_per_residue`;

> `tests/test_model.py::test_energetic_ceiling_components`); the methanol anabolism, which fixes no CO_2, and the 2 × P/O ATP per methanol (`src/pftea/thermo.py`, `anabolic`; `src/pftea/plant.py`, `methanol_yield`). The cost structure rests on these.

2.3 The fed-batch fermenter and its feasibility ceiling

Each production batch has two phases:

1. **Growth** from an inoculum of 0.88 g/L (5% of seed broth at 17.6 g/L, the density in Intelligen's lactoferrin flowsheet) at the maximum specific rate, slowed so that the oxygen uptake rate (OUR) never exceeds what the vessel can transfer and the heat never exceeds what it can remove, up to the production cell density X_{prod}.
2. **Production** for t_{prod} hours at a small growth rate and a constant specific productivity q_p. No product is made during growth, a simplification for constitutive promoters.

The initial liquid volume is solved so that the broth ends at 80% of the vessel, and q_p so that the harvest titer equals the target. A volume balance tracks the feeds (glucose at 500 g/L, about 42 wt%, Intelligen's feed; ammonia at 30 wt%, i.e. 268 g/L), O_2 uptake, CO_2 release and evaporation.

Oxygen transfer. The transfer coefficient follows van 't Riet's correlation for non-coalescing broths, $k_La = 0.002(P/V)^{0.7}U_s^{0.2}1.025^{T-20}$ (s^{-1}, SI units), as in Humbird's microbial fermenters, with a log-mean driving force between the bottom (hydrostatic pressure, fresh air) and the top (back-pressure, depleted gas). For each demanded OUR the model picks the air rate and agitation power that minimise compressor plus refrigeration-weighted agitator power, within 5,000 W/m³ and a superficial gas velocity of 0.1 m/s (grade D). In the base vessel the correlation alone would allow about 376 mol O_2/m³/h, but that extrapolates it well beyond the k_La of about 800 h^{-1} that Humbird gives as the practical upper end; the 25 kW/m³ heat-removal ceiling allows about 187. The design cap is Humbird's value for an air-sparged stirred tank, "an OTR approaching 200 mol O_2/m³-h is common", on his own basis, per m³ of cell-free liquid (his Eq. 4.2): 129 mol/m³/h per m³ of broth at the base cell density, 147 at 80 g/L and 100 at 150 g/L. Growth runs at it. (Version 2 used 150 per m³ of broth; 150 is Humbird's limit for bubble columns.) An industrial designer may plan on 100–120 per m³ of broth with air at these cell densities (review BE-7), and our aeration power is lower than NREL's and Intelligen's (Section 3).

Compressor, and a check on aeration power. The compressor raises the air adiabatically to the bottom pressure at 80% efficiency. Humbird's workbook formula puts $k = 1.4$ where $(k-1)/k = 0.286$ belongs and gives about 1.6× that power; we keep it only in the port of his workbook, and in our model it would add $0.02/kg (review H1). With his formula, our aeration reproduces his own 2021 aeration sheet: 0.50 kWh of compressor and agitator power per kg O_2 at 150 mol/m³/h in a 200 m³ vessel, against his 0.50; with the corrected formula, 0.39. Humbird, Davis and McMillan's (2017) process simulations imply about twice his 2021 figure (their Fig. 3 total less the chilling of reaction heat, our reading of a chart); their model could not hold dissolved O_2 down, as production plants do. Aeration electricity costs $0.15 per kg of protein in the base, so doubling it would add about as much; holding dissolved O_2 at 50% of air saturation adds $0.26/kg, and leaving out the temperature term of the k_La correlation, as the 2017 paper does, $0.07/kg.

Other limits. A design is **infeasible**, and never costed as if it worked, when it needs more O_2 or cooling than above, starts below 30% of the vessel volume (the lowest start in a published design, Intelligen's 30.5%), secretes more than 40 mg of protein per g of cells per hour (a judgment, grade D; the base runs at 4.7, and no published **K. phaffii** process is near 40; compiled maxima for proteins secreted by **K. phaffii** are below 1, Looser et al. 2015), exceeds the product's solubility, or has a yield above the energetic ceiling.

Why a ceiling exists. Each kg of BLG made costs about 133 mol O_2 and 9.2 kg of glucose across the batch, including growth and maintenance. The broth must end at 80% of the vessel, so every extra kg of product means more feed and a batch that starts emptier. In the base design (107 g/L cells, a 500 g/L feed) that, not O_2, is what caps the titer: at 70 g/L (72 h) the batch would have to start below 30% of the vessel. Longer production phases lower the ceiling (76 g/L at 36 h, 60 g/L at 120 h), because maintenance keeps consuming feed. The ceiling depends on the design choices behind it (Section 4.4).

What the fermenter model leaves out. Broth viscosity and foam do not lower k_La: the correlation is for water-like broths at every cell density and titer, while Humbird's own method stops a fed-batch at a viscosity

(2 cP) his broth model reaches near 80 g/L cells. At the high cell densities and titers of the best cases the O_2 limit is therefore generous (review FS-6, HM-4). Specific productivity is constant through the production phase, whereas real secreting cultures often lose productivity after two or three days; and no product is made during growth, which is conservative for constitutive promoters.

2.4 Downstream processing

Each protein has its own route (the specification sets it); all share cell removal, microfiltration and spray drying. The membrane train (Figure 2) follows Intelligen's OVA design, with evidence-graded performance (`data/evidence/dsp.yaml`, `ovomucoid.yaml`, `casein.yaml`, `lactoferrin.yaml`, `albumin.yaml`, `alac.yaml`):

- **Cell removal.** Dilute to 45 g/L cells, then two disk-stack centrifuge stages with re-dilution of the paste. Each paste holds 50% extracellular liquid by volume (Petrides: 40–60%), so a fraction $\varphi = eV_p/(V_1 - V_{wc})$ of the product is trapped per stage and the recovery is $1 - \varphi^2$ (96.9% in the base). The wash water is counted.
- **Microfiltration polish** at 20 L/m²/h (grade D, range 10–50) and 97.5% recovery (Behm et al. 2022).
- **Ultrafiltration and diafiltration.** Concentrate at 30 L/m²/h to 50 g/L, diafilter 3 volumes there at 5 L/m²/h (the only diafiltration flux on file, EVERY patent US 2025/0134130, measured on a redissolved precipitate at 10 °C, so probably conservative for clarified broth), then concentrate to the spray-dryer feed at the same low flux. Rejection 0.995. Skids cost $263 per m² of membrane installed (a dairy plant, Mercier-Bouchard et al. 2017), converted to a purchase cost before the installation factors are applied.
- **Dryer feed.** Dairy plants feed high-protein whey concentrates to the dryer at 27–37% solids by membranes, not evaporators (Tetra Pak); egg white is ultrafiltered to about 23% (Tsai et al. 1977). We use 30% for BLG, 23% for OVA, 23% for ovomucoid (which dissolves clear at 300 g/L; EVERY's own patents dry it from 5–6% solids, the low end of its range) and 27% for lactoferrin, 20% for albumin (sold as 20% solutions, Recombumin) and 30% for α-lac (by analogy with BLG: the calcium-bound α-lac of commerce is soluble from pH 2 to 9, Layman et al. 2018; at 14.2 kDa it needs a tighter ultrafiltration membrane than BLG does, and that membrane's lower flux is not modelled). Sodium caseinate is so viscous that dairy plants dry it from about 20% solids (Tetra Pak), so casein evaporates the most water per kg.
- **Spray dryer** at 4.87 MJ per kg water evaporated (industry survey, grade A); purchase cost from a supplier-based SuperPro model (Yver et al. 2012).
- **Isolate only:** ammonium-sulfate precipitation (350 g/L; 89% recovery), washed basket centrifuges; the dryer is fed the 20%-solids cake.
- **β-casein: concentration, then isoelectric precipitation.** Casein is least soluble near its isoelectric point, which is how dairy acid casein is made (pH 4.6). Formo recovers its PF α_{s1}-casein the same way, at pH 4 with phosphoric acid, from an extract of its cells (GRN 1312). Recombinant β-casein, without its phosphates, has a higher isoelectric point (5.5 by zeta potential), but it is least soluble at pH 4.5, where it behaves like milk β-casein: 22% of it stayed dissolved against 21% for the native protein, while at pH 4 half stayed dissolved (Chezan et al. 2026, a dilute, unheated assay). The model therefore precipitates at pH 4.5. The microfiltered broth is first ultrafiltered, without diafiltration, to the casein concentration of skim milk (26.8 g/L), so that dairy data apply to a similar feed. It is then acidified with H_3PO_4 at 5.65 kg/m³ (milk's buffer capacity per m³ from pH 6.6 to 4.6, an analogy, grade D; range 2.3–17.3), heated to 90 °C for 20 min (in Standing Ovation's patent recombinant β-casein precipitated only partly at 40 °C and fully at 90 °C) with 90% regenerative heat recovery (a guess), recovered at 95% (dairy recovers essentially all its casein; grade D; the assay's 78% is the low end), washed in 3 countercurrent decanter stages with 13 L of water per kg, redissolved with 2% NaOH by weight (Tetra Pak) and spray-dried as sodium caseinate (a design choice: drying the curd as acid casein, as Formo sells it, would be cheaper). Decanters are costed on the disk-stack centrifuge curve (no decanter quote on file); the heater on the HTST sterilizer curve.
- **Lactoferrin: cation-exchange capture** between microfiltration and ultrafiltration, as in the dairy process and every commercial PF lactoferrin: binding capacity 75 g/L of resin and 16 column volumes of buffer per cycle (EVERY patent), elution at 1 M NaCl, 83.2% step recovery, and resin at $1,250/L (Petrides) lasting 1,000 cycles (industrial lactoferrin columns, Pires et al. 2025). The column is loaded with dilute microfiltered broth, 26 column volumes per cycle, so a cycle takes 2.1 h at 20 column volumes per hour, the flow of Intelligen's design. The column hardware per litre of resin is a guess (grade E).
- **Albumin: cation-exchange capture** on the same equipment, with the column data of the industrial recombinant-albumin process (Delta Biotechnology's, now Sartorius's Recombumin; US 5,728,553 and US

6,638,740, grade B). Albumin (isoelectric point about 5) binds a cation exchanger when the clarified broth is brought to pH 4.5, where most yeast proteins and pigments do not, and elutes at pH 5.5 in dilute acetate (priced as NaCl), then a 2 M NaCl strip and 0.5 M NaOH: loading 40 g/L, 34 column volumes of buffer per cycle (the patent's maximum step volumes), and a recovery of 56% over the step, the midpoint of the patent's "anticipated" 46–66% (every public recombinant-albumin train recovers 46–68% overall; one dye-affinity capture step recovered 86%, Zhu et al. 2021, the high end). The capture eluate still holds a 45 kDa albumin fragment, yeast antigens and pigment, which the pharmaceutical train removes with dye affinity, gel filtration and anion exchange for 99.9% purity. For the cell-culture grade (96%) we take the patent's minimal claimed train, the capture followed by an **anion-exchange polish**, loaded at 30 g/L with no loss (the patent reports about 110% monomer yield over it, as dimers revert), at the capture column's resin price, resin life and buffer volumes (grade D); whether two columns suffice is a judgment, and capture alone is shown as the optimistic case (audit R12). The acid for the pH step, the octanoate of the patent's recipe and protease losses while broth waits at pH 4.5 are not costed.
- **Cleaning in place** (7.3 kg solution per kg product, Intelligen; its chemicals are inside Humbird's maintenance charge) and **spent-cell disposal** ($458/t dry cells in the US, a landfill tipping fee; $211/t in the EU, an anaerobic-digestion gate fee; both per wet tonne converted at the 15% dry solids of the model's centrifuge paste, with no dewatering or inactivation costed). Sale as feed or fertiliser would make it free or a credit.

The target protein share of the concentrate powder is 85% on a dry basis (grade D; the GRN 1056 minimum is 56%), and 90% for α-lac (the GRN 763 isolate; whether membranes alone reach it past yeast protein and mannans is as open as for BLG); of the caseinate, 93%; of the lactoferrin powder, 90%; of the albumin powder, 96%. The base overall recovery is 90% for BLG, 90% for ovomucoid, 87% for β-casein, 74% for lactoferrin, 50% for albumin and 90% for α-lac.

OVA solubility. Intelligen stops its ultrafiltration at 50 g/L, "its practical solubility limit", without a source; vendor data sheets give 40–50 mg/mL as quality-control concentrations. OVA dissolves at 200 g/L in water (Wang L et al. 2025) and is processed at about 110 g/L in ultrafiltered egg white (Tsai et al. 1977). We therefore do not treat 50 g/L as a limit, and show it as a sensitivity. No data exist at pH 5, the fermentation pH, which is close to OVA's isoelectric point.

2.5 Capital: vessel prices and two accounting conventions

Fermenter prices. Fermenters are the largest equipment item. The base prices them from a vendor quote: NREL 2011's 300 m³ stainless fermenter with internal cooling coils and an agitator, $0.99M in 2007 dollars, scaled with exponent 0.7, i.e. $1M per 200 m³ in 2025 dollars. It is an industrial enzyme-plant vessel, the closest public analogue of a food-grade PF fermenter. It is also at the vessel price level of Humbird's own baker's-yeast workbook, whose 200 m³ bubble column costs about $0.8M (2018), and consistent with Intelligen's food-PF design (Section 3). Version 2 used Humbird's quote curve for the aseptic stirred tank he designed for cultured meat: 316L, full vacuum, built to the ASME bioprocessing standard. That vessel costs $3M per 200 m³ (2025 dollars), 2.9× the vendor quote. Humbird himself separates aseptic industrial fermentation, which tolerates a few foreign organisms, from aseptic animal-cell culture, which does not. We keep his aseptic vessels as a named high case ("aseptic vessels", +$9.8/kg on the base BLG). Seed fermenters (4.7 and 0.47 m³, 11 of each so that a batch can start every few hours) are priced on the same curve, scaled far below its quote. They are 2.6% of fermenter cost; Humbird's aseptic quotes, which do cover that size, are 2.2× the curve there and would add $0.18/kg.

Vessel size (version 4). Capital buys whole fermenters, so a small plant in large vessels pays for idle volume. The designer therefore picks the cheapest feasible plant over 50, 100, 150 and 200 m³ vessels, at most 200 m³ (the base vessel; published food-PF designs use 100–150 m³), on each product's base case, and keeps that plant in its scenarios, tornado and Monte Carlo; the scale curve re-sizes at each output. The commodity plants at 10 kt/yr keep 200 m³ (larger vessels would be cheaper still: 500 m³ is a sensitivity); lactoferrin's 250 t/yr plant takes 100 m³ vessels, as in Intelligen's design, and the 1 kt/yr plants 100 m³ (albumin) and 100 m³ (α-lac).

Pioneer plants (version 4). The base is an nth-of-a-kind plant. RAND's study of 44 pioneer process plants found that their estimates averaged 0.78 of the actual capital cost, excluding inflation and other external causes (106 estimates; Merrow et al. 1981), and that they produced 72% of design capacity in months 7–12 after start-

up (94% for plants with no new process steps). We show a pioneer case with capital × 1/0.78 = 1.28 (the reciprocal of RAND's mean ratio; the mean of the reciprocals, the average overrun, would be about 1.36 by a first-order estimate, ours: please check), all capital except the resin's first fill, separately and never in the base: $24.4 (BLG), $24.6 (OVA), $24.6 (ovomucoid), $25.5 (β-casein), $192 (lactoferrin), $234 (albumin), $39.7 (α-lac) (Table 1). RAND's regression would predict less cost growth for a plant of commercially proven unit operations; the start-up shortfall is not modelled.

Price-anchored capital (Humbird method; base).

- Other equipment comes from Humbird's cost curves (Aspen ACCE quotes, 2018 dollars). Costs are escalated to 2025 with a factor of 1.33 (CEPCI and the BLS machinery price index; grade B, range 1.23–1.52). The BLS index for heavy-gauge metal tanks rose 1.92× over the same years; escalating the fermenters with it would add $2.3/kg.
- The equipment list is abbreviated (it omits CIP and feed tanks per fermenter, sterile filter housings, sugar storage, biomass inactivation, powder packing and effluent pretreatment), so we add 20% for missing items, the allowance of Humbird's yeast workbook for its own abbreviated list (his fuller cell-culture sheets use 10%).
- Installation, buildings, engineering, fees and contingency follow the "commodity aerobic" facility class that Humbird tabulates (a food-grade plant with the process in a sanitary building). None of his own sheets uses that class: his cultured-meat sheets use the cGMP biopharmaceutical class and his yeast workbook a flat Lang factor of 4.5. The effective Lang factor (investment over equipment cost) here is 6.5, high because buildings are costed by floor area while the fermenters are now cheaper.
- Maintenance is 4% of total capital investment (TCI) per year, including CIP chemicals, and insurance 5%, Humbird's whole fixed-cost rule. There is no separate plant-overhead, quality-control or local-tax line, so the 9% stands for all of them; insurance alone is usually well under 1%.
- What the anchor supports: with these factors and fixed costs Humbird's yeast workbook lands in the cost band implied by the market price (Section 3), and a lighter, NREL-like set does not. Capital-related charges are less than half of the yeast cost, so the check is loose: it favours Humbird's set but cannot pin it.

Design-report capital (NREL-style; sensitivity). The convention of NREL's 2011 design report (whose lead author is also Humbird): the same fermenter quote; NREL's installation factors (1.7 for fermenters, 1.8 otherwise) plus indirect costs of 60% of direct cost; no separate warehouse; maintenance of 3% of installed process-area cost and insurance of 0.7% of fixed capital.

On the base BLG plant the two conventions differ by $4.7/kg: $3.3 of it comes from the fixed-cost rule alone, and $1.4 from the installation and indirect factors.

A Netherlands plant costs 1.09× the same plant in the US Midwest (IEAGHG 2018, same plants costed in both places), applied to capital in both conventions.

2.6 Operating costs, labour and money

Operating costs. Raw materials, utilities and consumables come from the mass and energy balance and the prices in Appendix A: EIA, Eurostat and BLS statistics, trade unit values and published designs (grades A–B).

Labour. Humbird's operator-attention model: operators per fermenter by the fraction of time a batch needs attention, fixed staffing for continuous areas, 4.5 shifts per position, one supervisor per ten staff at 1.4× salary, and a 100% labour burden (Humbird). The base plant employs 124 people, operators and supervisors only; salaried staff are inside the fixed-cost rule (Section 2.5).

Headline metric. MSP_{CCF}: annual operating cost plus a capital charge, divided by annual output. The charge is the capital recovery factor $i/(1-(1+i)^{-n}) = 0.146$ per year at $i = 7.5\%$ and $n = 10$ years (Humbird). It is a cost with a return on capital, not a market price. We also compute the cost of production (COP: operating cost plus depreciation, no return) and an NREL-style discounted-cash-flow price MSP_{DCF} (10% IRR, 30 years, 21% tax, 7-year MACRS, construction over three years, working capital). For base BLG these are $18.9 and $20.7.

2.7 Evidence grades, scenarios and uncertainty

Every parameter and evidence item is graded (Table 3). The four scenarios combine grades differently and are never averaged:

- **academic:** the best public strain performance, read in full text. BLG: 13.88 g/L in **K. phaffii** (Yao et al. 2026), whose production phase runs on ethanol with glutamate feeding; the model has one carbon source and prices it as glucose (per electron the two cost about the same), so this case is flagged: ethanol is more reduced than glucose, so the O_2 demand and heat per kg of product are higher than modelled, and the feasibility check is not conservative for this case. The modelled batch ends at 88 g/L cells against about 120 in Yao's run, because the model's volume balance dilutes; it is a cost estimate for 13.88 g/L in 108 h, not a simulation of that run. On glucose, Li et al. (2026) reach 5.14 g/L in 132 h. Their feed table (supplementary Table S4) gives the only public check on a BLG yield on glucose: 1–2% overall, depending on whether its feed rates refer to the starting or the current broth volume (our arithmetic; the volume profile is not reported). Our model of that run gives 2.5%, so it is, if anything, optimistic on sugar there. OVA: 2 g/L in **T. reesei** (Aro et al. 2023), with no growth in production. That is the sum of full-length and clipped OVA, and the purified product was mostly the clipped 36 kDa form, so intact OVA was probably well under 2 g/L; a 1 g/L variant is shown. The **T. reesei** production phase uses the yeast maintenance correlation, which is a lower bound on its sugar use. Ovomucoid: the best peer-reviewed titer is 0.93 g/L in a 60 h glycerol fed-batch without methanol, at 176 g/L of cells in a 2.5 L vessel (Kim et al. 2025). EVERY's patent example, 30.19 g/L at about 140 h (B; vessel, cell density and feeding not stated), is shown as a variant; it is the only traceable number above 1 g/L (a 3.2 g/L figure in a 2025 review has no traceable source). β-casein: no company discloses a titer, and no secreted casein titer has been published for any yeast or fungus. In **K. phaffii**, β-casein made with methanol reached 0.7–1.0 g/L inside the cells, with only 1/20,000 of that secreted (Choi and Jiménez-Flores 1996), and a recent patent family found that **K. phaffii** secretes β-casein only once a milk kinase (Fam20C) is co-expressed, without giving a titer (NewMilkBuzz, EP 4 623 075). The academic case costs the 1.0 g/L as if secreted, an analogy. A patent example in which **E. coli** makes 3 g/L of β-casein inside its cells (Standing Ovation, quantified by eye on a gel) is shown as a variant. Lactoferrin: 2.8 g/L of porcine lactoferrin in **K. phaffii** without methanol (Yen et al. 2024), an analogy for bovine; 5.4 g/L of human lactoferrin with methanol, measured by ELISA only (Lv et al. 2023), is a variant. Albumin: the highest recombinant-albumin titer published, 17.47 g/L after 192 h of methanol induction in a 5 L vessel (Zhu et al. 2021), is per litre of supernatant; the cells fill half the culture, and the paper's 21.84 g in 2.5 L gives 8.74 g/L of whole broth, the model's basis. The same group's 8.86 g/L of supernatant in 96 h (Zhu et al. 2018), scaled by the same ratio to 4.4 g/L, is a variant, as is the best methanol-free titer, 1.5 g/L in an 8 m³ **S. cerevisiae** sucrose fed-batch (Delta's patent, B; costed as **K. phaffii** on glucose). Version 4 costs every methanol-induced academic case on methanol (albumin; lactoferrin, Lv et al.; β-casein, Choi and Jiménez-Flores) instead of as glucose. α-Lactalbumin: the best titer in any host is 0.60 g/L of human α-lac, His-tagged, in the supernatant of a **K. phaffii** culture in a 3 L bioreactor, fed ethanol for over 90 h at pH 3, which suppressed proteases (Wang X et al. 2025); it is converted to whole broth with the model's cell volume at the base cell density (×0.64, D; only OD was reported) and costed on glucose. Their cells grew poorly at pH 3, so the conversion probably over-discounts, while the authors say their assay underestimates the titer through proteolysis, by an amount not given. Bovine α-lac, the dairy protein, has reached 2 mg/L in yeast (Viaene et al. 1991): the protein unfolds and sticks to cells below pH 5.5, so solubility and protease control pull the culture pH in opposite directions. Yield and maintenance keep the base values: no measured product yield exists.
- **industry:** 42 g/L, the average that Hawkwood computed over private companies' models of unspecified secreted proteins for GFI (2025) (grade C), with everything else at each product's base case. It is not a measured titer for any of the seven, and least informative for casein, lactoferrin and α-lac, whose public titers are one to two orders of magnitude lower.
- **required:** the inverse problem: the titer needed for $10, $20 and $50/kg (lactoferrin $100, $200 and $400; albumin $10, GFI's target, $50, inside the bulk-albumin band, and $100; α-lac $25 and $130, the ends of its band per kg of α-lac ($25–129) rounded, and $60) at each production time.
- **physical limit:** a best case at assumed limits, not a floor set by physics. Every lever moved from the base: product yield 0.535 g/g (95% of BLG's energetic ceiling); the lowest measured maintenance in both phases; no growth in production; a 600 g/L glucose feed (Shemesh and Fishman 2024; 730 g/L has been fed in a 2.6 L laboratory vessel, Heyland et al. 2010, which would raise the ceiling somewhat further); 400 m³ vessels; O_2 transfer capped at 250 mol/m³/h per m³ of cell-free liquid (head pressure, not costed; grade D) and heat removal at 40 kW/m³ (external exchangers, not costed; grade D); an 8 h turnaround; a 90% concentrate; contract sugar; and 100 kt/yr (lactoferrin: 1,588 t/yr, the larger of two world-market estimates; α-lac: about

19 kt/yr, its market; for albumin, 100 kt/yr is GFI's high 2030 demand scenario). The yield is the same for every protein; it is 96% of β-casein's ceiling, the lowest. A search over cell density and production time puts the titer at its feasibility ceiling, where cost is lowest, and reports which limits bind. Secretion stays capped at 40 mg/g/h, and the result is shown against that cap from 5 to 120 mg/g/h.

Grade	Meaning
A	Measured, primary source or official statistic (peer-reviewed data, EIA, Eurostat, BLS, USDA)
B	Design value or patent example from a named source (NREL, Intelligen, Humbird, patents)
C	Company claim, press release, industry survey, secondary transcription
D	Analogy, judgment or calculation from graded inputs
E	Guess; listed in Section 6

Table 3: Evidence grades (`data/evidence/README.md`).

Uncertainty. Two tools:

- **Tornado.** Each parameter with a graded range moves to its low and high value, one at a time. An end that breaks a physical limit is moved to the last feasible value, so no bar quotes the cost of an impossible plant. The accounting convention and the fermenter price basis are shown as their own (structural) bars.
- **Monte Carlo.** Triangular draws over each scenario's uncertain parameters, with host-specific ranges. Draws that break a physical limit, or for which no plant can be built, count as infeasible; we report the feasible fraction, the limit each infeasible draw breaks, and percentiles over feasible plants and over all draws with a cost. Draws are independent: no data exist to set correlations, and the feasibility filter is the only coupling. Structural choices (conventions, vessel price basis, minimum fill, secretion ceiling) are not sampled; they are shown one at a time. Version 4 adds the global counterpart of the tornado: the rank correlation (Spearman's ρ) of each sampled input with the cost over the feasible draws. With independent inputs, ρ^2 approximates each input's share of the spread in the cost's ranks.

3 Calibration

The structure was frozen first. Each anchor then changes only its own process inputs, prices, cost year and financing, and misses are recorded in `docs/VALIDATION.md`. `tests/test_calibration.py` pins each recorded value to within five points, so a change that moves an anchor cannot pass silently.

Anchor (type)	Quantity	Ours	Theirs	Diff.
1. Humbird baker's yeast (software + reality)	cost, $/kg dry cells (2018)	1.890	1.890	+0.00%
2. NREL 2011 on-site cellulase (consistency)	glucose, kg/kg protein	3.87	4.17	−7.1%
	O_2, mol/kg protein	83.1	84	−1.0%
	electricity, kWh/kg	4.9	9	−45%
	cost, $/kg protein (2007), price-anchored	7.81	4.24	+84%
	cost, design-report (NREL's own) capital	6.08	4.24	+43%
3. Intelligen 2025 OVA (consistency)	cost, $/kg powder, price-anchored	42.00	43.52	−4%
	total capital investment	$114M	$92M	+24%
	glucose, kg/kg powder	10.6	13.0	−19%
	electricity, kWh/kg powder	9.6	27.1	−65%
4. Voutilainen 2021 recombinant case (consistency)	sugar, kg/kg protein (by construction)	6.28	6.15	+2.2%
	cost, €/t protein, their DCF, price-anchored	16,345	9,007	+81%
	their fixed costs, price-anchored capital	12,329	9,007	+37%
	their fixed costs, design-report capital	11,423	9,007	+27%

Table 4: Calibration anchors. Anchor 1 is Humbird's own workbook, ported line by line; anchors 2–4 are other models run on their own inputs. Diagnoses: `docs/VALIDATION.md`.

What the anchors say.

1. **The port is exact.** It reproduces Humbird's baker's-yeast cost to 0.004%, and that cost sits on his market-implied $1.90/kg. The market band itself depends on how his 33% margin is read: $1.47–$1.88/kg if it is a share of the price, $1.65–$2.11 if it is a mark-up on cost. With a lighter, NREL-like capital set (investment 3.2× listed equipment, fixed costs 1.9% a year) the workbook gives $1.37/kg, below both bands; with Humbird's factors but NREL-like fixed costs, $1.62; with our PF capital charge (7.5% over 10 years instead of his 10% over 20), $2.00. The anchor favours Humbird's factors and fixed costs, loosely. It says nothing about fermenter prices beyond its own bubble column.
2. **Sugar and O_2 agree with the design that states them independently** (NREL, −7% and −1%). Voutilainen's sugar is matched by construction, and Intelligen's expression phase is, too, because our product yield is derived from it; its growth phase uses 0.40 g/g, below our 0.53 effective and the measured 0.50–0.56. **Electricity does not agree**: ours is 45–65% lower, because our aeration is power-optimised while NREL sizes agitators for peak load and Intelligen aerates at 1.2–1.6 vvm and runs a fertiliser section; a real plant may need more aeration power than ours. These are consistency checks, not calibration fits.
3. **Capital now agrees; fixed costs do not.** With vendor-quote fermenters our investment for Voutilainen's plant is within +2% of theirs, and for Intelligen's +24% above (their Lang factor is 4, ours higher). Their total costs are still 1.8× lower than ours (NREL, Voutilainen), mainly because their fixed costs are 1.5–2% of capital per year instead of 9%, and their capital charges lighter. With their conventions our model still lands above theirs, by +27% (Voutilainen) and +43% (NREL); the rest is mostly labour, buildings and cleaning.
4. **Intelligen's total is within −4%** (their cost recast into our capital charge and fixed costs), with compensating differences: our capital is higher and our plant needs 6 whole fermenters where theirs has five; our glucose and electricity are lower.

We keep Humbird's factors and fixed costs as the base because his workbook with them lands in the yeast market band and a lighter set does not; that is a loose test, and the design-report convention is shown everywhere as a named structural sensitivity, not as an error band. NREL itself notes that enzyme producers quoted costs about 47% above its estimate.

4 Results

4.1 Base case

Cost item, $/kg protein	BLG	OVA	ovomu-coid	β-casein	lactofer-rin	albumin	α-lac
capital charge (14.6%/yr)	$7.14	$7.20	$7.23	$7.50	$54.86	$45.37	$12.96
maintenance	$1.96	$1.98	$1.98	$2.06	$15.06	$12.46	$3.56
insurance (Humbird's 5%, all overheads)	$2.45	$2.47	$2.48	$2.57	$18.83	$15.57	$4.45
glucose	$4.11	$4.10	$4.10	$4.21	$13.96	$7.40	$4.11
labour	$1.55	$1.55	$1.55	$1.77	$20.77	$6.24	$4.74
spent-cell disposal	$1.60	$1.60	$1.59	$1.64	$6.40	$2.88	$1.60
electricity	$0.81	$0.81	$0.83	$0.79	$2.83	$1.52	$0.84
media, ammonia and DSP chemicals	$0.64	$0.64	$0.64	$0.86	$11.37	$12.08	$0.64
gas, steam, water, membranes, resin	$0.91	$0.93	$0.94	$0.65	$23.72	$111.96	$0.91
MSP_{CCF}	**$21.17**	**$21.28**	**$21.35**	**$22.05**	**$167.79**	**$215.48**	**$33.81**
per kg powder	$17.10	$17.18	$15.41	$19.38	$143.46	$196.51	$28.91
target protein, % of powder (dry)	85%	85%	76%	93%	90%	96%	90%
overall recovery	90%	90%	90%	87%	74%	50%	90%
plant output, t/yr	10,000	10,000	10,000	10,000	250	1,000	1,000
titer, g/L	30	30	30	30	10	30	30
production fermenters	36 × 200 m³	36 × 200 m³	36 × 200 m³	37 × 200 m³	8 × 100 m³	13 × 100 m³	8 × 100 m³
total capital investment	$490M	$494M	$496M	$515M	$94M	$311M	$89M
glucose, kg per kg protein	10.3	10.3	10.2	10.5	34.9	18.5	10.3
electricity, kWh per kg	9.4	9.4	9.7	9.2	32.8	17.7	9.8
natural gas, MJ per kg	14	20	23	21	15	22	13
spent cells, kg dry per kg	3.5	3.5	3.5	3.6	14.0	6.3	3.5
staff per kt/yr	12.4	12.3	12.3	14.1	165.8	49.9	37.8

Table 5: Base case: **K. phaffii**, 72 h production (lactoferrin 83 h), US Midwest, vendor-quote fermenters, price-anchored capital. BLG, OVA, ovomucoid and α-lac: UF/DF concentrate; β-casein: UF then isoelectric precipitation to sodium caseinate; lactoferrin: cation exchange; albumin: cation exchange then an anion-exchange polish. Fermenters are sized to each plant (Section 2.5).

Capital-related charges (capital charge, maintenance and insurance) are 55% of the BLG price; glucose is 19% and spent-cell disposal 8%.

- **Why so much capital.** At 30 g/L the plant needs 36 production fermenters of 200 m³. Each batch occupies a vessel for about 121 h including turnaround and yields 4,299 kg of protein.
- **Why glucose is modest.** The overall yield of protein on glucose is only 0.097 g/g, because the cells, their maintenance and the product reaction all consume sugar; at $0.40/kg glucose it is still only 19% of the price.
- **Why the commodity proteins differ so little.** BLG, OVA, ovomucoid and β-casein share the fermentation; they differ only in their downstream route, purity and recovery. Ovomucoid costs +$0.1 against OVA on the same membrane route: its cleared product is a less pure concentrate (76% against 85% of dry matter), so more solids are dried per kg of ovomucoid. β-casein costs $0.9 more than BLG: the precipitation plant and its heater add capital, the caseinate dries from a dilute feed, and the acid is a new chemical line, while diafiltration is saved. Lactoferrin costs 8× BLG: one third of the titer, a longer batch, a cation-exchange step and a plant 40 times smaller.
- **Why albumin costs 10× BLG at the same titer.** Its columns are the cost: resin alone is $104 per kg (a $1,250/L resin lasting 1,000 cycles at 40 g of albumin per litre loaded is $31 per kg loaded on the capture column, and about two kg are loaded per kg sold; the polish column adds the rest at its 30 g/L loading), and with the capture step's 56% recovery only half of what is fermented is sold. The plant is also ten times smaller than BLG's. Resin life is the softest number: the base takes the one industrial measurement, over 1,000 cycles on whey lactoferrin, for a column fed pigmented yeast broth; at 125 cycles the base costs $941/kg.

- **Why α-lac costs more than BLG on the same route.** Its plant is ten times smaller: at BLG's 10 kt/yr it would cost $21.2/kg, as BLG does ($21.2). On the membrane route the model does not charge for purity: the target share only sets how much other dry matter travels to the dryer with the protein, so α-lac's range, 70% to 96% of dry matter, moves it only from $34.0 to $33.8. Whether membranes alone reach the isolate's 90% past yeast protein and mannans is the open question, not what it costs.

The product made per m³ of fermenter per hour is the lever the rest of the paper explores.

4.2 Q1: cost against titer, with every public and claimed data point

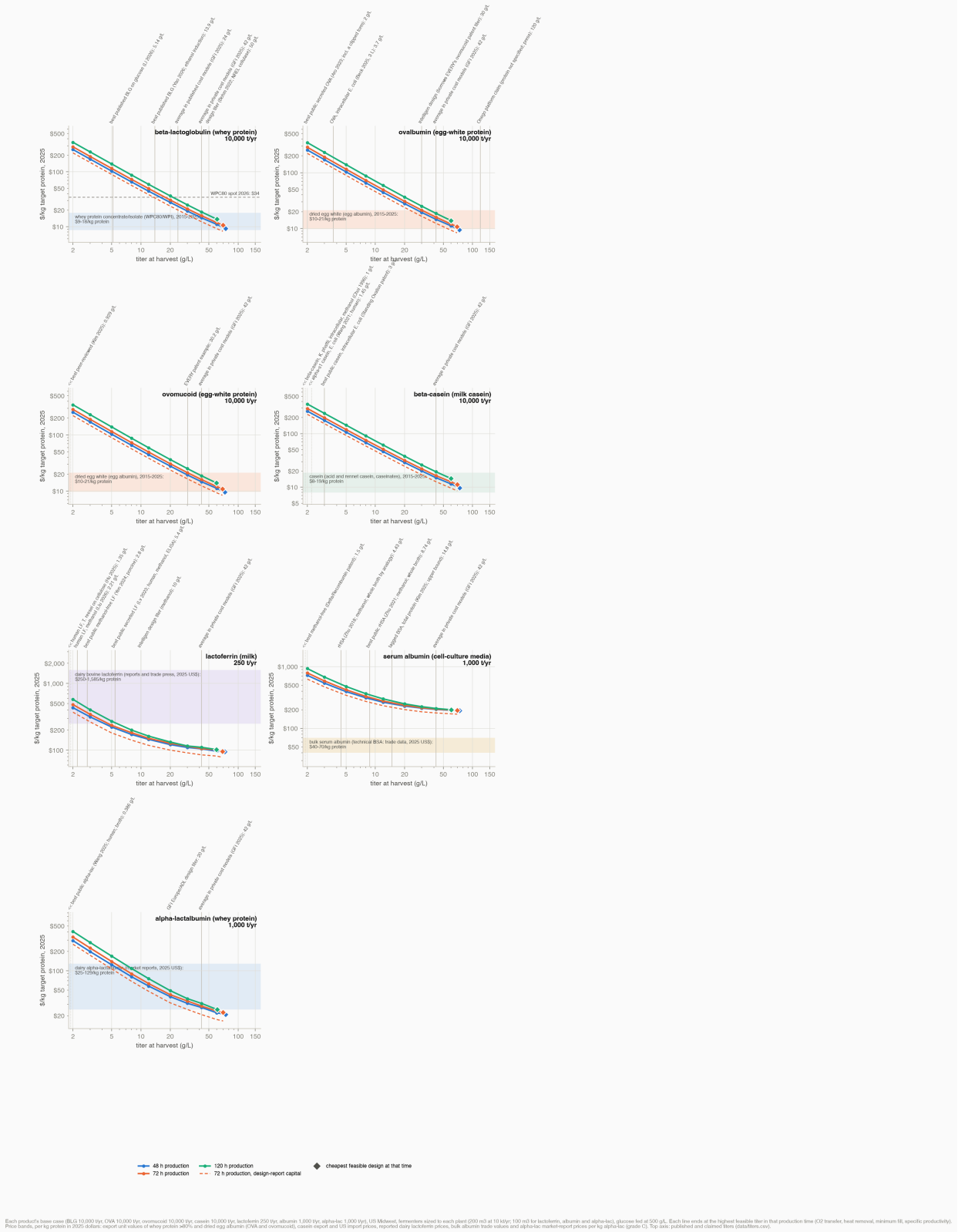


Figure 3: Minimum selling price against harvest titer for three production times, one panel per protein on its own base plant. Vertical lines mark the best public titers, the average titers assumed in published and private cost models (GFI 2025), and design or claimed titers; "<<" marks titers below the axis, drawn at its left edge. Lines end at the feasibility ceiling of the fermenter. Bands: each incumbent's price range.

Cost falls roughly as 1/titer until the ceiling (Figure 3). At the best public titers and at the titer private cost models assume:

Protein	Best public titer	$/kg there	$/kg at 42 g/L	Incumbent, $/kg protein
β-lactoglobulin	13.9 g/L	$43.0	$15.8 ($12.3)	$8.7–$17.8
ovalbumin	2.0 g/L	$227	$15.9 ($12.4)	$9.7–$21.2
ovomucoid	0.9 g/L	$479	$16.0 ($12.4)	$9.7–$21.2
β-casein	1.0 g/L	$803	$16.4 ($12.6)	$7.9–$18.7
lactoferrin	2.8 g/L	$413	$106 ($85.1)	$250–$1,585
serum albumin	8.7 g/L	$424	$205 ($177)	$40.0–$70.0
α-lactalbumin	0.4 g/L	$1,848	$28.2 ($20.8)	$25.0–$129

Table 6: The academic and industry scenarios against each incumbent (price-anchored capital; design-report in brackets). The best public titers and their caveats are in Table 1.

- **Best public academic strains** leave each commodity protein at a multiple of its incumbent's band: BLG 2.4–5.0×, OVA 10.7–23.3×, ovomucoid 22.5–49.1×, β-casein 43.0–101.5×. The patent examples are far closer: EVERY's 30 g/L ovomucoid costs $23.7/kg (1.1–2.4× the egg-white band, the only public titer near the design point), and Standing Ovation's 3 g/L casein $198. BLG on glucose (Li et al. 2026, 5.14 g/L) costs $107/kg. Lactoferrin at its best methanol-free titer costs $413/kg, inside its reported price range; at 5.4 g/L (Lv et al. 2023, methanol, ELISA) $279. Albumin at the record titer (8.7 g/L of whole broth after 192 h of methanol induction) costs $424/kg, at Zhu et al.'s 2018 titer $514, and at the best methanol-free titer (1.5 g/L) $997. The model's methanol use in the record case, 19 kg per kg of albumin made, is 3.1× less than the 58 kg that Zhu et al. fed, because their cells kept growing on methanol, to about 500 g/L of wet cells, while the model's barely grow in production: optimistic for this case.
- **α-Lactalbumin's strains are the furthest behind.** Its best public titer, 0.6 g/L of human α-lac in the supernatant (0.39 g/L of broth) on ethanol (Wang X et al. 2025), costs $1,848/kg on 650 fermenters, 14× the top of its reported band; the next best are 0.12 g/L of human α-lac in a **Bacillus subtilis** flask, after a His-tag purification (Zhu et al. 2024), and 0.11 g/L of human α-lac in **K. phaffii** (Deng et al. 2023, abstract). The dairy (bovine) protein has reached 2 mg/L in yeast, and 1 mg/L in **B. subtilis** (Wang et al. 2024). The GFI Europe/Arthur D. Little α-lac case assumes 20 g/L at 0.13 g/g, about 33× the best public titer in the supernatant, and costs it at $24/kg (membranes, food grade) and $100/kg (with chromatography, above 99% pure). No company discloses a titer; 21st.BIO and Novonesis announced a bovine α-lac programme in 2025.
- **After α-lac, casein has the weakest public titers.** No company discloses one, and no secreted casein titer has been published for any yeast or fungus. In **K. phaffii** the best is 1.0 g/L inside the cells ($803/kg costed as if secreted, on 1,553 fermenters); the best peer-reviewed casein titer in any host, which a 2026 review calls the highest reported, is 1.45 g/L of human α_{s1}-casein in **E. coli**. The base and industry cases for casein therefore borrow titers from other proteins.
- **The titer private cost models assume (42 g/L)** brings the commodity proteins to $15.8 (BLG), $15.9 (OVA), $16.0 (ovomucoid), $16.4 (β-casein) per kg: all four inside their incumbents' bands, if a strain secretes about 7 mg/g/h for three days, about three times the best demonstrated (Section 4.4).

The incumbents. Official trade unit values form a consistent series, deflated to 2025 dollars with the US consumer price index:

- **Whey protein >80%** (WPC80 and WPI mixed; US and EU exports): $9–$18 per kg protein in 2015–2025, taking the traded product as 80% protein, the minimum for its trade code. If the mix averaged 88% (more isolate), the band would be $8–$16.
- **Dried egg albumin** (EU exports; 81.1% protein, USDA): $10–$21.
- **Casein and caseinates** (EU exports; US import prices of acid and rennet casein, USDA Dairy Market News; 85.4% protein as sold, Southward): $8–$19, and $9.1–$10.1 in 2025.
- **Bulk serum albumin.** Albumins other than egg and milk albumin share one trade code, which mixes feed-grade plasma at $3–5/kg with high-grade albumin at hundreds of dollars. The US and Dutch export unit values ($42–58/kg in 2023) match GFI's (2025) "$40 to $70/kg in public trade databases", which we read as technical bovine serum albumin: $40–$70 per kg (2025 dollars; grade C). Cell-culture recombinant albumin sells at $43,490–$54,800 per kg in 10–100 g packs (InVitria, 2026).
- **Dairy α-lactalbumin** has no trade code either. GFI Europe and Arthur D. Little (2025) give $15/kg for food grade and $123/kg for high purity, from online listings. Their own cost-in-use arithmetic shows these are

per kg of ingredient at 60% and 95% α-lac, so per kg of α-lac the band is $25–$129 (grade C; our conversion). A market-research page gives $31–86/kg for food grade, on an unstated basis; we show it as context only.

- **Dairy lactoferrin** has no trade code; reports and trade press give $250–$1,585 per kg (2013–2025, deflated; grade C), partly from a PF seller, and the price is volatile (about $200/kg nominal in the 2016 glut). Egg white is not always dearer than whey: per kg of protein it was cheaper than the average whey export price in 5 of 11 years (2017, 2019, 2021, 2024, 2025).
- **A 2026 whey spike.** WPC80 and WPI rose to about $34 per kg protein at the 2026 peak (USDA Dairy Market News narrative quotes, not a formal series). At that price the base case ($21.2) would undercut whey, as would the industry benchmark ($15.8). Against 2015–2025 prices, the industry benchmark is inside the band with either capital convention.

Other protein ingredients (Figure 4). Per kg of protein, plant protein concentrates and isolates trade far below whey and egg white ($3.8–$10.6 for pea and soy, GFI 2025), and casein, gelatin and collagen sit around $10–$12. PF proteins do not compete with plant proteins on price per gram of protein; they compete with the dairy and egg ingredients whose functions (gelling, foaming, whipping, curd formation) they reproduce.

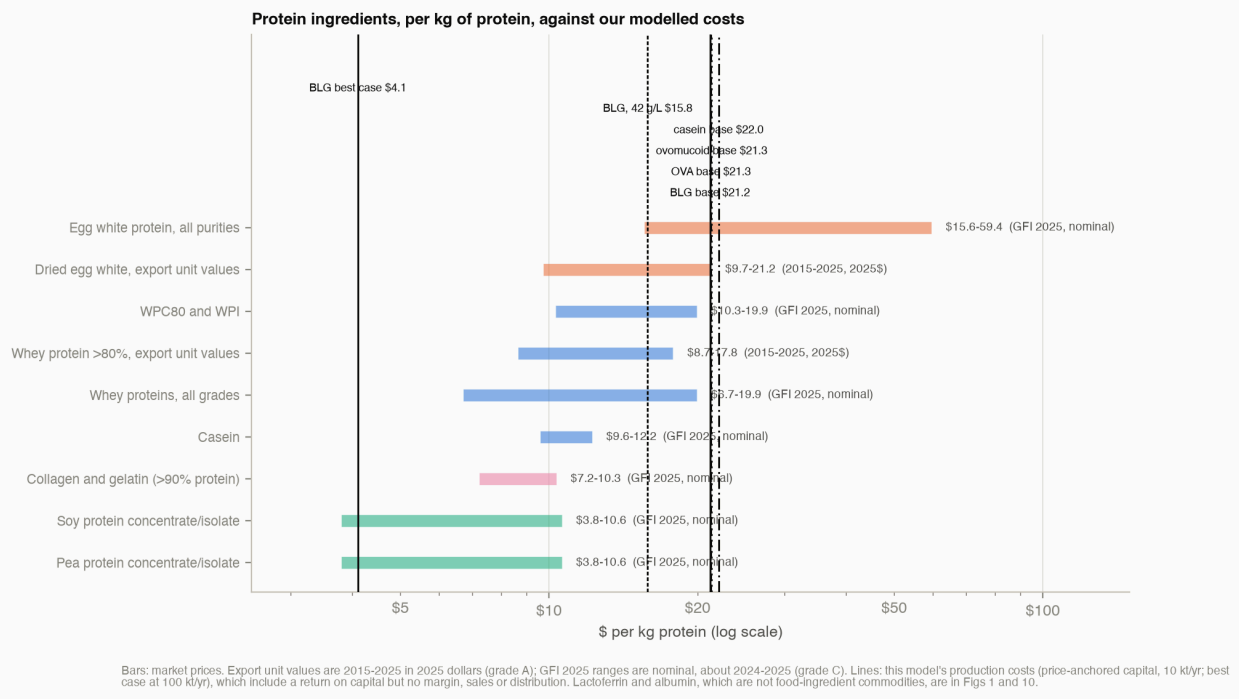


Figure 4: Protein ingredients per kg of protein against this model's costs. Bars: market prices (export unit values, 2015–2025 in 2025 dollars; GFI 2025 ranges, nominal). Lines: modelled production costs.

Parity in production cost is necessary but not sufficient for commercial parity: PF proteins also compete on functionality and price per unit of function.

4.3 Q2: what drives the cost

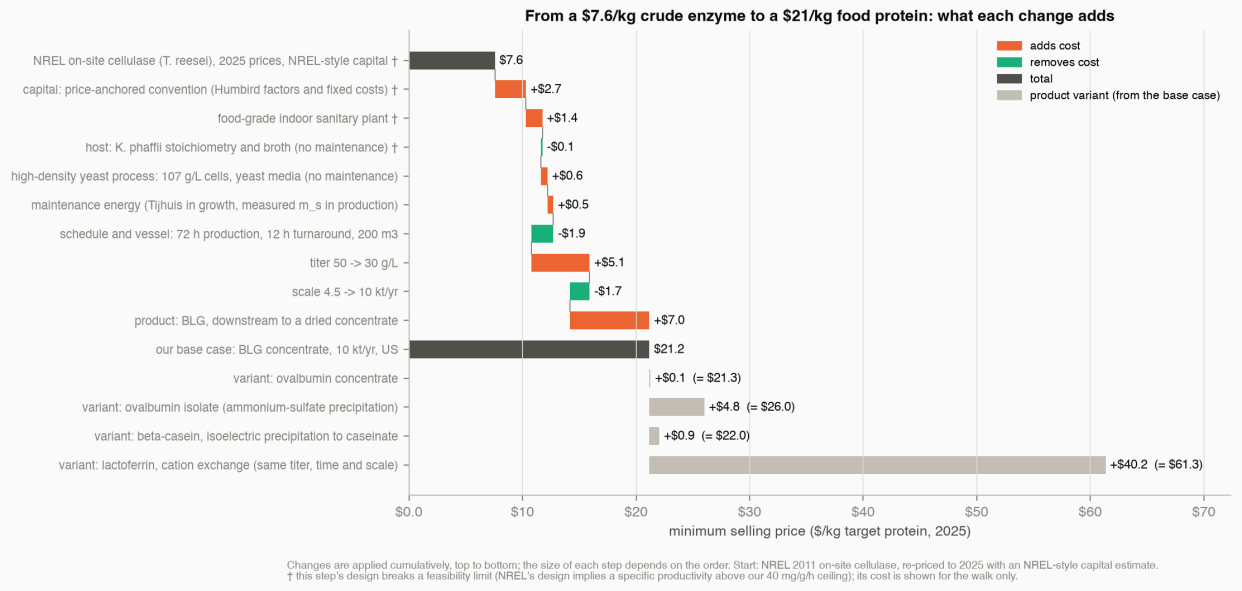


Figure 5: From NREL's on-site cellulase to our base case, one change at a time. The start is NREL's 2011 design re-priced to 2025 with design-report capital.

Why a food protein costs more than an industrial enzyme. Figure 5 walks from an NREL-style crude enzyme ($7.6/kg) to the base BLG concentrate ($21.2/kg). The steps:

- capital: price-anchored convention (Humbird factors and fixed costs): +$2.7
- food-grade indoor sanitary plant: +$1.4
- host: K. phaffii stoichiometry and broth (no maintenance): −$0.1
- high-density yeast process: 107 g/L cells, yeast media (no maintenance): +$0.6
- maintenance energy (Tijhuis in growth, measured m_s in production): +$0.5
- schedule and vessel: 72 h production, 12 h turnaround, 200 m3: −$1.9
- titer 50 -> 30 g/L: +$5.1
- scale 4.5 -> 10 kt/yr: −$1.7
- product: BLG, downstream to a dried concentrate: +$7.0

The same plant making the other products, at the same titer, time and scale: OVA concentrate +$0.1; the OVA isolate, which adds precipitation and a second recovery loss, +$4.8; β-casein by isoelectric precipitation +$0.9; lactoferrin with cation exchange +$40.2. The order of the steps changes their sizes, not the total. The first 4 designs of the walk (marked † in Figure 5) break our secretion ceiling: NREL's design makes 50 g/L of cellulase with only 6.6 g/L of cells, which implies about 117 mg/g/h. Their costs are shown for the walk only.

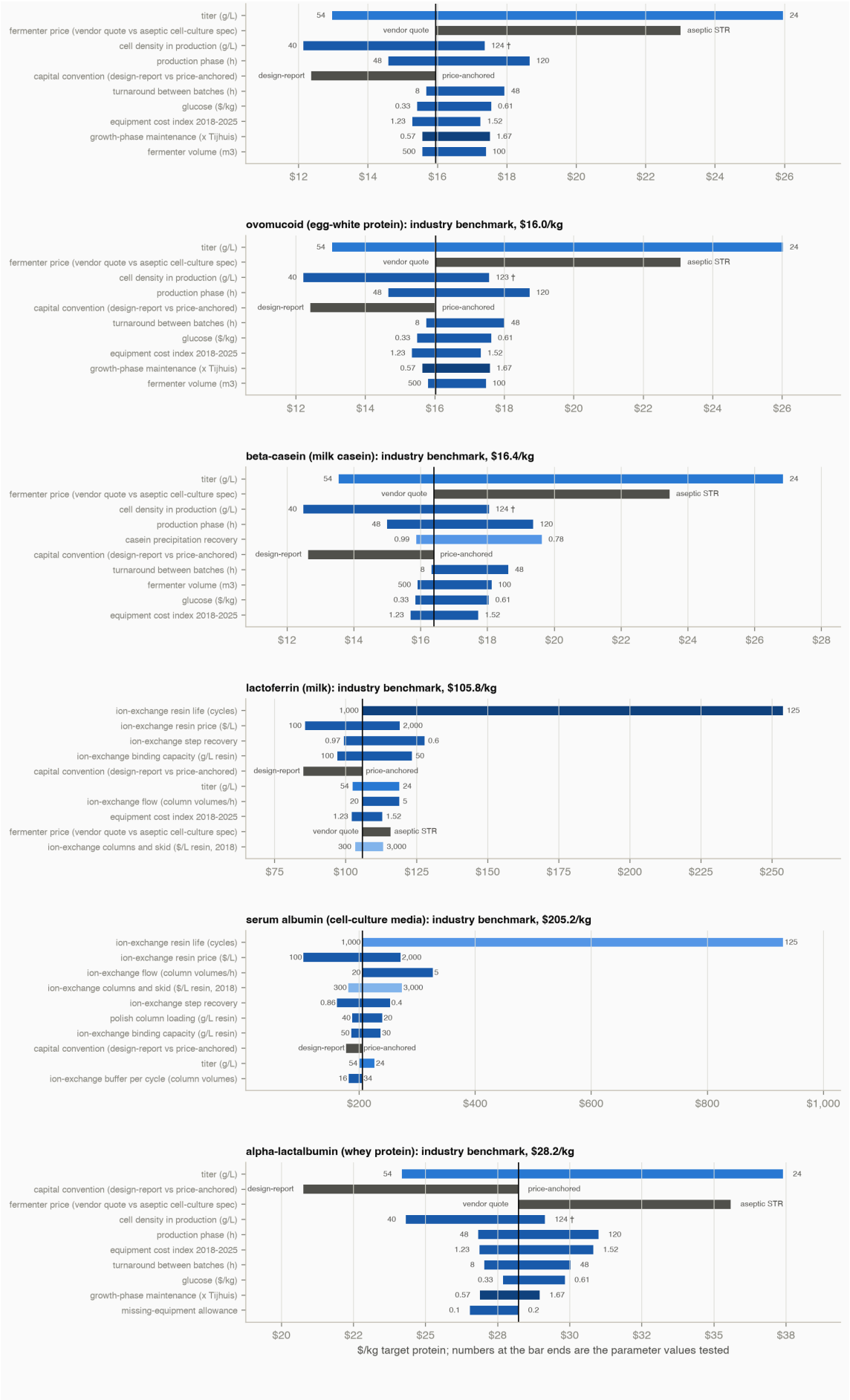


Figure 6: One-at-a-time sensitivity around each protein's industry benchmark (42 g/L, on its own plant), coloured by the evidence grade of each parameter's range; the ten largest swings per protein. † marks an end moved to the feasibility boundary.

Tornado (Figure 6). Around the BLG industry benchmark the largest swings are:

1. titer (g/L): $12.9–$25.9 (grade C)
2. fermenter price (vendor quote vs aseptic cell-culture spec): $15.8–$22.9 (a convention)
3. cell density in production (g/L): $12.0–$17.3 (grade B), one end truncated at the feasibility boundary
4. production phase (h): $14.5–$18.5 (grade B)
5. capital convention (design-report vs price-anchored): $12.3–$15.8 (a convention)
6. turnaround between batches (h): $15.6–$17.8 (grade B)
7. glucose ($/kg): $15.3–$17.5 (grade B)

What sets fermenter productivity leads: titer, production time and cell density decide how much protein a vessel makes per hour and so how much capital each kg carries. Three points deserve emphasis:

- **Yield matters through the ceiling, not the sugar bill.** At a fixed titer the product yield moves the price by $0.7 (rank 20), because most sugar goes to making and maintaining cells. But it sets the sugar feed (and O_2) each gram of protein needs, and so the ceiling: at 72 h the ceiling is 48 g/L at 0.28, 57 g/L at 0.35, 63 g/L at 0.40, 70 g/L at 0.47, 73 g/L at 0.50, 76 g/L at 0.54 g/g.
- **Vessel size.** 100–500 m^3 moves the benchmark by $1.8 (rank 10).
- **The structural choices.** The fermenter price basis (vendor quote against aseptic cell-culture vessels) moves it by $7.1 (rank 2), and the accounting convention by $3.5 (rank 5).
- **The minimum starting volume** does not appear: raising it above 0.35 makes the benchmark design infeasible, so its bar is truncated to almost nothing; Section 4.4 shows what it does to the ceiling.

The other proteins share the ranking at the top, because they share the fermentation. Their three largest swings:

- **ovalbumin** (industry benchmark $15.9/kg): titer (g/L) $13.0–$26.0; fermenter price (vendor quote vs aseptic cell-culture spec) $15.9–$23.0; cell density in production (g/L) $12.1–$17.4.
- **ovomucoid** (industry benchmark $16.0/kg): titer (g/L) $13.0–$26.0; fermenter price (vendor quote vs aseptic cell-culture spec) $16.0–$23.1; cell density in production (g/L) $12.2–$17.6.
- **β-casein** (industry benchmark $16.4/kg): titer (g/L) $13.6–$26.9; fermenter price (vendor quote vs aseptic cell-culture spec) $16.4–$23.5; cell density in production (g/L) $12.5–$18.0.
- **lactoferrin** (industry benchmark $106/kg): ion-exchange resin life (cycles) $106–$254; ion-exchange resin price ($/L) $85.8–$119; ion-exchange step recovery $99.4–$128.
- **serum albumin** (industry benchmark $205/kg): ion-exchange resin life (cycles) $205–$931; ion-exchange resin price ($/L) $104–$271; ion-exchange flow (column volumes/h) $205–$327.
- **α-lactalbumin** (industry benchmark $28.2/kg): titer (g/L) $24.2–$37.4; capital convention (design-report vs price-anchored) $20.8–$28.2; fermenter price (vendor quote vs aseptic cell-culture spec) $28.2–$35.6.

Their own downstream inputs: for β-casein they rank lower, the largest being casein precipitation recovery ($3.8, grade D), for lactoferrin ion-exchange resin life (cycles) ($148, grade A), and for albumin, where the column inputs lead, ion-exchange resin life (cycles) ($725, grade D, rank 1).

Few strain-performance ranges are grade A; they rest on design values and company-reported averages, because no public fed-batch dataset for any of the seven reports titer, rate and yield together.

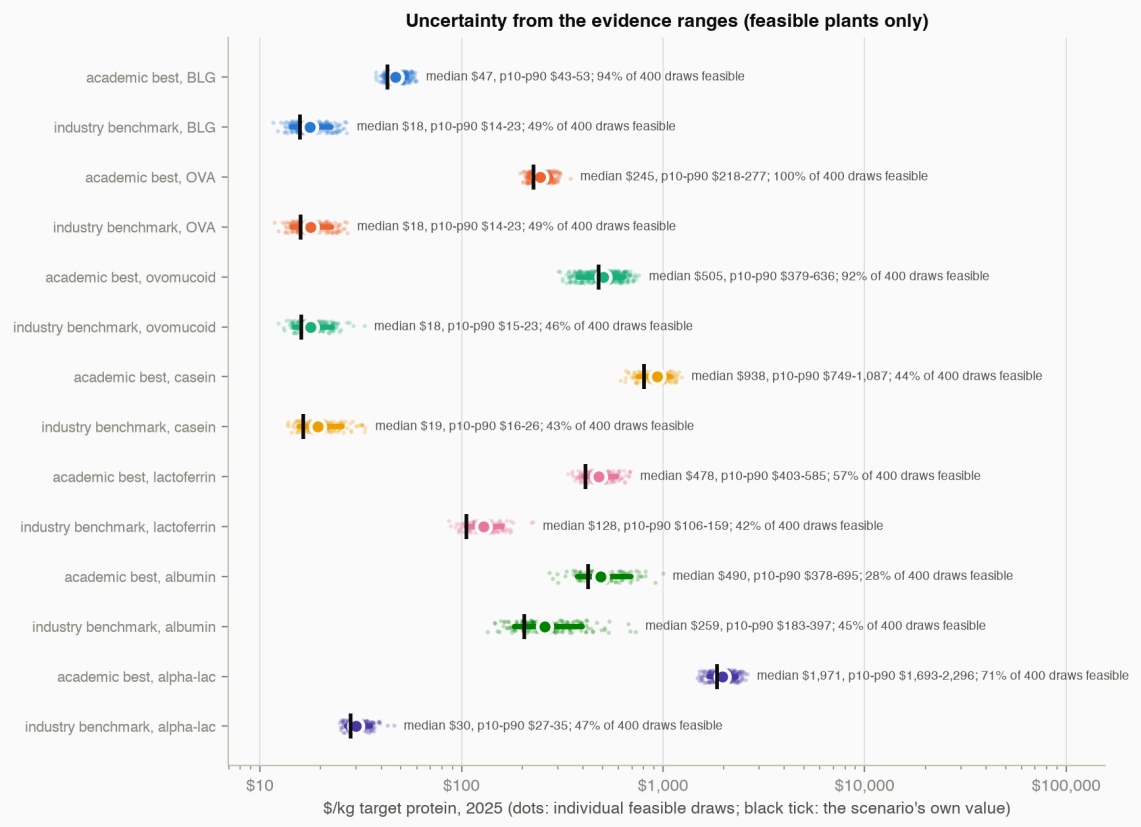


Figure 7: Monte Carlo over each scenario's evidence ranges (triangular), feasible plants only; the black tick is the scenario's own value.

Monte Carlo (Figure 7). Over the evidence ranges the industry benchmark's median is $18 for BLG (p10–p90 $14–$23) over feasible plants, and $19 ($15–$24) over all draws with a cost. 49% of draws are feasible. The rest fail as follows: 164 on the initial fill, 22 because no batch could be built (e.g. yield above its ceiling), 18 on oxygen transfer (OUR). The failing draws are those with many cells at the start of production (111 against 83 g/L on average) and fast growth in production (0.0090 against 0.0049/h): they need so much feed that the batch would have to start below 30% full. The scenario values sit at these percentiles of their own feasible distributions: industry BLG 29, industry OVA 29, academic BLG 13, academic OVA 22. For the academic strains the p10 is $43 (BLG) and $218 (OVA). For the other proteins the industry medians (p10–p90, share of draws feasible) are $17.9 for ovomucoid ($14.8–$22.9, 47%); $19.4 for β-casein ($15.7–$25.5, 43%); $128 for lactoferrin ($106–$159, 42%); $259 for albumin ($183–$397, 45%); $30.0 for α-lac ($27.0–$35.4, 47%).

Protein	Largest rank correlations with cost, industry benchmark (ρ)
β-lactoglobulin	titer (g/L) −0.80; liquid in centrifuge paste +0.27; production phase (h) +0.26
ovalbumin	titer (g/L) −0.80; liquid in centrifuge paste +0.27; cell density in production (g/L) +0.27
ovomucoid	titer (g/L) −0.79; production phase (h) +0.35; cell density in production (g/L) +0.29
β-casein	titer (g/L) −0.77; cell density in production (g/L) +0.37; equipment cost index 2018-2025 +0.26
lactoferrin	ion-exchange resin price ($/L) +0.52; ion-exchange resin life (cycles) −0.46; ion-exchange binding capacity (g/L resin) −0.42
serum albumin	ion-exchange resin life (cycles) −0.56; ion-exchange resin price ($/L) +0.55; ion-exchange columns and skid ($/L resin, 2018) +0.26
α-lactalbumin	titer (g/L) −0.77; production phase (h) +0.32; equipment cost index 2018-2025 +0.29

Table 7: The global counterpart of the tornado: Spearman's ρ between each sampled input and the cost over the feasible Monte Carlo draws of each industry benchmark. The sign says which way the input moves the cost; ρ^2 approximates its share of the spread. With about 150–250 feasible draws per case, $|\rho|$ below about 0.15 cannot be told from zero and close second and third entries cannot be ordered. Full lists: SI workbook, sheet MonteCarlo_rank.

Which uncertainties matter jointly (Table 7). Ranked by their correlation with cost across the draws, the titer comes first for 5 of the 7 proteins; for lactoferrin and albumin the column's inputs lead instead, as in the tornado.

4.4 Q3: what must be true for price parity

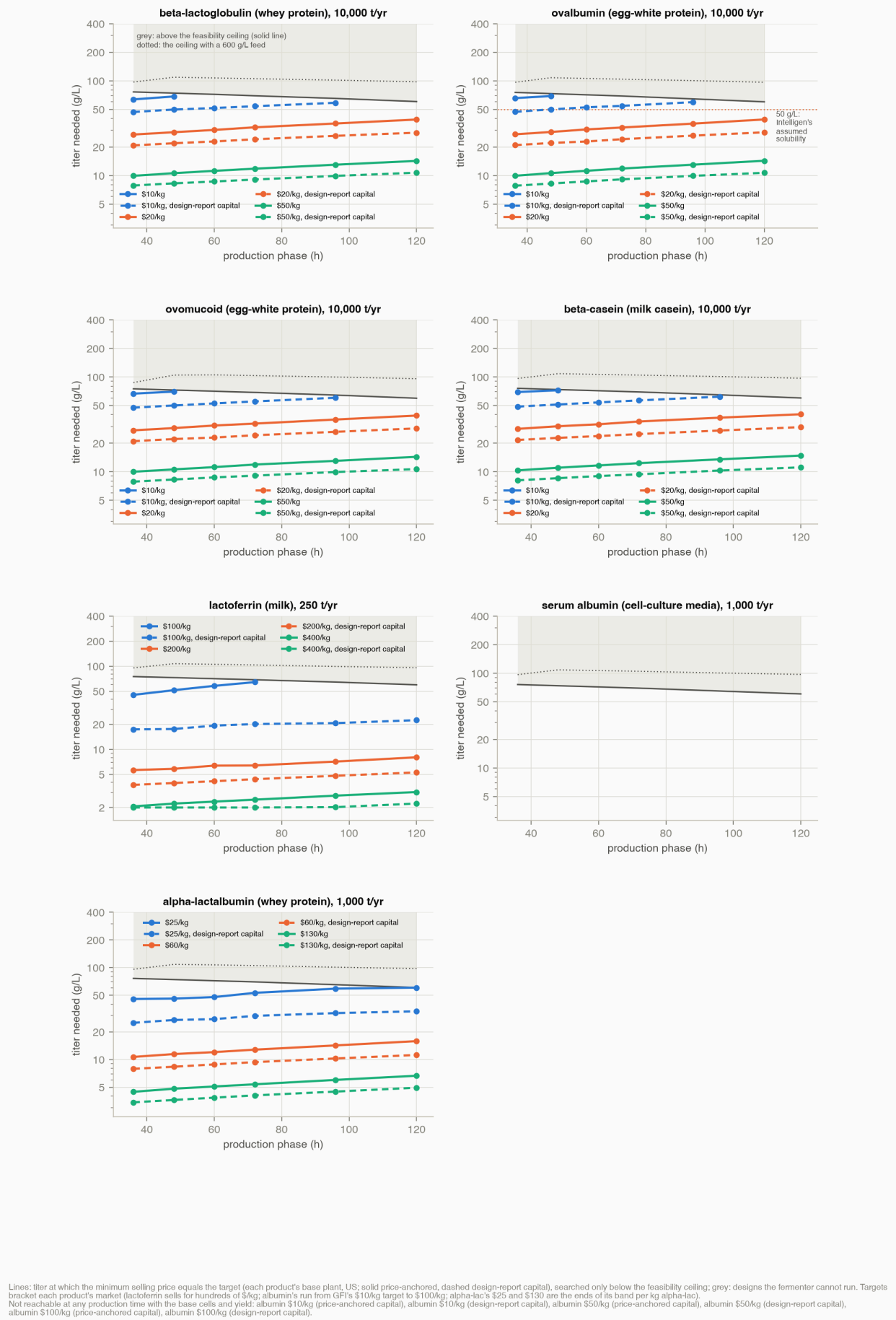


Figure 8: Titer needed for a target price, against production time, at the base cell density, one panel per protein on its own plant; targets bracket each market. Grey: designs the fermenter cannot run with a 500 g/L glucose feed. Dotted: the ceiling with a 600 g/L feed.

Production phase (h)	36	48	60	72	96	120
Ceiling, 500 g/L feed (base)	76	74	72	70	65	60
Ceiling, 600 g/L feed	97	109	107	105	102	98
$10/kg, price-anchored	64	68	—	—	—	—
$20/kg, price-anchored	27	29	30	32	35	39
$50/kg, price-anchored	10	11	11	12	13	14
$20/kg: implied q_p (mg/g/h)	8	7	6	5	4	4
$10/kg, price-anchored, 600 g/L feed	67	70	73	78	86	92
implied q_p (mg/g/h)	23	19	16	15	13	11
$10/kg, design-report	47	50	52	54	59	—
$20/kg, design-report	21	22	23	24	26	28
$50/kg, design-report	8	8	9	9	10	11

Table 8: BLG titer (g/L) needed for each target price at the base cell density (107 g/L), searched only below the feasibility ceiling; grey rows give the specific productivity that titer implies (the base design runs at 4.7). — means the target is not reachable at that production time (`results/analyses.json`, key `required`).

Figure 8 and Table 8 answer the inverse question at the base cell density:

- **$50/kg** needs 10–14 g/L depending on production time: 1.9–2.8× the best public BLG titer on glucose.
- **$20/kg** needs about 29–32 g/L in 48–72 h with price-anchored capital (design-report: 22–24 g/L), a specific productivity of 5–7 mg/g/h.
- **$10/kg** needs 64–68 g/L with price-anchored capital. With a 600 g/L feed it needs 67–92 g/L in 36–120 h, at 11–23 mg/g/h. With design-report capital it needs 47–59 g/L in 36–96 h.

Protein	Targets ($/kg)	Lowest	Middle	Highest	Ceiling
β-lactoglobulin	10, 20, 50	—	32	12	70
ovalbumin	10, 20, 50	—	32	12	69
ovomucoid	10, 20, 50	—	32	12	69
β-casein	10, 20, 50	—	34	12	70
lactoferrin	100, 200, 400	64	6	2	69
serum albumin	10, 50, 100	—	—	—	70
α-lactalbumin	25, 60, 130	53	13	5	70

Table 9: Titer (g/L) each protein needs for its three target prices with a 72 h production phase, price-anchored capital and each product's base plant, against the feasibility ceiling; — means not reachable below the ceiling. Lactoferrin's targets bracket its reported price range.

The other proteins follow BLG (Table 9): OVA, ovomucoid and β-casein need about the same titers, β-casein slightly more because its downstream costs a little more. Lactoferrin reaches $200/kg at 6 g/L, against a best public methanol-free titer of 2.8 g/L. For albumin, with its columns, cost flattens out as titer rises: at 72 h none of its targets ($10, $50, $100/kg) is reached at any titer below the ceiling. GFI's $10/kg is out of reach of any titer with column purification. α-Lac, on BLG's route but a 1 kt/yr plant, reaches $25/kg, the low end of its reported band, at 53 g/L (a secretion rate of about 9 mg/g/h, against 4.7 in the base design); $60/kg needs 12.8 g/L and $130/kg, the high-purity price, 5.4 g/L, against a best public titer of 0.39 g/L of human α-lac per litre of broth.

The gate is secretion. Cell density is a free design choice, and at a fixed titer fewer cells cost less; what fewer cells cannot escape is the specific productivity. Table 10 lets the model choose cell density, production time and titer freely, with everything else at the base (yield, maintenance, 200 m³ vessels, 10 kt/yr, price-anchored capital), for four secretion ceilings. With secretion at 40 mg/g/h the cheapest BLG design costs $7.1/kg; at 10 mg/g/h it costs $12.0. The base design runs at 4.7 mg/g/h, Intelligen's design value; the best public BLG fed-batch implies about 1 (Yao et al. 2026: 13.88 g/L in 108 h at roughly 120 g/L cells). Fungal cellulase processes make 0.2–0.4 g protein per litre per hour (Ellilä et al. 2017), which at the 20–50 g/L of cells such processes run is roughly 5–20 mg/g/h (our arithmetic), for native enzymes, not animal proteins.

Secretion ceiling (mg/g/h)	$/kg BLG	Cells (g/L)	Production (h)	Titer (g/L)	Limits that bind
40	$7.1	70	63	98	O2 transfer, specific productivity
20	$8.7	90	81	87	O2 transfer, specific productivity, minimum initial fill
10	$12.0	100	90	62	O2 transfer, specific productivity
5	$18.6	100	90	36	O2 transfer, specific productivity
2.1 (best demonstrated)	$36.2	114	90	19	O2 transfer, specific productivity

Table 10: What process design alone can do: the cheapest BLG design with base biology, vessel, scale and price-anchored capital, when cell density, production time and titer are free, for five secretion ceilings (a coarse search; review SC-1). 2.1 mg/g/h is the best secretion rate we found demonstrated in a **K. phaffii** glucose fed-batch (Prielhofer et al. 2018).

How firm is the ceiling? It is a design ceiling, not a law of physics (Table 11). In the base it comes from the vessel's volume: the minimum starting volume (30%, the lowest start in a published design), the glucose feed (500 g/L, Intelligen's) and the cell density at the start of production (107 g/L) are all design choices. A 600 g/L feed lifts the 72 h ceiling to 105 g/L (minimum starting volume); 80 g/L of cells lifts it to 97 g/L; a 20% start to 121 g/L, still set by the minimum starting volume. Only with no volume limit at all does another limit take over: secretion ceiling at 127 g/L. The other way, a 35% start brings the 72 h ceiling down to 44 g/L, and at 40% 19 g/L (NREL's design starts at 50%). The ceiling also rises with the product yield (Section 4.3) and with lower production-phase maintenance: at 72 h it is 73 g/L at 0.0023, 70 g/L at 0.00437, 64 g/L at 0.008, 51 g/L at 0.0161 g/g/h.

Design choice	36 h	72 h	120 h
base	76 g/L (minimum starting volume)	70 g/L (minimum starting volume)	60 g/L (minimum starting volume)
glucose feed 600 g/L	97 g/L (secretion ceiling)	105 g/L (minimum starting volume)	98 g/L (minimum starting volume)
80 g/L cells at the start of production	75 g/L (secretion ceiling)	97 g/L (minimum starting volume)	91 g/L (minimum starting volume)
minimum fill 0.20	90 g/L (secretion ceiling)	121 g/L (minimum starting volume)	114 g/L (minimum starting volume)
minimum fill 0.35	52 g/L (minimum starting volume)	44 g/L (minimum starting volume)	33 g/L (minimum starting volume)
minimum fill 0.40	27 g/L (minimum starting volume)	19 g/L (minimum starting volume)	6 g/L (minimum starting volume)
minimum fill 0.50 (NREL)	—	—	—
no minimum fill	90 g/L (secretion ceiling)	127 g/L (secretion ceiling)	152 g/L (O2 transfer)
no minimum fill, O2 cap 150 (bubble column)	75 g/L (O2 transfer)	112 g/L (O2 transfer)	138 g/L (O2 transfer)

Table 11: BLG titer ceiling (g/L) by production time, and the limit that fails just above it. Base: 107 g/L cells, a 30% minimum start, a 500 g/L glucose feed, and an O_2 cap of 200 mol/m³/h per m³ of cell-free liquid.

4.5 Q4: the physical-limit best case

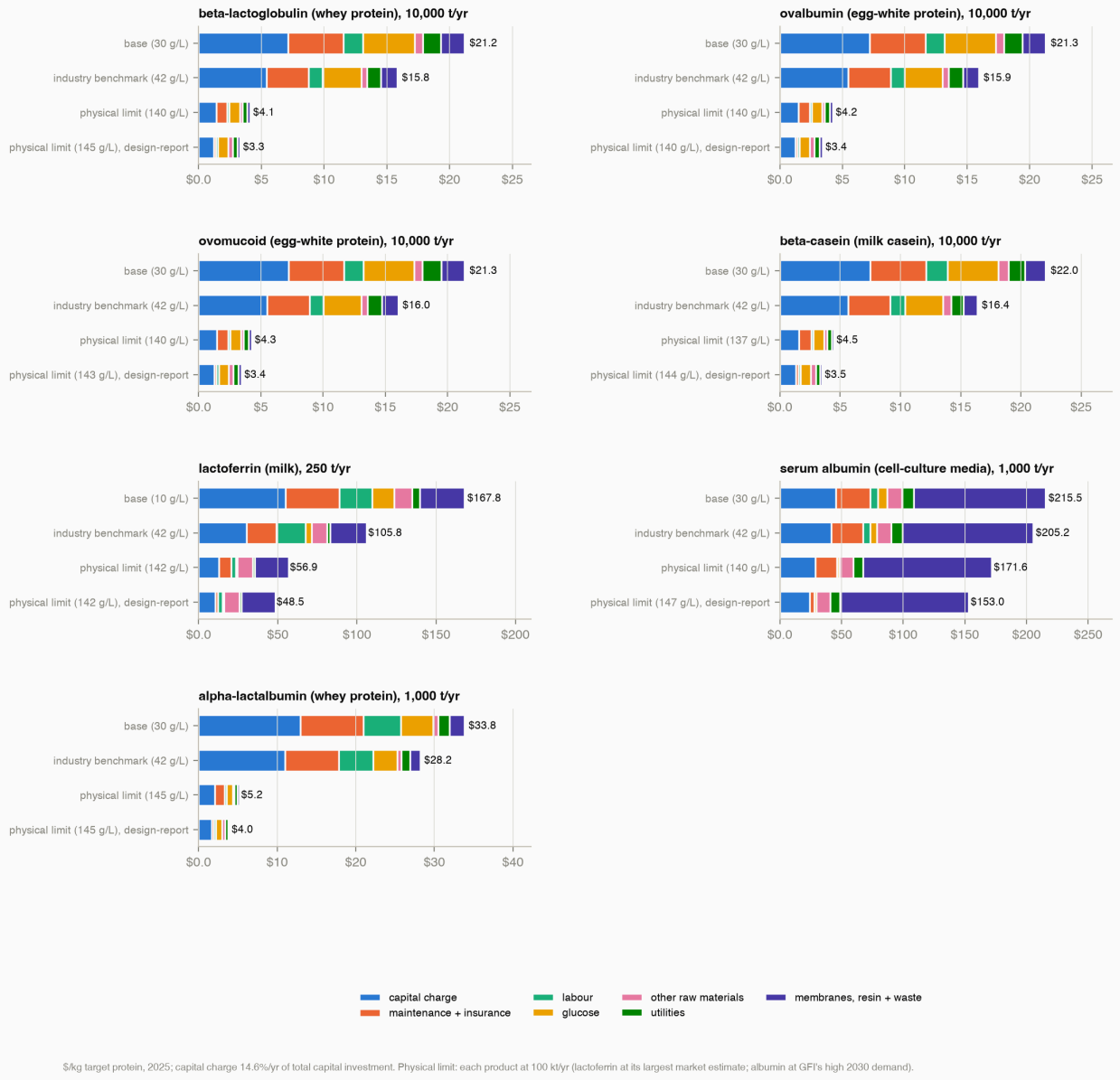


Figure 9: Cost structure from today's benchmarks to the physical-limit best case, one panel per protein (each on its own scale). The best academic strains are off these scales.

The search (Figure 9) finds the cheapest feasible design once every lever listed in Section 2.7 is at its assumed best. For BLG it has 83 g/L cells, a 84 h production phase and 140 g/L, at a specific productivity of 40 mg/g/h. The limits active there are O2 transfer, specific productivity, minimum initial fill.

Protein	Price-anchored	Design-report	Cells (g/L)	Production (h)	Titer (g/L)	Limits that bind
β-lactoglobulin	$4.1	$3.3	83	84	140	O2 transfer, specific productivity, minimum initial fill
ovalbumin	$4.2	$3.4	83	84	140	O2 transfer, specific productivity, minimum initial fill
ovomucoid	$4.3	$3.4	83	84	140	O2 transfer, specific productivity, minimum initial fill
β-casein	$4.5	$3.5	83	80	137	O2 transfer, specific productivity
lactoferrin	$56.9	$48.5	77	96	142	O2 transfer, specific productivity, minimum initial fill
serum albumin	$172	$153	83	84	140	O2 transfer, specific productivity, minimum initial fill
α-lactalbumin	$5.2	$4.0	77	100	145	O2 transfer, minimum initial fill

Table 12: The physical-limit best case of each protein: 100 kt/yr (lactoferrin 1,588 t/yr; α-lac about 19 kt/yr; for albumin, GFI's high 2030 demand scenario), 400 m³ vessels, secretion up to 40 mg/g/h, $/kg target protein (2025).

It costs $4.1/kg BLG with price-anchored capital and $3.3 with design-report capital; for OVA $4.2 and $3.4, for β-casein $4.5 and $3.5, for lactoferrin at its market's size $56.9 and $48.5, for α-lac at its market's size $5.2 and $4.0, and for albumin $172 and $153, with its columns unchanged (Table 12).

- **It is set by an assumed secretion rate.** The binding biological limit is the 40 mg/g/h ceiling, a judgment (grade D) about 9× the base design and above any published **K. phaffii** value. Table 13 shows the case at other ceilings.
- **Capital still dominates:** 56% of the BLG case. Glucose ($0.84) is the largest non-capital item.
- **Engineering levers are part of it.** 400 m³ vessels, 100 kt/yr, contract sugar and the 600 g/L feed are not biology. With the base 500 g/L feed the BLG case costs $4.5; with aseptic cell-culture vessels, $5.7.
- **OVA and the 50 g/L question.** If Intelligen's 50 g/L were a real solubility limit at fermentation pH, the OVA case would cost $8.3 instead of $4.2.

Secretion ceiling (mg/g/h)	$/kg BLG	Cells (g/L)	Production (h)	Titer (g/L)
2.1 (best demonstrated)	$24.5	150	90	25
5	$12.1	150	90	53
10	$8.1	150	81	81
20	$5.4	114	90	118
40 (assumed)	$4.1	83	84	140
120 (NREL's design, implied)	$3.4	45	72	162

Table 13: The BLG physical-limit case at other secretion ceilings (price-anchored capital; the sensitivities use a coarser search). The base design runs at 4.7 mg/g/h.

The case is not a forecast and not a lower bound: design-report accounting, continuous operation or further relaxed design choices would all go lower. It is what the model gives once every assumed limit is at its best at the same time. Its yield sits at 95% of an energetic ceiling that leaves out transport, secretion and turnover costs. Its O_2 limit assumes water-like broth at 83 g/L cells and 140 g/L of dissolved protein, where viscosity and foam would lower k_La. And its heat removal (up to 25–40 kW/m³, i.e. about 16 MW per 400 m³ vessel) needs external heat-exchange loops that are not costed. For β-casein the O_2 limit is more generous still: at 137 g/L the broth holds casein close to the concentration above which dairy caseinate solutions become "very viscous" (about 150 g/L, Southward), which the model does not represent. The best cases are all four below their incumbents' bands, lactoferrin's is below its reported range, α-lac's below its reported band, and albumin's is above the bulk-albumin band and above GFI's $10/kg.

4.6 Q5: claimed titers against the fermenter's limits

Claim or benchmark	Titer	Modelled as	At its own time: $/kg	Design-report	Shortest feasible time	q_p there	$/kg there
average titer in private companies' cost models (GFI 2025)	42 g/L	*K. phaffii*, 107 g/L cells	$15.8 (72 h)	$12.3	13 h	38	$12.3
average titer in published cost models (GFI 2025)	24 g/L	*K. phaffii*, 107 g/L cells	$25.9 (72 h)	$20.1	12 h	21	$19.9
Behm 2022 BLG design titer (T. reesei, 120 h)	50 g/L	*T. reesei*, 35 g/L cells	$13.5 (120 h)	$9.9	48 h	40	$10.2
T. reesei homologous secreted protein (Behm 2022, grade C)	100 g/L	*T. reesei*, 35 g/L cells	infeasible at 72 h	—	143 h	40	$8.7
Onego platform claim (protein not specified; tested as OVA; press, grade C)	120 g/L	*T. reesei*, 35 g/L cells	infeasible at 72 h	—	not reachable	—	n/a

Table 14: Claimed and benchmark titers in a 200 m³ vessel: the cost at the claim's own production time (Behm's design: 120 h; otherwise 72 h), and the shortest production phase in which the titer is feasible (O_2 transfer, heat removal, minimum starting volume, 40 mg/g/h secretion ceiling), with the specific productivity and price-anchored cost there. The shortest-time columns are a best case: secretion runs at or near its ceiling. The Onego figure is a platform claim for an unnamed protein, tested as OVA.

No casein or lactoferrin company discloses a titer, and EVERY's ovomucoid titer is a patent example (the academic case), so the claims tested are for BLG, OVA or unnamed proteins. Of the 5 claims and benchmarks, 4 are feasible if the production phase is right (Table 14). The price of a high titer is time or secretion rate: at its own 72 h the 42 g/L average costs $15.8, the industry benchmark of Section 4.2; squeezed into the shortest feasible production phase it costs $12.3, but only by secreting 38 mg/g/h. A claimed titer alone therefore does not settle the cost. The check is physical; whether a strain can sustain the implied productivity is a separate, biological question. Onego platform claim (protein not specified; tested as OVA; press, grade C) cannot be reached at any production phase up to 20 days in our model: either the O_2 demand exceeds what the broth can take up, or the batch would have to start below 30% of the vessel because the feed fills it first. Such claims need better O_2 transfer than our assumptions (for **T. reesei**, a k_La half that of yeast broth, grade D), a denser feed, or a different process (continuous, repeated fed-batch or cell recycle), none of which is modelled here.

4.7 Q6: scale and siting

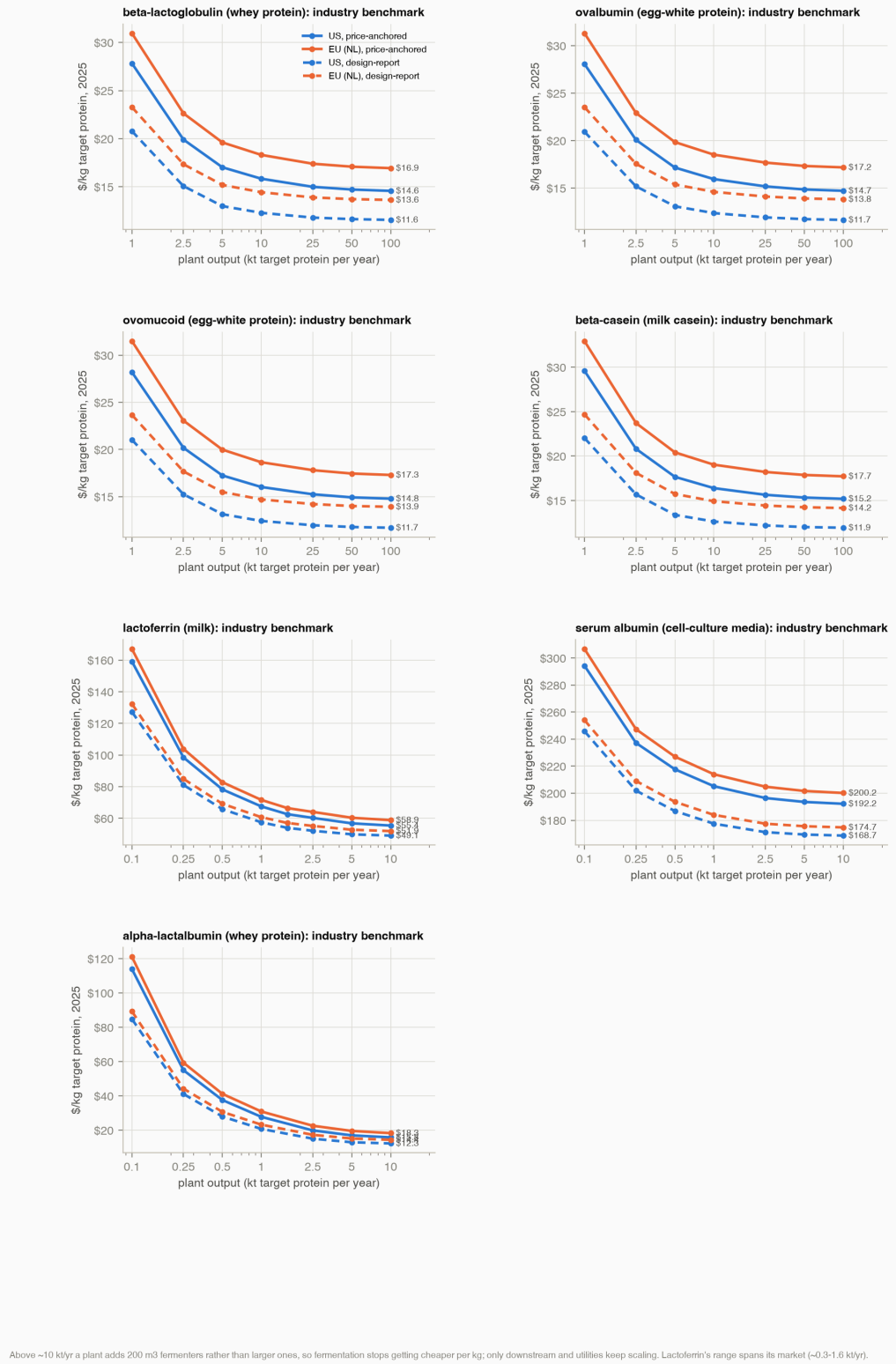


Figure 10: Each protein's industry benchmark against plant output, US Midwest and north-west Europe, both capital conventions. Lactoferrin's range spans its market; albumin's and α-lac's run from 0.1 to 10 kt/yr.

Scale (Figure 10). Going from 1 to 10 kt/yr cuts the price by $12.0, and going from 10 to 100 kt/yr saves a further $1.3. Beyond about 10 kt/yr the plant adds 200 m³ fermenters rather than larger ones, so only the downstream train and utilities keep scaling; larger vessels are a separate lever (Section 4.3). OVA, ovomucoid and β-casein follow the same curve. For lactoferrin scale is the main lever: its industry benchmark costs $98 at 250 t/yr (the scale curve re-sizes the vessels at each output; Table 1 keeps the base plant), $63 at 1,588 t/yr

(the larger market estimate) and $55 at 10 kt/yr, a plant the market could not absorb. Albumin's base case (30 g/L; Figure 10 plots its industry benchmark) costs $308 at 100 t/yr, $215 at 1 kt/yr and $203 at 10 kt/yr: past about 1 kt/yr its resin, a cost per kg that does not scale, dominates. α-Lac's industry benchmark follows BLG's curve (the same plant at the same output), re-sizing its vessels at each output: $114 at 100 t/yr, $27.8 at 1 kt/yr and $15.8 at 10 kt/yr (Table 1 keeps the base plant's 100 m³ vessels: $28.2 at 1 kt/yr).

Siting. A Netherlands plant costs $3.1/kg more than one in the US Midwest (base BLG). One at a time, US-level inputs would remove:

- glucose ($0.61 against $0.40/kg): $2.2;
- electricity and gas ($0.146 against $0.086/kWh; $19.6 against $4.8/GJ): $0.8;
- construction cost (1.09×): $1.0;
- wages: nothing (Dutch and Iowa operator pay are within a few per cent)
- spent-cell disposal (anaerobic digestion in the EU against US landfill): −$0.9, i.e. the EU is cheaper here;
- water and wastewater: +$0.0.

The one-at-a-time effects need not add up to the gap exactly, because the inputs interact. For the other proteins the Netherlands adds $3.2 (OVA), $3.3 (ovomucoid), $3.3 (β-casein), $13.8 (lactoferrin), $10.1 (albumin), $3.9 (α-lac) per kg on their base cases.

4.8 Q7: milk, egg and cultivated-meat equivalents

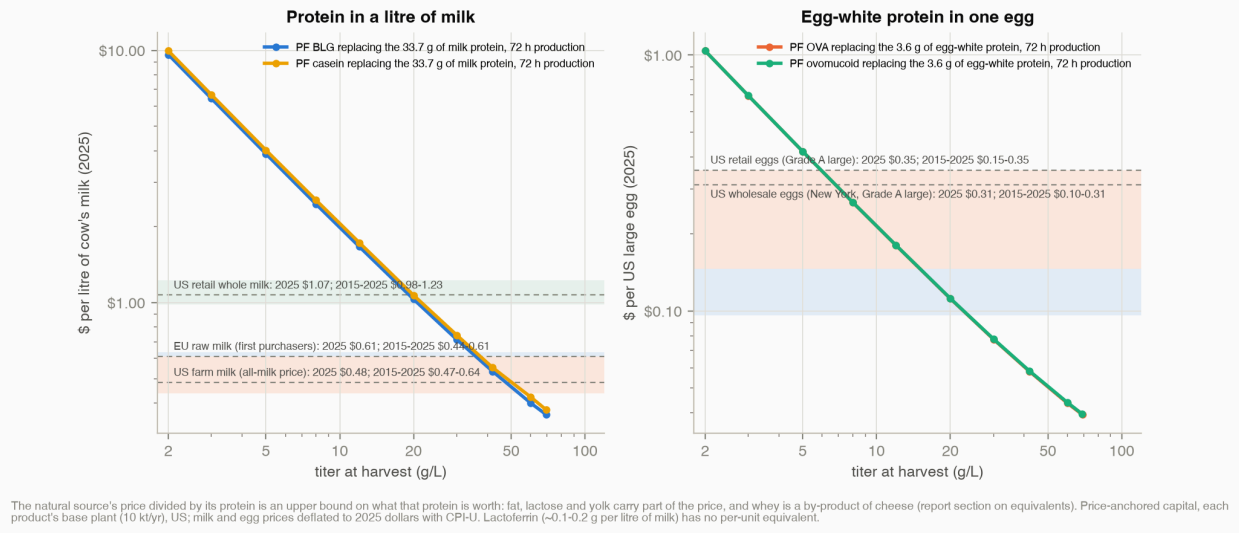


Figure 11: The cost of replacing the protein of a litre of milk with PF BLG or β-casein (left) and the egg-white protein of an egg with PF OVA or ovomucoid (right), against titer (72 h production, price-anchored capital), beside milk and egg prices in 2025 dollars. Lactoferrin (0.1–0.2 g per litre of milk) has no such unit; albumin's, per kg of cultivated meat, is in Table 15.

A litre of cow's milk holds 33.7 g of protein, 3.3 g of it BLG and about 27 g casein; a large egg holds 3.64 g of egg-white protein, 1.96 g of it OVA (`data/evidence/milk_egg.yaml`). At the base cost, the PF protein needed to replace all the protein of a litre of milk costs $0.71 as BLG and $0.74 as β-casein, against $0.48 for the litre at the US farm gate in 2025 and $1.07 at retail. Replacing one egg's white costs $0.077 as OVA and $0.078 as ovomucoid, against $0.311 per egg wholesale in 2025, a year of avian-influenza prices ($0.096–$0.311 over 2015–2025).

Dividing a source's price by its protein gives the price per kg of protein at which PF would match it: US farm milk sold its protein at $14–$19/kg in 2015–2025, and wholesale eggs their egg-white protein at $26–$86/kg. With a 72 h production phase (ceiling 70 g/L), BLG matches the 2025 farm-milk protein price at 48 g/L and β-casein at 50 g/L, and OVA matches the lowest egg-white protein price of the decade at 24 g/L, ovomucoid at 24 g/L.

These are upper bounds on what the protein is worth: charging the whole price of milk or eggs to their protein makes them look dearer per gram than they are, since fat and lactose carry much of the milk price and the yolk part of the egg's. Bought as the incumbent ingredient, the same protein is much cheaper: the white of one egg, as dried egg white, cost $0.035–$0.077 in 2015–2025. Whey is a by-product of cheese: the whey protein of

a litre of milk (6.6 g) at WPC80/WPI prices was worth 11%–21% of the US farm milk price in the same year, so replacing whey protein cuts milk output far less than the "litre of milk" framing suggests. Casein is different: it is about four-fifths of milk protein and what cheese is made of, so a cheese maker buying PF casein instead of milk removes demand for the milk itself. For animal welfare, egg white (OVA, ovomucoid) is the strongest lever per kg of target protein: it involves roughly 50–60 times more animal-years than whey BLG (about 0.6 kg OVA per hen-year against about 35 kg BLG per cow-year). A cow yields about eight times more casein than BLG (about 280 kg a year at the same milk yield), so per kg casein involves fewer cow-years than BLG; but displacing casein removes demand for milk, while displacing whey, a by-product, mostly lowers the whey price. These figures are our order-of-magnitude arithmetic, not model outputs, and they ignore substitution: one kg of PF protein sold need not displace one kg of the animal product.

Albumin, per kg of cultivated meat. Albumin's unit is the meat its medium grows. GFI's (2023) demand scenarios imply about 1.13 g of albumin per litre of medium (our arithmetic from its tonnages), and 8–42 litres of medium per kg of cultivated meat (GFI 2023, after CE Delft 2021), so 9–47 g of albumin per kg of meat:

Albumin at	$/kg albumin	8 L/kg (9 g)	13 L/kg (15 g)	42 L/kg (47 g)
GFI's target	$10.0	$0.09	$0.15	$0.47
bulk bovine albumin, low	$40.0	$0.36	$0.59	$1.90
UF/DF only (optimistic route)	$34.3	$0.31	$0.50	$1.63
industry benchmark (42 g/L)	$205	$1.86	$3.01	$9.74
base case (30 g/L)	$215	$1.95	$3.17	$10.23
best public titer (methanol)	$424	$3.83	$6.22	$20.11
research catalogue (InVitria, 100 g)	$43,490	$393	$639	$2,064

Table 15: Albumin's cost per kg of cultivated meat (2025 $), at GFI's implied 1.13 g/L and three medium uses. Model rows: price-anchored capital, 1 kt/yr.

At the base cost, albumin adds $1.95–$10.23 per kg of meat, a material share of a meat that must sell for a few dollars per kg; at GFI's $10/kg it would add $0.09–$0.47, which is negligible, and at catalogue prices hundreds of dollars (Table 15). That is the case for keeping albumin in media; the case against is that the leading firms are removing it (Section 6). Its leverage on animals is indirect: if cultivated chicken displaced chicken one for one, a kg of albumin would grow 21–111 kg of meat (our arithmetic), conditional on cultivated meat reaching scale with albumin in its medium.

4.9 Q8: the downstream route, the product and its cost



Figure 12: Each protein's base route against its alternative: cost per kg of target protein (filled) and per kg of powder (hollow), with the target protein's share of the powder's dry matter. Stars: Intelligen's own lactoferrin model, as costs of production with no return on capital: compare them with our cost of production in Table 16, not with the dots, which include one. Bands: market prices per kg protein.

The specification fixes the route, and the route can change what is sold more than what it costs (Figure 12, Section 2.4):

- **BLG, OVA and ovomucoid.** A UF/DF concentrate (85% target protein, dry basis; ovomucoid 76%, as EVERY's cleared product) against an ammonium-sulfate isolate (99%): the isolate adds +$4.8/kg for BLG, +$4.7 for OVA and +$4.7 for ovomucoid. EVERY's ovalbumin product adds chromatography (GRN 1104) and its ovomucoid product does not (GRN 967). α-Lac's concentrate is its isolate grade (90%), as the dairy isolate and GFI Europe's membrane-only case are; precipitating it with ammonium sulfate adds +$7.6/kg. GFI Europe's above-99% grade adds a chromatography step, not modelled; in their numbers it raises the cost from $24 to $100/kg.
- **β-casein.** The only public **K. phaffii** casein process, Eden Brew's (FSANZ A1342), clarifies the broth and uses membranes alone. Its powder is 22% protein, of which about a quarter is β-casein, and 68% yeast carbohydrate (cell-wall mannans and glucans): about 6% β-casein. In our model that route costs $1.69/kg of powder but $30.6/kg of β-casein, against $22.0/kg by precipitation. That comparison holds only at low titer: the variant keeps Eden Brew's composition, and at 30 g/L of casein the broth could not carry the yeast carbohydrate that composition implies (several hundred g/L; audit R10-5), so a 30 g/L strain's membrane-only powder would be purer and cheaper than shown. At 3 g/L the two routes cost about the same per kg of casein. What holds at any titer is that precipitation gives a dairy-casein grade product and the membrane route an ingredient of a different composition. Casein's own physics makes the precipitation route possible: it is insoluble near its isoelectric point, and, having almost no folded structure, survives heating to 90 °C, which denatures host proteins. Of the precipitation inputs with the weakest evidence, the acid dose and the heat recovery move the base little, but the recovery does not: at the ends of their ranges acid to precipitate

casein (kg H3PO4/m3) 2.3: $21.9; acid to precipitate casein (kg H3PO4/m3) 17.3: $22.4; casein precipitation recovery 0.78: $26.4; casein precipitation recovery 0.99: $21.2; precipitation heater heat recovery 0.5: $22.1; precipitation heater heat recovery 0.95: $22.0; casein curd wash water (L/kg) 11.0: $22.0; casein curd wash water (L/kg) 36.0: $22.5. No PF casein recovery is published; it is the input to measure first.

- **Lactoferrin.** Cation exchange (90% lactoferrin) against UF/DF alone (Intelligen's case a, purity by assumption): $168 against $111/kg. Resin life is the largest downstream lever: at 125 cycles (biopharma practice) the base costs $316.
- **Albumin.** Two columns (96% albumin) cost $215/kg, the capture column alone $143, and UF/DF alone $34.3; but albumin is only 77% of the protein in the best public broth (Zhu et al. 2021), so membranes alone make a different, below-specification product, as for casein. The columns' inputs decide where between: a polymeric resin at $100/L on both columns gives $114, resin at $2,000/L $281, a resin lasting 125 cycles $941, loading 50 instead of 40 g/L $196; the capture recovery at 40% or 86% gives $270 or $170. Inducing the base design with methanol instead of glucose adds +$6.0/kg. A purification route as cheap as a food protein's, not a better strain, is what albumin near $10/kg needs; on today's evidence the base sits at the favourable end of resin life and purification, and mid-range on recovery.

$/kg lactoferrin (2025)	Intelligen a	Ours, their design a	Intelligen b	Ours, their design b
Raw materials	$101.2	$14.6	$123.5	$22.2
Labour	$41.6	$19.6	$47.2	$24.7
Facility (maintenance, insurance, depreciation)	$91.6	$61.3	$122.5	$69.4
Laboratory, QC and QA	$11.6	$0.0	$13.2	$0.0
Consumables (resin, membranes)	$29.8	$0.6	$104.8	$81.6
Waste treatment	$6.5	$5.9	$11.1	$5.9
Utilities	$17.1	$3.1	$18.7	$3.0
Cost of production	**$299.3**	**$105.2**	**$441.0**	**$206.8**
Our base design		$97.1		$150.6
Total capital investment	$126M	$73M	$150M	$75M

Table 16: Intelligen's lactoferrin model against ours, in SuperPro's operating-cost categories. Intelligen's figures are its 2024 costs (Tables 2–5 of the report) per kg of lactoferrin, deflated to 2025; "ours, their design" runs our model with the design inputs their report states (scale, six 100 m³ fermenters, diavolumes, column flow, loading and buffers, step recovery, resin at $1,000/L lasting 150 cycles, electricity at $0.10/kWh) and our own unit prices. Our facility line is maintenance, insurance and ten-year depreciation. `data/pf/scenarios/lactoferrin.yaml`, variants `lf_intelligen_a` and `lf_intelligen_b`; review L.

At its design titer PF lactoferrin costs $130–$168 per kg (design-report to price-anchored capital), below the reported range for dairy lactoferrin in 2025 dollars (Section 4.2).

Against Intelligen's own model. Intelligen published the design our case takes its titer from (Da Gama Ferreira et al. 2026): six 100 m³ fermenters, glycerol growth and methanol induction to 10 g/L, and two downstream routes. Our base case runs that titer on glucose; costed on methanol, as their strain is fed, it is $177/kg instead of $168. No methanol-free strain is public near 10 g/L: the best is 2.8 g/L of porcine lactoferrin ($413/kg). Its unit production costs, per kg of lactoferrin in 2025 dollars, are $299 without chromatography and $441 with cation exchange. They include depreciation and no return on capital, so they compare with our cost of production, $97 and $151: theirs are 3.1× and 2.9× ours. Their design inputs explain a small part of the gap. With their output, vessels, diavolumes, column flow and loading, buffers, step recovery, resin price and life, and electricity price, our cost of production becomes $105 and $207 (Table 16); the largest single input is their short resin life (150 cycles against our 1,000). The rest is their unit costs, category by category: raw materials (glycerol at about $1.8/kg, and a part of the line that the report does not itemise), capital (their investment is 2.0× ours for the same design; their equipment list is not published), labour, membranes, utilities, and a quality-control line that we do not have. We keep our sourced prices; theirs are SuperPro inputs without a stated basis. Our cost sits below the reported price range and Intelligen's inside it, but only at the design titer: at the best public methanol-free titer the cost is $413/kg. No PF lactoferrin price has been disclosed, so this is a cost comparison only.

4.10 Feedstock against feed: the metabolic floor and the animal

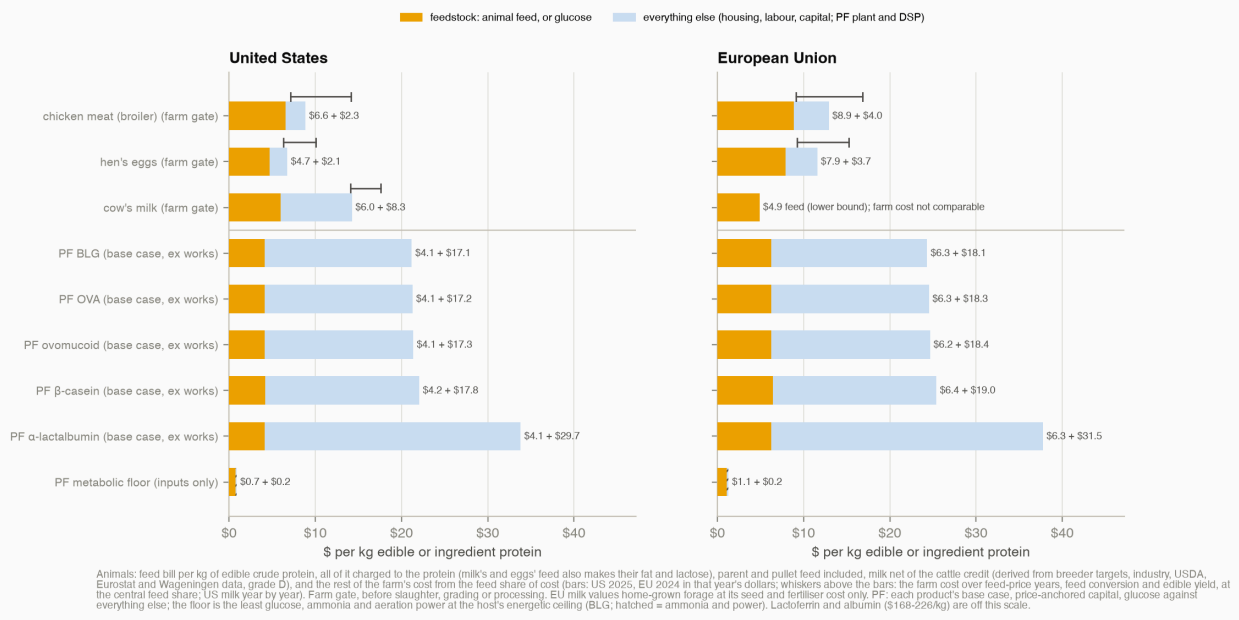


Figure 13: What a kg of protein costs, split into feedstock (animal feed, or glucose) and everything else: the incumbent animals at the farm gate, each PF food protein's base case ex works, and the metabolic floor, in the US and the EU. Animal rows charge all the feed to the protein and are derived from industry, USDA, Eurostat and Wageningen data (grade D); see the text for what each side includes.

Chickens, hens and cows are also machines that turn crop calories into protein, and their main cost is feed. This section puts PF on the same footing: what it must spend on feedstock at the least, what it spends today, and what the animals spend.

The floor. A cell must supply a protein's carbon and nitrogen, and pay in ATP for building it: chaining the amino acids, making them from glucose and ammonia, and reducing sulfate. Element and electron balances fix how much glucose that takes once the ATP is paid by respiring glucose (the energetic ceiling of Section 2.2; `src/pftea/floor.py`). For BLG it is 1.77 kg of glucose, 0.19 kg of ammonia and 11.7 mol of O_2 per kg of protein, releasing 6.8 MJ of heat. At our prices those inputs cost $0.88 per kg in the US ($0.71 of it glucose) and $1.28 in the EU; the other food proteins are within a few per cent. No **K. phaffii** process can do better: the floor leaves out transport, secretion, folding and protein turnover, all of which only add to it. It does rest on the host's P/O ratio (1.48 mol ATP per mol O, a genome-scale-model value; none is measured), so a host that made more ATP per O_2 would need slightly less. The same accounting bounds the fermenter: with every mole of O_2 and every joule going to product, a vessel at our O_2-transfer and cooling limits could make at most 13 kg of protein per m^3 per hour (cooling binds first; 8–17 over our range of cooling limits, O_2 transfer taking over at the top of it). Real cells cannot approach it: at our secretion ceiling and packing limit (40 mg/g/h at 150 g/L) they would make at most 6 kg/m^3/h, and their own respiration takes most of the O_2. The physical-limit case reaches 1.3 kg/m^3/h, the base case 0.27. So the capital floor is set by the cells, by how fast they secrete and how much O_2 they burn staying alive, not by the vessel.

Today against the floor. The base case uses 10.3 kg of glucose per kg of BLG, 5.8× the floor. Most of the difference is the yeast: the batch grows 3.5 kg of cells per kg of protein before and while it produces, and keeps them alive; the product is made below the ceiling yield, and 10% of it is lost downstream (Section 2.2, Section 2.4). The glucose costs $4.11/kg, 19% of the base cost. At the physical-limit best case (Section 4.5) it is 2.7 kg ($0.84), 1.5× the floor.

The animals. A US broiler eats about 22 kg of feed per kg of edible protein, counting its parents' feed and only the meat that is eaten; a laying hen about 20 kg per kg of egg protein (`data/evidence/feed.yaml`, our arithmetic from breeder targets, industry averages and USDA yields, grade D). In 2025 that feed cost $6.6 per kg of chicken protein, $4.7 per kg of egg protein and $6.0 per kg of milk protein (US dollars; 2020–2025 range in Figure 13). These charge all of the feed to the protein; milk's and eggs' feed also makes their fat and lactose. Broiler feed is no dearer per unit of energy than glucose (2–3 US cents per MJ for both;

`broiler_feed_me_ross_specs`), so the 7–9× gap between the feed bills and the glucose floor is the animal's conversion loss: a broiler's feed carries about 18 kg of glucose-equivalent energy per kg of edible protein against the floor's 1.8. US broiler feed is priced as delivered compound feed (corn and soybean meal plus about $53 per tonne for milling and delivery, from Wageningen's US costings), and milk's bill is net of the cull cows and calves sold with it (10% of milk and cattle sales, credited by value). Dividing by the feed's share of the farm's costs, on the same scope (74% for broilers and 70% for eggs, Wageningen's model farms with parent and pullet feed added; 42% for milk, of USDA costs that include land and unpaid labour) gives a farm-gate cost of about $8.9, $6.7 and $14.2 per kg of protein. This scales the non-feed costs with the feed price, which may understate them: the Egg Industry Center's 2023 survey puts eggs' non-feed costs at $4.2 per kg of protein, against our $2.1.

In the EU feed is dearer: in Wageningen's 2017 costings broiler feed cost €31.1 per 100 kg in the EU against €24.1 in the US, and Eurostat's 2024 EU price was €39.6 (mean of Germany, France and the Netherlands). EU broilers convert feed about as well as US ones (1.63 against 1.83 kg per kg live weight in 2017, at a lighter bird), EU hens slightly worse (2.00 against 1.96 kg per kg of eggs in 2021). So the EU feed bills are higher, mainly because the feed is dearer: $8.9 per kg of chicken protein and $7.9 per kg of egg protein (2024 dollars), and the farm-gate costs $12.9 and $11.6. EU milk's feed bill, $4.9, is a lower bound: the EU farm accounts value home-grown forage only at its seed, fertiliser and spray costs, where USDA values it at market prices, so the two milk figures are not comparable and no EU farm cost is shown for milk. EU glucose is dearer too ($0.61/kg against $0.40), and the comparison comes out the same: PF's glucose floor is $1.08 per kg, its base-case glucose $6.3, against $7.9–$8.9 of poultry feed.

What it means. Charging the animal's whole feed bill to its protein, PF at the base case spends no more on feedstock than the animals: $4.1 of glucose per kg of BLG against $4.7–$6.6 of feed. That framing flatters PF for milk and eggs: their feed also makes fat and lactose, and charged by energy content their protein's share of it would fall below PF's glucose (we have not costed this split). Chicken meat, which is mostly protein, would stay above. What holds either way is that glucose is only 19% of PF's cost. Its handicap is the rest, the plant, its capital and the purification: $17.1 (BLG), $17.2 (OVA), $17.3 (ovomucoid), $17.8 (β-casein), $29.7 (α-lac) per kg, against $2–$8 for the farm. The room PF has is large on paper, since its feedstock floor is 7–9 times below the animals' feed bills, but it is taken up by the conversion plant, which is why fermenter productivity, not sugar, drives the cost (Section 4.3). Four further cautions. The animal is costed at the farm gate, as a live bird, a shell egg or raw milk, before slaughter, grading or processing, and its protein comes in a whole food with fat and other value; PF's is a purified ingredient ex works. The delivered-feed adder for US broilers rests on Wageningen's 2015–2017 US prices; the US egg bill uses the Egg Industry Center's feed formula on national prices, which may be 5–8% low. Milk is credited for its beef, but spent hens (about 1% of egg costs) and offal are not; the feed shares are for 2017–2021, applied at other years' feed prices; and pullet feed is counted to 17 weeks only. And the feed bills are our derivations, grade D.

4.11 Casein for rennet cheese: α_{s1}- plus κ-casein

Our β-casein product is an acid-casein or caseinate grade. A rennet cheese is different: chymosin sets the curd by cutting κ-casein, the casein on the micelle surface, so a casein without κ makes no rennet curd. With fully recombinant, unphosphorylated proteins a patent example made rennet curds at 16.7–25% κ (New Culture, US 2023/0404098, grade B); with dairy caseins, curds were firmest near 19% κ (Antuma et al. 2025), and milk casein is 12% κ (Runthala et al. 2023). We take 19%, and show 16.7%–25%. Formo makes a stretched-curd analogue from α_{s1}-casein alone, by acid and calcium and without rennet (WO 2026/068856), so κ is needed for rennet cheese, not for every cheese.

We cost the blend as two strains and two plants (`data/pf/scenarios/cheese.yaml`): α_{s1}-casein at the base design point on β-casein's precipitation route, and κ-casein from its own strain, each plant sized for its share of 10,000 t/yr of casein. How κ would be recovered is not known: no source measures it, and unphosphorylated κ (computed isoelectric point 5.85) is the casein least likely to precipitate with acid or calcium. Two cases bracket it: κ **co-precipitated** with α_{s1}, as milk κ is, costed on the precipitation route at a 78% recovery (optimistic, grade E), and κ on **its own cation-exchange column**, on lactoferrin's column data (D). Per kg of blend at κ share s,

$$c_{\text{blend}} = (1-s)c_{\alpha s1} + s c_{\kappa}(T_{\kappa}).$$

For the blend to cost at most 10% more than α_{s1} alone (a threshold we chose), $s(c_{\kappa} - c_{\alpha s1}) \le 0.1 c_{\alpha s1}$, i.e. $c_{\kappa} \le c_{\alpha s1}(1 + 0.1/s)$: at s = 0.19, κ may cost up to 1.53× α_{s1}. One strain secreting both in proportion would avoid the second plant; it would cost what α_{s1} costs at the combined titer, provided κ is secreted at its share of it, which no public strain shows.

> **Check (reader):** The blend cost is a mass-weighted average of two plants' costs; the 10% threshold and the factor $1 + 0.1/s$ follow from it in one line.

κ-casein titer (g/L)	κ co-precipitated, $/kg	Blend, $/kg	κ by column, $/kg	Blend, $/kg
0.25 (best published)	$2,888	$567	$2,923	$573
1.00	$724	$156	$764	$163
2.00	$366	$87.6	$404	$94.8
5.00	$152	$47.0	$187	$53.6
10.00	$80.1	$33.3	$116	$40.1
20.00	$44.0	$26.4	$79.7	$33.2
30.00	$33.4	$24.4	$67.8	$31.0
42.00	$26.2	$23.0	$60.6	$29.6
α_{s1}-casein alone		$22.3/kg on 8,100 t/yr; $22.1 on 10,000 t/yr (β-casein: $22.0)		

Table 17: Rennet-cheese casein at 19% κ: κ-casein made by its own strain and plant (1,900 t/yr), on the two routes that bracket its unknown recovery, and the blend with α_{s1}-casein at its base case (price-anchored capital, US). κ titers from 0.25 g/L, the best published (a **K. lactis** flask in a patent), are costed as **K. phaffii** on glucose in 72 h.

- **α_{s1}-casein costs what β-casein costs.** At the base design point it is $22.3/kg ($17.2 with design-report capital) and $17.1 at 42 g/L, against $22.0 for β-casein: the two differ in their formula by about 1%. At the best peer-reviewed bovine α_{s1} titer, 1.13 g/L inside **E. coli** cells in a 15 mL flask (Wang et al. 2021), costed as if secreted, it is $522/kg, and at half of a patent example's 5 g/L α_{s1} + β mix (Standing Ovation, EP 4 533 954) $237/kg.
- **κ sets the blend's cost at today's κ titers.** The best published κ titer is about 0.25 g/L, secreted by **Kluyveromyces lactis** in a flask and measured by Western blot (Redbiotec, WO 2023/002061); the best peer-reviewed one is 4.14 mg/L, secreted by **Aspergillus oryzae** (Panchanawaporn et al. 2026), below what the model's batch can resolve. At 0.25 g/L the blend costs $567–$573/kg, 25–26× α_{s1} alone. No company discloses a κ titer; κ from milk sells only as a laboratory reagent, at $1,080/g.
- **The κ titer the blend needs.** Within 10% of α_{s1} alone, co-precipitated κ needs about 29 g/L (28–29 g/L over the κ range), roughly α_{s1}'s own design titer, because κ's plant is smaller and its recovery lower; on its own column no titer up to 42 g/L is enough. That design titer is itself assumed: the best peer-reviewed bovine α_{s1} titer is 1.13 g/L.

So casein for rennet cheese is gated by κ-casein: its best titer is over 100× below the design point, and nothing public says how a recombinant, unphosphorylated κ would be recovered.

4.12 Heme: a cost-in-use note

Soy leghemoglobin (LegH) is the heme protein of Impossible Foods' beef analogue. The US lists it as a colour additive at up to 0.8% of the uncooked analogue (21 CFR 73.520); Impossible uses 0.45% in beef and 0.25% in pork analogues (FSANZ A1186). What it sells, LegH Prep, is a frozen liquid concentrate of a cell lysate: 6–9% LegH, at least 65% of its protein LegH (80% on average in five batches), LegH 38–51% of its dry matter, made inside **K. phaffii** cells and released by bead milling (FDA GRN 737). The model has secreted-protein routes only, so this note estimates the cost from the best published **secreting** processes on the membrane route at LegH Prep's purity (43% of dry matter; `data/pf/scenarios/heme.yaml`). It is an estimate, not a bound. Leaving out lysis and debris removal makes it too low; counting the drying of a product sold frozen makes it too high; the published titers are per litre of supernatant, converted here to whole broth with the model's cell volume (×0.64, D); and one strain's heme loading is unknown.

Process	150 t/yr, $/kg LegH	1,000 t/yr, $/kg LegH	$/kg beef analogue at 0.45% (0.25%–0.8%)
Chen et al. 2025: 6.0 g/L of supernatant (3.9 g/L of broth) in 102 h, methanol, hemin fed; heme loading not measured	$243	$173	$1.09 ($0.61–$1.94)
Shao et al. 2022: 3.5 g/L of supernatant (2.2 g/L of broth), methanol, no hemin; 93% heme-loaded	$372	$290	$1.68 ($0.93–$2.98)
the base design point (30 g/L, 72 h, glucose)	$91.7	$35.4	$0.41 ($0.23–$0.73)

Table 18: Secreted leghemoglobin on the membrane route at LegH Prep's purity (price-anchored capital, US; ex-works cost in use at the 150 t/yr cost). An estimate for comparison, not the commercial route (intracellular, lysed, sold frozen).

- **Cost per kg of LegH.** The best published secreted titer (Chen et al. 2025) costs $243/kg on a 150 t/yr plant, about the size of the LegH market GFI Europe and Arthur D. Little estimate from trade data (of LegH or of the preparation; the source does not say), and $173 at 1 kt/yr. Their own cost range for heme proteins is $24–33/kg in their text and $32–122/kg in their chart (the report gives both); the $275/kg they list as the "conventional" price is a customs value, most likely of Impossible's own shipments.
- **Per kg of heme-loaded protein.** Only globin that carries its heme gives colour and flavour. Shao et al. measured 93% loading, so their cost barely changes; Chen et al. did not measure it, and their feed carried hemin, a blood-derived heme source not costed here, covering at most about 41% of what full loading would need (our arithmetic).
- **Per kg of burger.** At 0.45% the best secreted process adds $1.09 per kg of analogue, an ex-works cost equal to 7% of US ground beef's retail price ($15.26/kg, BLS, August 2026). At the design point on a 1 kt/yr plant it would add $0.16.

Heme from today's best published secreting strains would therefore cost about $1–2 per kg of analogue at the factory gate (7%–11% of retail ground beef), and up to twice that at the legal maximum; at a commercial design point, cents. Whether it is worth that cost is a question about taste that we found no consumer study isolating, and Impossible launched in the UK in September 2026 without it (Green Queen 2026).

4.13 Transferrin: a cost-in-use note

Transferrin carries iron into cells; serum-free media add it at 10–20 mg/L (Essential 8, B8, Beefy-9). It is the second-largest recombinant protein in GFI's (2023) projections of cultivated-meat media after albumin, but far behind it (about 2.4% of the volume), and Wildtype's FDA file shows it dropped from that company's process. This note costs recombinant human transferrin on albumin's plant and two-column route (Section 4.9), the route Novozymes uses for both on the same yeast platform, with transferrin's own column data from its patents: capture recovery 70%, polish 93% (`data/pf/scenarios/transferrin.yaml`). The polish needs a borate salt that is not costed and is not a permitted food additive in the EU, so it would have to be washed out. Transferrin has 19 disulfide bonds and secretes poorly from **K. phaffii** (20–30 mg/L); the best published titer is 2.29 g/L of supernatant from **S. cerevisiae** on sucrose in a 10 L vessel (Finnis et al. 2010), costed here as **K. phaffii** on glucose (1.47 g/L of whole broth).

Process	10 t/yr, $/kg rTf	100 t/yr, $/kg rTf	$/kg cultivated meat at 0.37 g/kg (0.075–1.19), 100 t/yr
Finnis et al. 2010: 2.29 g/L of supernatant, **S. cerevisiae**	$1,939	$949	$0.35 ($0.07–$1.13)
the base design point (30 g/L, 72 h, glucose)	$1,218	$314	$0.12 ($0.02–$0.37)
catalogue price today (InVitria Optiferrin, 10 g)	$217,716	$217,716	$80.55 ($16.33–$259.08)
Humbird's at-scale estimate ($12/g)	$12,000	$12,000	$4.44 ($0.90–$14.28)
GFI's assumed cost	$1,000	$1,000	$0.37 ($0.08–$1.19)

Table 19: Recombinant transferrin on albumin's route (price-anchored capital, US) and its cost per kg of cultivated meat at GFI's implied 28 mg/L × 13 L of medium per kg (range: Humbird's 10.7 mg/L medium to GFI's 42 L/kg). Prices in the last three rows are not model results.

- **Per kg of transferrin.** At the best published titer it costs $949/kg on a 100 t/yr plant and $1,939 at 10 t/yr; at the design point, $314 and $1,218. Tens of tonnes a year would really be made in campaigns on a shared plant, so the dedicated 10 t/yr plant here overstates its cost.
- **Per kg of meat.** Made at that scale, transferrin adds $0.35 per kg of cultivated meat at today's best titer and $0.12 at the design point, against $81 at today's catalogue price. Unlike albumin's (Table 15), its cost is set almost entirely by whether it is made at food scale at all, not by how well: its dose is about 40 times smaller.

4.14 Ovalbumin as a binder: a cost-in-use note

Plant-based burgers and nuggets need a binder to hold the mince together and keep it juicy when cooked. Most use methylcellulose (MC), a cellulose ether; some vegetarian products use egg white. PF egg-white protein is cleared for this use in the US: ovalbumin up to 10% of a meat analogue (FDA GRN 1249), and egg-white protein at 3–8%, "based upon current usage rates of egg white protein" (GRN 1104). EVERY has promoted its ovalbumin as a replacement for MC (AgFunderNews 2024). This note asks what each binder costs per kg of finished analogue: the use level times the price (`data/pf/scenarios/binder.yaml`, evidence in `data/evidence/binder.yaml`). The protein binders are dosed at 5% of an 80%-protein product (3%–8%), i.e. 4.0% protein.

Binder	Use level	Price	$ per kg of analogue
methylcellulose	1.5% (0.7%–3%)	$4.45/kg ($2.6–$26)	$0.07 ($0.02–$0.79)
dried egg white (2015–2025)	4.0% protein	$9.7–$21.2/kg protein	$0.39–$0.85
PF ovalbumin, base case (30 g/L)	4.0% protein	$21.3/kg protein	$0.85 ($0.51–$1.36)
PF ovalbumin, industry benchmark (42 g/L)	4.0% protein	$15.9/kg protein	$0.64 ($0.38–$1.02)
PF ovalbumin, physical-limit best case	4.0% protein	$4.2/kg protein	$0.17 ($0.10–$0.27)

Table 20: Binder cost per kg of plant-based meat analogue. Methylcellulose price: India's import unit values for its methylcellulose tariff line, 2025 (all origins; China-origin low, US-origin high), the only official line that isolates it. Dried egg white: EU export unit values per kg protein, 2025 dollars. PF ovalbumin: this model, price-anchored capital, US. Ranges are the use-level ranges, except dried egg white's, which is its price range at the central dose.

- **Against methylcellulose.** At the base case, ovalbumin costs 13 times as much per kg of analogue as methylcellulose at central values, and even at the physical-limit best case about 2.5 times as much (within methylcellulose's own range if food-grade methylcellulose is the dear, US-origin grade). A protein binder cannot match a cellulose ether on cost; it has to win on function or label. The one like-for-like test found used 5% ovalbumin against 0.7% methylcellulose and found the ovalbumin burger harder when cooked (EVERY's patent US 12,096,784, Example 45, grade B), so the dose at which ovalbumin merely matches methylcellulose is lower than 5% but not measured. EVERY's public claim that its cost in use equals methylcellulose's (AgFunderNews 2024) is not supported by these numbers.
- **Against egg white.** At the base case, PF ovalbumin costs about what dried egg white does at the same protein dose ($0.39–$0.85 per kg of analogue over 2015–2025). This is the substitution that matters for animals: replacing methylcellulose spares none, replacing egg white in vegetarian products does. How much egg white those products use is not public; Quorn's mince lists at most 9% rehydrated egg white (under about 1% egg-white protein; label, grade C).

5 Discussion

5.1 Answers, in brief

- **Q1.** Today's best public titers give $43.0 (BLG), $227 (OVA), $479 (ovomucoid), $803 (β-casein) per kg: not competitive. Only a company's patent example comes close: EVERY's 30 g/L ovomucoid, $23.7/kg. The titer private cost models assume gives $15.8 (BLG), $15.9 (OVA), $16.0 (ovomucoid), $16.4 (β-casein) ($12.3 (BLG), $12.4 (OVA), $12.4 (ovomucoid), $12.6 (β-casein) with design-report capital): all four inside their incumbents' bands, at about 7 mg/g/h of secretion over 72 h (three times the best demonstrated). Lactoferrin at the design titer of Intelligen's methanol-induced process, costed on glucose, ($168) is below its reported range; at the best public methanol-free titer ($413) inside it. Albumin costs $215 at the base titer and $424 at the record titer, above the bulk-albumin band and far below catalogue prices. α-Lac costs $33.8 at the base titer on a 1 kt/yr plant, inside its reported band, but $1,848 at its best public titer (0.6 g/L of supernatant, 0.39 of broth).

- **Q2.** Fermenter productivity (protein per hour of vessel time) drives the cost of BLG, OVA, ovomucoid, β-casein and α-lac, followed by the capital basis (vessel prices and fixed-cost rule); sugar and energy prices come third. For lactoferrin and albumin the column (resin price, life and loading, and the capture recovery) leads (Section 4.3). The product yield acts mainly through the feasibility ceiling. The strain parameters are the weakest evidence: design values (B) and survey averages (C), and for casein and lactoferrin titers borrowed from other proteins or species.
- **Q3.** With price-anchored capital, $20/kg BLG needs 29 g/L at 48 h or 32 g/L at 72 h at the base cell density, and OVA, ovomucoid and β-casein about as much (Table 9). What $10/kg needs above all is secretion: with today's biology and capital, process design alone reaches $7.1 at 40 mg/g/h and $12.0 at 10. Albumin at GFI's $10/kg needs a route without the columns' cost, not a better strain.
- **Q4.** The physical-limit best case is $4.1 (BLG), $4.2 (OVA), $4.3 (ovomucoid), $4.5 (β-casein) and $56.9 (lactoferrin, at its market's size) and $172 (albumin) per kg, bound by O2 transfer, specific productivity, minimum initial fill; at a secretion ceiling of 10 mg/g/h the BLG case is $8.1. α-Lac's, at its market's size, is $5.2.
- **Q5.** 4 of 5 claimed titers are feasible in our fed-batch model, some only with long production phases or secretion near the ceiling. No casein or lactoferrin titer has been claimed.
- **Q6.** Scale matters up to about 10 kt/yr for the commodity proteins and is the main lever for lactoferrin, whose market caps its plant; albumin's cost flattens past about 1 kt/yr. EU siting adds about $3.1/kg to BLG.
- **Q7.** Replacing the protein of a litre of milk costs $0.71 as BLG and $0.74 as β-casein, while the litre costs $0.48 at the farm gate; bought as WPC80/WPI the same whey protein is a fraction of that. Albumin at the base cost adds $1.95–$10.23 per kg of cultivated meat.
- **Q8.** The route sets the product: precipitated β-casein is 93% of its powder, membrane-only β-casein (Eden Brew) about 6%, in a powder that is mostly yeast carbohydrate. Intelligen's lactoferrin model costs 2.9× ours to produce. Albumin by UF/DF alone would cost $34.3 against $215 with its columns, but membranes alone reach only 77% albumin.
- **Feedstock.** PF's least feedstock is 1.8 kg of glucose per kg of protein at the host's energetic ceiling, $0.71 in the US: several times below the animals' feed bills ($4.7–$6.6 per kg of protein with all the feed charged to protein, grade D). The base case's glucose ($4.1) is 19% of its cost and no more than that feed; its gap is the conversion plant ($17/kg against $2–$8 on the farm; Section 4.10).
- **Notes.** Casein for rennet cheese needs κ-casein at about 19% of it; at today's best κ titer (0.25 g/L, a patent example) the blend costs $567–$573/kg, and co-precipitated κ must reach roughly α_{s1}'s own design titer for the blend to cost within 10% of α_{s1}-casein alone (on its own column, no titer up to 42 g/L is enough; Section 4.11). Secreted leghemoglobin at the best published titer costs $243/kg on a 150 t/yr plant, $1.09 per kg of beef analogue (Section 4.12). Recombinant transferrin made at food scale would add $0.35 per kg of cultivated meat at today's best titer, against $81 at catalogue prices (Section 4.13). As a binder in plant-based meat, PF ovalbumin costs about what dried egg white does but 13 times methylcellulose per kg of analogue (Section 4.14).

5.2 Why published PF costs are lower

Published PF cost estimates for food proteins span roughly $12–39/kg: GFI Europe and Arthur D. Little (2025) give $24–33/kg for their "P1" molecules today and $12/kg for α-lactalbumin after improvements; Intelligen (2025) $39/kg for an OVA isolate at 1 kt/yr. Of the designs we can rebuild from their own inputs, our model comes within −4% of Intelligen's OVA once theirs is recast into our capital charge; NREL's cellulase and Voutilainen's recombinant case come out 1.8× lower. With vendor-quote fermenters our capital matches theirs; the gap is mostly accounting: fixed costs of 1.5–2% of capital a year instead of Humbird's 9%, and lighter capital charges. Intelligen's lactoferrin model runs the other way: its cost of production is 2.9× ours, through its unit prices for raw materials, labour and consumables and a capital estimate 2.0× ours for the same design (Table 16). Published costs differ from ours in both directions, for reasons that can be traced line by line.

Neither convention is "correct". Humbird's reproduces the cost of one fermented commodity in his own workbook; the design-report one matches how process-design studies cost greenfield enzyme plants. We read our base as a central estimate on Humbird's method and design-report capital as an optimistic one, not as a confidence interval.

5.3 For funders

In order of expected cost reduction per unit of evidence gained (our judgment):

1. **Secretion rate without growth.** How much BLG or OVA a strain secretes per gram of cells per hour when it barely grows, over a production phase of two to five days at 1–10 m³. It sets both today's cost and the best case; the one near-zero-growth study we found saw secretion fall with growth, and every compiled glucose process secretes fastest at 0.1–0.2/h (Looser et al. 2015; Prielhofer et al. 2018).
2. **Public fed-batch data.** Titer, productivity and yield measured **together** for any of these proteins; no public dataset reports all three, and the product yield is currently derived, not measured. For casein even the titer is missing: no secreted casein titer above 1 g/L is public in any yeast or fungus, and every company keeps its titer confidential.
3. **The design ceiling.** The workable starting volume, the most concentrated feed and O_2 transfer at production scale and cell density move the titer ceiling itself.
4. **Real capital and fixed costs.** Published capital, staffing and running costs of recent PF plants (for example Liberation Labs' 600 m³ Richmond plant) would settle the fermenter price basis and the fixed-cost rule.
5. **Downstream data.** Diafiltration flux at scale (one patent value, 5 L/m²/h, sizes most of the membrane area), dryer-feed solids for PF proteins, and for casein the solubility of the unphosphorylated protein against pH and calcium, which sets the precipitation route.
6. **Function.** Whether unphosphorylated PF casein forms curd and melts like dairy casein in cheese. The evidence is mixed (New Culture's patent reports near-normal cheese yields; dephosphorylated casein forms micelles poorly), and the cost model cannot answer it.
7. **Albumin's purification and its demand.** What purity cultivated-meat cells actually need from albumin, whether a cheaper route than chromatography (membranes, precipitation, a high-capacity polymeric resin) reaches it, and the capture recovery at scale; and whether media will keep albumin at all, which the leading firms' choices currently argue against.
8. **α-Lactalbumin and κ-casein strains.** Both are about four orders of magnitude short of a commodity design point in public data (bovine α-lac 2 mg/L in yeast; κ-casein 4 mg/L peer-reviewed). For α-lac the culture pH is the known conflict: in culture, without its calcium, the protein unfolds and sticks to cells below pH 5.5, while proteases need low pH to be suppressed. For κ, whether a recombinant κ co-precipitates with α_{s1} decides whether rennet casein can use the cheap route at all.

On the evidence here, PF BLG at the titer private cost models assume (which needs about 7 mg/g/h of secretion) would undercut whey in a spike year such as 2026. With design-report capital ($12.3) it would have undercut whey in 8 of the 21 annual export prices of 2015–2025; with price-anchored capital ($15.8) in 3. High-value proteins such as lactoferrin cost less than most reported incumbent prices at a design titer. OVA's position against egg white depends on the year, because egg-white prices swing more than whey's. β-casein sits close to BLG in cost but faces a cheaper incumbent than egg white; it is the protein whose displacement would most directly reduce milk demand (Section 4.8), and the one with the least public strain data.

> **Check (reader):** A judgment, not a model output, written by the AI assistant and not yet endorsed by the author (please revise): the probability that a PF plant built with 2026 technology makes BLG below $10/kg protein is about 0.15, with low confidence (unchanged from version 2: the cheaper vessels raise it, the secretion requirement lowers it). The model reaches $10 only if cells secrete several times faster than any published **K. phaffii** process, sustained over days, or with the lighter design-report accounting and high titers together.

5.4 For EU policy

At Dutch sugar, energy, wage and construction prices the base BLG plant costs $3.1/kg more than in the US Midwest. Cheaper capital is a large lever: a capital charge at 4% over 20 years instead of 7.5% over 10 cuts $3.9 from the EU cost; of that, a 20-year recovery alone cuts $2.5 and a 4% rate alone $1.2. US-level glucose would remove $2.2 and US-level energy $0.8. These are cost effects on one plant; the model says nothing about which instruments would deliver them.

5.5 What would change these conclusions

- A strain that secretes ≥10 mg/g/h of any of these proteins over days at near-zero growth, or evidence that it cannot, would move both today's cost and the best case more than any other single result.

- A peer-reviewed fed-batch at ≥40 g/L in ≤48 h with a measured yield would test the derived product yield and the ceiling; for casein, any secreted titer above a few g/L.
- The ceiling rests on design choices: a 30% minimum starting volume, a 500 g/L glucose feed and 107 g/L of cells. Industrial practice with lower starts, denser feeds or fewer cells lifts it; then secretion and O_2 transfer set it.
- Published capital and running costs of recent PF plants would settle the fermenter price basis and the fixed-cost rule.
- Measured OVA solubility at pH 5, and protease control for OVA in **T. reesei**.
- For albumin: a published purification for cell-culture-grade albumin without affinity or a $1,250/L resin, or a capture recovery near 80%; and a methanol-free titer above a few g/L.
- For α-lac: a secreted bovine titer above 1 g/L, and whether membranes alone reach the 90% isolate.
- For rennet casein: a secreted κ-casein titer above a few g/L, or one strain secreting α_{s1} and κ together.

6 Limitations

- **Strain parameters.** Few are grade A. The product yield is derived from a design's lumped stoichiometry, not measured; specific productivities are design values or survey averages, and the base production phase assumes secretion without growth (Section 2.3). 40 titer rows in `data/titers.csv` are not yet verified in full text.
- **Carbon sources.** Methanol-induced production phases are modelled (version 4), with a measured maintenance scaled to near-zero growth (D) and a product yield set at the glucose yield's share of its energetic ceiling (D); no product yield on methanol is measured. The ethanol-fed BLG case is still priced as glucose, which understates its O_2 and heat. Methanol's safety equipment is inside the missing-equipment allowance.
- **The feasibility limits are design choices**: the minimum starting volume (30%), the feed concentration (500 g/L glucose) and the cell density, which together set the base ceiling; the O_2 cap and its basis (200 mol/m³/h per m³ of cell-free liquid); the heat-removal ceiling (25 kW/m³, grade D); and the secretion ceiling (40 mg/g/h), a judgment: about 2.5 times the highest **measured** specific secretion rate we found for any protein (15–16 mg/g/h, **A. niger** glucoamylase, chemostat and pH auxostat; `data/evidence/kinetic.yaml`), and 2–8 times what industrial cellulase processes imply at assumed cell densities (5–20 mg/g/h, Section 4.5; native enzymes).
- **Directions of the simplifications.** k_La ignores viscosity and foam (optimistic at high cell density and titer); specific productivity is constant for up to 120 h (optimistic for long production phases); no product is made during growth (conservative for constitutive promoters); the energetic ceiling leaves out transport, secretion and turnover (optimistic).
- **OVA** at fermentation pH: solubility is unmeasured (50 g/L is shown as a sensitivity), and **T. reesei** clips much of it.
- **β-casein** (added in version 3; AI-audited only): its titers are borrowed (the base design point, an **E. coli** patent example for the academic case); the acid dose is milk's buffer capacity applied per m³ of broth (grade D), the precipitation recovery an analogy with dairy (D) and the heater's heat recovery a guess (E); decanters are priced on the disk-stack centrifuge curve; phosphorylation, micelle formation and the casein's function in cheese are outside the model; and at the physical-limit titer the broth's viscosity would lower O_2 transfer, which the model ignores. Whether secreted casein stays dissolved at fermentation pH is contested (Perfect Day's patent says caseins precipitate in the fermenter; Eden Brew's dossier says its β-casein stays soluble), and so is whether **K. phaffii** secretes β-casein at all without a co-expressed milk kinase (NewMilkBuzz). The base case assumes a secreting strain at the titer of the other proteins; no public one exists.
- **Continuous processes** and cell recycle are not modelled.
- **Chromatography** is one capture column for lactoferrin, and a capture plus an anion-exchange polish for albumin, with guessed column hardware (the flow rate is Intelligen's design value) and no itemised buffer tanks (inside the missing-equipment allowance). Lactoferrin's academic titer is porcine lactoferrin, and its market size and prices are grade C and disagree by about five times between sources.
- **Albumin** (added in version 4; AI-audited only, audit R12): no food- or media-grade specification exists, and whether two columns reach a cell-culture grade is a judgment; its column data are from a 1990s pharmaceutical patent (maximum step volumes, and recovery and step tables from different examples); resin

life is borrowed from whey lactoferrin, the favourable end for a yeast-broth feed; protease losses at pH 4.5 and the acid, acetate and octanoate are not costed; its methanol use is well below what the record run fed (Section 4.2); the methanol yield leaves out the ATP of assimilating methanol's carbon (about +$1/kg); its 2018 titer is scaled to whole broth by analogy; the bulk-albumin price is a technical grade, not a cell-culture one; the plant size is a design choice; and its demand depends on media that the leading firms are moving away from (Wildtype's FDA file; GFI 2026). No priors were registered for albumin before its first model outputs.

- **α-Lactalbumin** (added in version 4; AI-audited only): its best titer is human α-lac, measured in the supernatant after a His-tag purification, on ethanol priced as glucose; its market prices are online listings and market-research pages (C), converted to a per-kg-α-lac basis by us; its dryer feed is borrowed from BLG (D), and the tighter ultrafiltration membrane its 14 kDa size needs is not costed; the plant size is a design choice. No priors were registered for it.
- **The rennet-cheese, heme, transferrin and binder notes** use products outside the main list, each run on an existing route by analogy: κ-casein on two routes that bracket its unknown recovery (co-precipitation at an assumed 78%, E; its own column on lactoferrin's data, D), and leghemoglobin on the membrane route although the commercial product is an intracellular lysate (no lysis or bead mill is modelled, and the drying is extra). The κ share for unphosphorylated recombinant caseins rests on one patent example. The supernatant titers of α-lac and LegH are converted to whole broth with the model's cell volume (D). Hemin fed to the best LegH strain is not costed. Transferrin runs on albumin's route with a borate polish whose borate is not costed, from an **S. cerevisiae** titer at 10 L costed as **K. phaffii**; its use level is derived from GFI's tonnages (D). The binder note's methylcellulose price is a trade unit value that mixes grades, and the dose at which ovalbumin binds as well as methylcellulose is not measured.
- **The feed comparison** (Section 4.10) sets PF ex works against the animal at the farm gate. Its feed bills are our arithmetic (grade D) on breeder targets, industry averages, USDA and Eurostat statistics and Wageningen costings: US broiler feed as corn and soybeans plus a delivered-feed adder, US layer feed by the Egg Industry Center's delivered-feed formula, EU poultry feed at Eurostat complete-feed prices, and milk at USDA and EU farm-account feed costs. All the feed is charged to the protein (milk's and eggs' also makes fat and lactose); milk is credited for its beef only; layer-breeder feed is left out; and the EU milk bill is a lower bound not comparable with the US one.
- **Not costed:** start-up and commissioning, working capital (except in the DCF price), failed batches, QC release testing, packaging, dewatering and inactivation of spent cells, and external heat-exchange loops for the largest vessels.
- **EU wastewater** is priced as in the US; the Dutch levy is per kg of oxygen demand, which is not modelled.
- **Guesses (grade E)** remain for: `hosts.treesei.mu_max_per_h`, `dsp.cex_hardware_usd_per_L_resin`, `dsp.acid_heat_recovery`, `utilities.sterilizer_peak_factor`, `utilities.boiler_peak_factor`, `labour.fermenters_per_continuous_crew`, `capital.dsp_area_factor`, `capital.compressor_building_m2_each`.

7 Scoring the pre-registered priors

The priors were written in `docs/SCOPE.md` by the AI assistant (Claude) on 2026-09-26, before any model output existed, and before the author had seen them. They were committed to git together with the first outputs, so the repository cannot prove the order; the session record shows it but is not public.

Prior (2026-09-26)	Stated	Outcome (v4.1)	Score
1a. Intelligen configuration, $/kg OVA (80% interval)	$38–60	$44.7/kg OVA	inside
1b. Base case, $/kg protein (80% interval)	$28–50	$21.2 (BLG), $21.3 (OVA)	outside
1 (superseded). Purified BLG/OVA at ~30 g/L and 1–10 kt/yr	$20–40	10 kt/yr: $21.2 concentrate, $26.0 OVA isolate; 1 kt/yr: $44.7 (Intelligen configuration)	inside at 10 kt/yr, outside at 1 kt/yr
2. P(physical floor < $10/kg)	0.65	BLG $4.1 at the assumed 40 mg/g/h (yes); at 10 mg/g/h $8.1 (yes)	conditional: Brier 0.12 if secretion reaches 40 mg/g/h, 0.12 if only 10
3. P(a public BLG/OVA yield on sugar turns up)	0.20	no measured yield; design values exist (Behm 0.223, Intelligen 0.28 lumped)	Brier 0.04 (measured); 0.64 if design values count
4. P(first-try calibration within tolerance): yeast / NREL / Intelligen	0.6 / 0.6 / 0.5	yes / no (+136%) / yes (−3.6%)	Brier 0.16 / 0.36 / 0.25
5. P(public v1 by 2027-04-30 / 2027-07-31)	0.5 / 0.8	open	—

Table 21: Pre-registered priors and outcomes. Prior 4 is scored on the first run (2026-09-26), before the changes recorded in `docs/VALIDATION.md` and `docs/AUDIT.md`. Prior 3 meant a measured strain yield; both readings are scored. Prior 2 is scored once (BLG and OVA are one event), and its outcome depends on the secretion ceiling chosen during the audit, so it is scored conditionally.

The base-case interval prior missed low: after the review moved the fermenters from aseptic cell-culture prices to an industrial vendor quote, the base case ($21.2) fell below the $28–50 interval; with aseptic vessels it would be $30.9, inside. The superseded prior was too optimistic at 1 kt/yr, as its replacement said. The calibration prior was about right on average; the NREL miss came from capital conventions, which it did not anticipate.

8 Reproducibility

```
.venv/bin/python -m pytest                      # model, calibration and invariant tests
.venv/bin/python scripts/run_analyses.py        # all results -> results/analyses.json
.venv/bin/python scripts/make_figures.py        # docs/figures/*.png
typst compile --root . report/pf_tea_report.typ # this document
```

Inputs are in `data/pf/base.yaml` (every value with a range, grade and source), with scenario overrides in `data/pf/scenarios/`. The evidence files are `data/evidence/*.yaml`, with verbatim quotes and locators. Calibration cases are in `data/calibration/`. The audit is `docs/AUDIT.md`.

References

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A Parameters (generated from `data/pf/base.yaml`)

Parameter	Value	Range	Gr.	Source
plant.product	blg			blg | ova
plant.spec	concentrate			concentrate (GRN 1056-type, UF/DF + dry) | isolate (adds precipitation) | cex (adds cation-exchange capture; lactoferrin) | crude (whole broth)
plant.annual_target_protein_t	10000	1000–100000	C	gfi2025costs Table 2: private-sector volumes 2,500-25,000+ t/yr; SCOPE base
plant.operating_days	330	320–350	B	NREL 2011 uses 8,410 h/yr (96%); Humbird A.1 350 d; Intelligen ~342 batches x 24 h; 330 d allows for campaigns and cleaning
plant.location	us_midwest			us_midwest | eu_nw
strain.host	kphaffii			
strain.titer_g_L	30	5–60	B	intelligen_ovalbumin2025 design basis 30 g/L; Every2024-HY ovomucoid 30.19 g/L (patent Ex. 9); gfi2025costs public avg 24, private avg 42. For scale: the highest secreted titer in the Looser 2015 review's compilation is 19.55 g/L (glucose oxidase, methanol/ mannitol, 168 h, 100 -> 245 g/L cells; Table 4). Tornado ends beyond the feasibility boundary are truncated to it
strain.methanol_frac_prod	0		B	0 = methanol-free (the food-PF base: GRN 1056, 1219, 1284; Intelligen's ovalbumin design induces with 0.28 kg methanol per kg product beside glucose, not modelled). 1 = a methanol-only induction phase (K. phaffii AOX1 processes: the academic albumin, lactoferrin (Lv 2023) and beta-casein (Choi 1996) cases). Intelligen's lactoferrin design co-feeds glucose and methanol in a split it does not state, so its variants keep 0. pf_ferm module docstring (review G1)
strain.t_prod_h	72	48–120	B	intelligen_ovalbumin2025 p.6: expression phase 72 h; NREL 2011: 96 h production
strain.X_prod_g_L	107	40–150	B	derived from intelligen_ovalbumin2025: cells at the start of expression = 92 g/L x 107 m3 at harvest (p.7) - 8/28 x 3,210 kg rOVA made (Eq. 2) = 8,927 kg, in 107 m3 less the expression-phase additions (11,464 kg glucose as 500 g/L solution 22.9 m3 + 30% ammonia 3.5 m3 - net O2/CO2 exchange 2.2 m3) = 82.8 m3 -> 108 g/L; feeding then dilutes it to 92 g/L at harvest. Range: K. phaffii glucose fed-batch 93-169 g/L at harvest (Shemesh 2024, A); patents 100-200 (Every2024-HY, B); hosts.yaml
strain.X_max_g_L	150	100–200	A	K. phaffii glucose fed-batch 93-169 g/L (Shemesh 2024); typical 150; patents up to 200 (hosts.yaml). Growth stops here. Heyland 2010 grew at constant rate only up to ~155 g/L, slowing above it (218 g/L after 65 h in 2.6 L), and states such densities are reachable only in small fermenters because O2 transfer and heat removal limit larger ones (hosts.yaml kphaffii_glucose_hcd_fedbatch_heyland2010)
strain.mu_prod_per_h	0.0014	0–0.02	B	derived from intelligen_ovalbumin2025 Eq. 2: 917 kg cells made on ~8,930 kg in 72 h (8 g per 28 g rOVA at 30 g/L x 107 m3; 92 g/L at harvest) -> ln(9,844/8,927)/72 = 0.00136/h. Range: 0 (non-growing production) to 0.02 (K. phaffii production-phase growth rates, hosts.yaml; D). Growth stops at X_max: from 107 g/L the high end reaches 150 g/L after ~17 h and the phase continues without growth
strain.Y_ps_g_g	0.465	0.28–0.535	B	derived so that the model reproduces intelligen_ovalbumin2025 Eq. 2 (expression phase: 100 g glucose -> 28 g rOVA + 8 g cells + 4 g HCP, maintenance included). Glucose per g product in

Parameter	Value	Range	Gr.	Source
				expression = 1/Y_ps + (8/28)/Y_x + m_s,prod/q_p = 100/28, with Y_x = 0.5705 g/g (the model's Heijnen growth yield for K. phaffii), m_s,prod = 0.00437 g/g/h (hosts.kphaffii.m_s_prod_g_g_h) and q_p = 3,210 kg / 675,213 kg.h = 0.00475 g/g/h (Intelligen) -> 1/Y_ps = 3.5714 - 0.5008 - 0.9192 = 2.151, Y_ps = 0.465 (audit R2-Q2/R6-C2: 0.28 is a lumped yield, not a reaction yield). The glucose of Intelligen's 4 g of host-cell protein is charged to the target (read with the HCP as product, the yield would be 0.53; review HM-6). The inversion is sensitive to m_s,prod: at 0.0023 it gives 0.39 (review FS-4). low = 0.28 (the lumped value read as a reaction yield, as in v1); high = 0.535 (0.95 x the energetic ceiling 0.564 of BLG with amino-acid synthesis and sulfate assimilation counted, product.energetic_ceiling; review FS-2/SC-2; D). Checks: Behm 2022 design 0.223 g/g overall (T. reesei BLG); NREL cellulase 0.24 g/g overall
`strain.q_p_max_mg_g_h`	40	5–40	D	judgment: a physiological ceiling on specific secretion, applied as a feasibility limit. No K. phaffii measurement is near it: Intelligen's design runs at 4.75 mg/g/h (low, B); the best public BLG fed-batch implies ~1 (Yao 2026, 13.88 g/L in 108 h at ~120 g/L cells); a VHH-secreting strain peaked at 0.64 and fell to 0.019 at near-zero growth (Rebnegger 2024, hosts.yaml); the Looser 2015 review compiles q_P,max of secreted proteins from 0.004 to 0.69, the glucose (GAP) ones 0.05-0.49 at mu 0.15-0.20/h, growth-coupled; the best glucose fed-batch on file secreted 2.1 near mu 0.1/h (Prielhofer 2018, hosts.yaml; the report's 'best demonstrated' case); fungal enzyme processes reach ~5-20 (Ellila 2017 productivities at 20-50 g/L cells; review FS-3). Our model of NREL's T. reesei cellulase design implies ~118 (a model output at 6.6 g/L cells, not physiological evidence; shown only as a sensitivity). Binds only in the physical-limit and required-titer analyses, where the report shows the floor at 5-120. Kinetic bracket (data/evidence/kinetic.yaml): the highest specific secretion rate we found published for one protein in any microbe is 15-16 (A. niger glucoamylase, 20 gene copies, chemostat and pH auxostat; Withers 1998, Swift 1998, A); a 0.25 share of K. phaffii translation at mu 0.2/h gives ~18-41 (derived, D; at the production phase's near-zero growth the ribosome growth law does not bound q_p). So 40 is a generous feasibility ceiling, ~2.5x anything demonstrated, not a target
`strain.X0_g_L`	0.88	0.88–15	B	derived: inoculum 5% v/v (Intelligen OVA p.6) x the seed density fermenter.seed_final_X_g_L = 17.6 g/L (Intelligen's lactoferrin flowsheet; one number for both, review FS-11). v2.1 used 3 (from a guessed 60 g/L seed). Humbird A.1 starts at 15 g/L (high)
`hosts.kphaffii.biomass_formula`	(table in file)		A	Carnicer et al. 2009 Microb Cell Fact 8:65, Table 4: Fab-producing K. phaffii CH1.665N0.134O0.602S0.0039 (evidence hosts.yaml; re-check table cell)
`hosts.kphaffii.mu_max_per_h`	0.2	0.15–0.28	A	hosts.yaml: mu_max ~0.2 1/h on sugar vs 0.11-0.12 on methanol. Glucose batch (Heyland 2010 Table I and p.360): 0.28 for the non-producing parent (high: a producer cannot outgrow it; also Looser 2015 Table 2, glucose) and 0.15-0.16 for the producing strain (low); both papers note the better producers grow more slowly

Parameter	Value	Range	Gr.	Source
hosts.kphaffii.temperature_C	30		B	intelligen_ovalbumin2025 p.6: 30 C
hosts.kphaffii.maint_factor	1	0.57–1.67	A	growth phase: multiplier on Tijhuis 1993 maintenance (0.0174 g glucose/g DCW/h at 30 C, Humbird A.1 method). Producing K. phaffii in glucose chemostats: 0.0147-0.0161 (Rebnegger 2016 citing Rebnegger 2014; Maurer 2006) = 0.85-0.93x. low = 0.0100 (non-producing strain, Rebnegger 2016) / 0.0174; high = 0.029 (5-copy lipase clone, Nieto-Taype 2020) / 0.0174
hosts.kphaffii.m_s_prod_g_g_h	0.00437	0.0023–0.0161	A	production phase (mu ~ 0): Rebnegger et al. 2024 Microb Cell Fact 23:44, VHH-secreting K. phaffii, glucose, 30 C, fitted over mu <= 0.05/h: 'm S (4.37 +/- 0.27 mg g-1 h-1)'. Maintenance falls at near-zero growth (Rebnegger 2016: retentostat 0.0031 vs chemostat 0.0100 for the same strain). low = 0.0023 (1-copy lipase clone, Nieto-Taype 2020); high = 0.0161 (Maurer 2006 chemostat, producing Fab strain). Not raised to Looser 2015 Table 2's 0.009-0.032 (growing cultures on methanol and glycerol; no glucose value) or to the 5-copy lipase clone's 0.029 (Nieto-Taype 2020, chemostat; used for the growth phase, maint_factor high): maintenance falls at near-zero growth (Rebnegger 2016). The base expression phase can carry at most ~0.006 with the product made at the energetic ceiling (results constants ms_prod_max_at_ceiling; review H2). Evidence: placeholders.yaml nietotaype2020 and ms_heyland2010_looser2015 entries
hosts.kphaffii.m_s_prod_methanol_g_g_h	0.005	0.004–0.016	D	methanol production phase (only if strain.methanol_frac_prod > 0). Measured on methanol in growing K. phaffii fed-batches: Dietzsch et al. 2011 Microb Cell Fact 10:14 (PMC3059269, Mut-S, HRP): 'the maintenance coefficient for this P. pastoris strain could be determined with around 0.5 mmol g-1 h-1 (which equals 0.016 g g-1 h-1)'; Jahic et al. 0.013 (as cited there). Derived (mine), to put methanol on the glucose base's footing (m_s_prod_g_g_h is a near-zero-growth value): x 0.31, the same-strain retentostat/chemostat ratio on glucose (Rebnegger 2016: 0.0031/0.0100). 0.016 x 0.31 = 0.0050; low = 0.013 x 0.31 = 0.0040; high = 0.016, the measured value without the near-zero fall (A). Check: per unit Gibbs energy, 0.016 g methanol (0.36 kJ/g/h) is 1.5x the glucose chemostat value 0.0155 (0.24 kJ/g/h), about the share of methanol's energy alcohol oxidase wastes; the base keeps that 1.5-1.7x
hosts.kphaffii.po_ratio	1.48		B	Chung et al. 2010 Microb Cell Fact 9:50: 'P/O ratio (1.48 mol ATP/mol O)', a genome-scale-model constraint (hosts.yaml kphaffii_po_ratio_chung2010); no measured P/O found
hosts.kphaffii.kla_factor	1		B	yeast broth treated as non-coalescing water-like (Humbird Fig1.3)
hosts.kphaffii.cell_removal	centrifuge			
hosts.treesei.biomass_formula	(table in file)		B	humbird2011 p.38: T. reesei cell mass CH1.645O0.445N0.205S0.005
hosts.treesei.mu_max_per_h	0.1	0.05–0.2	E	no sourced T. reesei mu_max found (hosts.yaml gap entry); fungi grow slower than yeast
hosts.treesei.temperature_C	28		B	humbird2011 Table 19: 28 C
hosts.treesei.maint_factor	1	0.5–2	D	as for yeast (Tijhuis in both phases; no T. reesei production-phase m_s found)
hosts.treesei.m_s_prod_g_g_h	—			no evidence: the production phase uses Tijhuis x maint_factor
hosts.treesei.po_ratio	1		B	Castillo et al. 2016 Biotechnol Biofuels 9:252: 'the P/O ratio was set to 1', a metabolic-model assumption

Parameter	Value	Range	Gr.	Source
				(hosts.yaml treesei_po_ratio_castillo2016); no measured value
hosts.treesei.kla_factor	0.5	0.3–0.8	D	viscous mycelial broth lowers kLa vs water-like correlations (humbird2011 used a fungal-specific correlation); evidence file
hosts.treesei.cell_removal	filter_press			
energetics.atp_per_peptide_bond	4		A	Martin et al. 2025 BBA Bioenerg 10.1016/j.bbabio.2025.149564: 'The cost assumed for protein synthesis at the ribosome is uncontested, 4 ATP per peptide bond' (hosts.yaml atp_per_peptide_bond)
energetics.atp_aa_synthesis_per_residue	0.29	0–1.2	A	Stouthamer 1973 as recomputed by Martin et al. 2025 (PMC7617979, section 1.2): synthesis of amino acids from glucose and NH4+ adds '1.4 mmol ATP for amino acid synthesis from glucose' per 4.78 mmol of amino acids, 'or average 0.28 ATP per amino acid' weighted by E. coli composition (0.29 = 1.4/4.78). high = the unweighted average over amino acids, 1.2 (same source, Table 2); low = 0 (the common amino acids yield net ATP). Membrane transport (~5 mmol ATP/g cells, Stouthamer) and secretion costs are not counted, so the ceiling stays an upper bound
energetics.sulfur_atp_per_S	3		B	textbook sulfate assimilation: ATP sulfurylase (ATP -> AMP + PPi, 2 ATP-equivalents) + APS kinase (1 ATP). Not re-read for this project. The 8 electrons of sulfate -> sulfide are charged separately at the host's P/O (4 NADH-equivalents), sulfur_electrons_per_S
energetics.sulfur_electrons_per_S	8		A	oxidation state +6 (sulfate) to -2 (cysteine/methionine sulfur)
energetics.methanol_respired_O	2		B	methylotrophic-yeast dissimilation (textbook; not re-read for this project): alcohol oxidase passes methanol's first 2 electrons to O2 as H2O2 (no ATP; catalase returns half the O2), formaldehyde and formate dehydrogenases each make 1 NADH, i.e. 4 electrons = 2 O atoms respired, so 2 x P/O ATP per methanol (2.96 at P/O 1.48) against 21.8 per glucose (3.63 per C-mol). Sets the energetic ceiling on methanol (plant.energetic_ceiling) and with it the product yield on methanol (plant.methanol_yield)
energetics.substrate_atp_per_glucose	4		A	textbook stoichiometry: 2 ATP in glycolysis + 2 GTP/ATP from succinyl-CoA in the TCA cycle per glucose. Oxidative phosphorylation adds P/O x 12 (24 electrons = 12 O atoms = 6 O2 per glucose), so 4 + 12 x 1.48 = 21.8 ATP per glucose for K. phaffii. Not re-read for this project
fermenter.volume_m3	200	100–500	B	engineering.yaml: industrial aerobic vessels 200-1,000 m3 (Humbird, B), lysine 500 m3 (B), baker's yeast 200 m3 (B); PF designs 136 m3 (Intelligen, B), 150 m3 (Liberation Labs, C), 80-120 m3 culture volume (GFI Europe, C); Humbird STRFERM quotes to 196 m3 (larger sizes extrapolate with capital.fermenter_exponent_above_table)
fermenter.sizing	cheapest			cheapest (base): the designer takes the cheapest feasible plant over size_choices_m3 up to volume_cap_m3 (scenarios.size_fermenter; review G2: capital is costed for whole vessels, so a small plant in 200 m3 vessels pays for idle volume) | fixed: volume_m3 as given. Sized on the product's base case; scenarios, tornado and Monte Carlo keep that plant; base-case variants with other prices (design-report or aseptic capital, EU, pioneer) and the scale curve re-size under their own inputs (audit R12-L7)

Parameter	Value	Range	Gr.	Source
fermenter.volume_cap_m3	200		B	the base vessel (volume_m3): the largest size the designer may choose; larger vessels are a sensitivity (volume_m3 high 500, physical limit 400)
fermenter.size_choices_m3	(table in file)		B	sizes spanning published food-PF designs: 100 m3 (Intelligen lactoferrin, 6 staggered vessels, lactoferrin.yaml), 136 m3 (Intelligen ovalbumin), 150 m3 (Liberation Labs, C), 80-120 m3 culture volume (GFI Europe, C), up to the 200 m3 cap; 50 m3 is a design choice (D) for plants below ~250 t/yr
fermenter.h_to_d	2		B	Humbird Equip Table10 STRFERM 200,000: 196.5 m3, D 5 m
fermenter.fill_max	0.8		B	Humbird Fig1.3!C8; NREL 2011 80%
fermenter.min_initial_fill	0.3	0.3–0.5	B	lowest start fill in a published design: Intelligen OVA starts the growth phase at 41,523 L in a 136 m3 vessel = 30.5% (p.6); NREL 2011 starts at 50% (p.37, high); Impossible US 9,938,327: 40% of the maximum working volume (= 32% of vessel). No design text states a physical minimum (impeller submergence). A batch that must start below this fill is infeasible
fermenter.pv_max_W_m3	5000	2500–5000	B	Humbird 2021 p.5 'an agitation rate up to ~5,000 W/m3' design limit in high-density fermentation (B); low = NREL 300 m3 T. reesei maximum 600 kW on 240 m3 of liquid = 2,500 W/m3 (B, derived)
fermenter.us_max_m_s	0.1	0.05–0.23	D	judgment: no design text states a superficial-velocity limit. Humbird's own 200 m3 aeration model at OUR 150 runs 0.050 m/s at the top (Fig1.3 row 197, B); NREL averages 0.033 m/s (B). high = Humbird 2021 p.5 'sparge rate up to ~2 vvm' in a 200 m3 vessel (8 m liquid height, top at 1.33 bar, 30 C) = 0.23 m/s (derived); Intelligen runs 1.6 vvm (B)
fermenter.otr_cap_mol_m3_h	200	150–250	B	Humbird 2021: 'an OTR approaching 200 mol O2/m3-h is common in an air-sparged STR fermentor' (text p.28), on his basis OUR = qO2 / cell-free liquid volume (Eq. 4.2, p.49); low = 'Bubble columns have a practical OTR limit of 150 mol O2/m3-h' (A.1, p.91); high = physical-limit scenario (head pressure, D). Per m3 of broth this is 129 at 107 g/L cells, 147 at 80 and 100 at 150 (review HM-3/BE-7: an industrial designer would use 100-120 per m3 of broth with air at ~100 g/L cells)
fermenter.otr_cap_basis	cell_free_liquid			cell_free_liquid (Humbird's basis, Eq. 4.2; base since the Phase F review) | broth (the cap applied per m3 of broth, the v2 base with 150)
fermenter.heat_per_o2_kJ	460	430–500	A	Roels 1983 via Almeida Benalcazar et al. 2025 (engineering.yaml heat_per_o2_roels_design_value); calorimetry 455 +/- 15, range 430-480 (Maskow et al. 2010, engineering.yaml oxycaloric_equivalent); Cooney/Shuler-Kargi 500 (high). Used only in the OTR_max heat check; the batch heat comes from formation enthalpies (~480-510)
fermenter.heat_removal_max_kW_m3	25	15–40	D	no production-scale ceiling is published. Evidence: 'several million kcal' per hour from 200-300 m3 SCP yeast fermenters = ~8-30 kW/m3 (US 3,986,934, B, derived); internal coils alone ~17 kW/m3 in a 600 m3 column (Almeida Benalcazar 2025, derived with its U 1.4 kW/m2/K), so external exchangers are needed; 42 kW/m3 claimed at pilot scale with tube baffles (US 4,670,397, B). 25 = 1.2 x the OTR-150 load (19.2 metabolic + ~1.2 agitation)
fermenter.turnaround_h	12	8–48	B	intelligen_ovalbumin2025 (5 fermenters, 24 h cycle, 108 h culture -> ~12 h); Humbird A.1 8 h; NREL 48 h
fermenter.feed_methanol_g_L	791		A	neat methanol, density 0.791 kg/L at 20 C (CRC Handbook); methanol is fed undiluted in K. phaffii fed-batches (e.g. Dietzsch 2011 'pure methanol' pulses)

Parameter	Value	Range	Gr.	Source
fermenter.feed_glucose_g_L	500	500–600	B	intelligen_ovalbumin2025 p.6: 500 g/L glucose solution (about 42 wt%; not '50 wt%', review FS-1/BE-2/HM-1). high = Shemesh 2024 K. phaffii fed-batches: 'Dextrose or glycerol as a carbon source was fed from a 60% (w/v) sterile solution' (600 g/L, A). Concentrated feeds are common in industry (Aro 2023: '55 % glucose'; Nielsen 2025 DoE up to 52 w/w%), and the titer ceiling depends on this choice (report Table 7). Heyland 2010 fed 730 g/L (73% w/v) to a 2.6 L lab fermenter; not used as the high, which already comes from lab fed-batches (Shemesh 2024), while no production-scale design on file feeds more than 500 g/L. A 730 g/L feed would raise the physical-limit ceiling somewhat further (not computed)
fermenter.nh3_feed_g_L	268		B	intelligen_ovalbumin2025 p.6: 'aqueous ammonia (30% w/w)' at 0.895 kg/L (interpolated between 0.91 at 25% and 0.88 at 35% w/w: GESTIS via Wikipedia 'Ammonia solution') = 268 g NH3/L (review BE-2: v2 counted it at 1 kg/L)
fermenter.seed_ratio	0.1		B	NREL 2011 p.39: seed vessels 10% of the next; Intelligen 5% inoculum
fermenter.seed_time_h	24	15–36	B	intelligen 15 h per seed step; NREL 2-day seed cycle incl. cleaning
fermenter.seed_final_X_g_L	17.6		B	Intelligen lactoferrin design (Feb 2026 report, flowsheet of case a): seed broth leaves the first two seed fermenters at 17.36 and 17.77 g/L biomass (15-16 h fed-batch steps, 5% transfers); the ovalbumin report (the base process) does not state it. v2.1 used 60 g/L (D, a guess)
fermenter.DO_top_frac	0.1258		B	Humbird 2021 A.1!C92 (the dissolved-O2 setpoint of his aeration method). Humbird, Davis & McMillan 2017 p.10: in practice fed-batch aeration and agitation hold DO near zero, where transfer is most efficient; their own simulations could not set DO, and imply about three times this model's aeration energy per kg O2. Review H1 reproduced their 200 m3 stirred-tank curve with this model's equations only by holding DO near 0.5 together with their geometry, 1 atm headspace, no kLa temperature term and a 72%-efficient compressor with line losses (a reconstruction, not a published value); the base sensitivity at 0.5 alone raises aeration energy by about half
fermenter.kla_temperature_factor	1.025	1–1.025	B	Humbird 2021 Fig1.3!C11:F11: van 't Riet kLa x 1.025^(T-20). low = Humbird, Davis & McMillan 2017 Eq. 2, the same correlation (0.002, 0.7, 0.2) with no temperature term (review H1)
fermenter.compressor_formula	adiabatic			adiabatic: P = (D/3600) R T_in (r^a - 1) / (a eta), a = (k-1)/k = 0.286, r = p_btm,max / 1 atm (the textbook form; review H1) | humbird_a1: A.1!BD as written, which puts k = 1.4 where a belongs (1.6x the power at r = 2.1)
fermenter.compressor_efficiency	0.8		B	Humbird 2021 A.1!BD: the compressor power is divided by 0.8
media.salts_kg_per_m3	22.3	14.7–38	B	derived from intelligen_ovalbumin2025 Table 3: 'Basal Salts' 816,341 kg/yr over 342 batches x 107 m3 final broth = 22.3 kg/m3 (audit R6-m1). low = Table 1 basal medium 38 g/L in the 41.5 m3 initial volume per 107 m3 final (14.7); high = Table 1 applied to the final broth (38, the v1 value)
media.salts_price_usd_kg	0.2	0.15–0.4	B	intelligen_ovalbumin2025 Table 3 'Basal Salts' $0.200/kg
media.h3po4_kg_per_kg_cells	0.047		B	intelligen Eq.(1): 1.87 H3PO4 per 40 cells
media.h3po4_price_usd_kg	0.85		B	intelligen Table 3: H3PO4 (85%) $0.850/kg

Parameter	Value	Range	Gr.	Source
media.antifoam_kg_per_m3	0.2	0.02–1	B	Behm 2022 Table 2: antifoam 0.2 g/L (T. reesei BLG); US 5,260,202 (Delta Biotechnology): 'Typical total usage rates ... 0.02-0.2 g per liter' in sensor-controlled yeast fed-batch (low); high = NREL 2011 corn oil 0.1% v/v, ~1.0 kg/m3 (P-420 13.4 kg/h over ~12.9 m3/h broth, derived)
media.antifoam_price_usd_kg	1.12	0.5–2	C	prices.yaml: Voutilainen 2021 design value EUR 1,000/t x 1.12 (no market price found); range is a guess
media.methanol_inducer_kg_per_kg_product	0			methanol co-fed as an inducer beside a glucose expression phase, costed only (its O2, ~10% of the batch's at Intelligen's 0.30 kg/kg, is not added). Used only by the Intelligen OVA calibration anchor, which states it (audit R12-M4). A methanol-fed production phase is strain.methanol_frac_prod instead
media.methanol_price_usd_kg	0.91		B	intelligen_ovalbumin2025 Table 3: $0.910/kg
dsp.dilution_to_g_L_cells	45		B	intelligen p.7: broth diluted to 45 g/L biomass before centrifugation; the first underflow is re-diluted to 45 g/L
dsp.centrifuge_stages	2		B	intelligen p.7
dsp.paste_extracellular_frac	0.5	0.4–0.6	B	Petrides 2015 s.11.3.1 p.11-11: 'the centrifuge paste contains 40 to 60% v/v extracellular liquid' (dsp.yaml centrifuge_paste_liquid_fraction). Wet cells taken as 30% dry matter (Humbird A.1!C67)
dsp.recovery_cell_removal_override	—			calibration anchors only: fixes the cell-removal recovery instead of the paste model
dsp.centrifuge_kWh_m3	1.35		B	Humbird SI Equip!AB40 'Power demand 1.35 kWh/m3' (dsp.yaml); Westfalia HSB400 ~1.4 (Milledge & Heaven 2011, C)
dsp.mf_flux_LMH	20	10–50	D	no primary-clarification MF flux on file for yeast broth: EVERY US2025/0134130's 10 +/- 2 LMH is a final bioburden filtration (audit R6-M1, low); microalgae MF 20-100 (Slegers, D)
dsp.mf_recovery	0.975		B	Behm 2022 (T. reesei BLG design): 'microfiltration assuming 100% fungus rejection and 97.5% permeate recovery' (dsp.yaml mf_permeate_recovery_fungal_behm)
dsp.uf_flux_LMH	30	20–50	B	intelligen p.8: 30 L/m2/h concentration flux; Petrides 20-50
dsp.uf_rejection	0.995		B	intelligen p.8: rejection 0.995 for rOvalbumin
dsp.diafiltration_conc_g_L	50		B	EVERY US2025/0134130: diafiltration of rOVA at 50 g/L (dsp.yaml); the protein is diafiltered at this concentration, then concentrated further
dsp.df_flux_LMH	5	3–7	B	EVERY US2025/0134130: DF at 50 g/L '5 +/- 2 LMH' (dsp.yaml); also used for the final concentration to the dryer feed (audit R3-F2)
dsp.diafiltration_volumes	3	2–6	D	salt and sugar washout to meet the protein spec (evidence file); EVERY uses 6-8 DV after resolubilising a precipitate
dsp.dryer_feed_solids_blg	0.3	0.27–0.37	C	whey protein practice (Tetra Pak Dairy Processing Handbook): WPC80 UF retentates reach 27-32% DM and are dried at ~36% ('~36% for WPC80', ch.19); WPI and >60%-protein WPC are brought to 35-37% DM by NF/high-concentration membranes, not evaporators (ch.18). value = UF-only midpoint; high = NF. The final concentration step is costed at the DF flux
dsp.dryer_feed_solids_ova	0.23	0.2–0.3	C	egg-white practice applied to rOVA (close analogy): egg white is ultrafiltered industrially from 11.5% to 23% solids (Tsai et al. 1977 J Food Prot 40:449, 24 plant batches, A as data); UF to 20-30% DM before spray drying (EP 2 926 674 B1 citing DD 290128, B). 23% solids is ~200 g/L protein; OVA dissolves at 200 g/L in water (Wang 2025, A), so the Intelligen/vendor '50 g/L practical solubility' is not a limit (audit P-8)

Parameter	Value	Range	Gr.	Source
dsp.dryer_feed_solids_ovd	0.23	0.2–0.3	C	analogy with egg white and OVA, with the same range (dsp.dryer_feed_solids_ova): egg white is ultrafiltered industrially to ~23% solids (Tsai et al. 1977). Ovomucoid dissolves clear at 300 g/L (US 12,096,784 Ex. 4), so no property limits the feed. EVERY's documented practice is a 5-6% feed (rOVD concentrated to 50-60 g/L by 5 kDa TFF before spray drying, US 12,096,784 Ex. 20; ovomucoid.yaml ovd_dryer_feed_solids, C), as it is for its rOVA; for OVA that case is kept as a named sensitivity, not as the range (audit P-8, R10-2)
dsp.dryer_feed_solids_alb	0.2	0.1–0.25	C	albumin is ultrafiltered to 20% w/v and sold so (Sartorius Recombumin Prime 'Liquid | 50 mL Vial | 20%w/v'; Elite 10% w/v), so a 20% UF retentate is routine (analogy as dryer feed, C). low = the 10% w/v product; high = 25% albumin solutions (pharma practice, D)
dsp.dryer_feed_solids_lf	0.27	0.1–0.37	D	analogy: no lactoferrin-specific value; whey-protein UF retentates reach 27-32% DM (Tetra Pak, C), so the UF-only low end is used; high as for BLG and egg white. low = Intelligen's lactoferrin design, which dries a ~10% feed and calls 100 g/L the protein's solubility limit (pp.7, 10; no citation given)
dsp.dryer_feed_solids_alac	0.3	0.27–0.37	D	analogy with BLG's whey practice (dsp.dryer_feed_solids_blg, same range): alpha-lactalbumin is 'highly soluble over pH 2-9' and heat stable (Layman 2018; alac.yaml alac_functional_properties_for_dsp), so nothing in the protein limits the feed. No alpha-lac-specific value; at 14.2 kDa it needs a tighter UF membrane (~5-10 kDa) than BLG, a flux penalty not modelled (E)
dsp.dryer_feed_solids_legh	0.156		B	LegH Prep, five 2016 batches: 12.6-18.4% total solids, mean 15.6% (GRN 737; heme.yaml legh_prep_solids_batches). The commercial product is this liquid, frozen; here it is spray-dried from it (the drying cost is then an overstatement)
dsp.dryer_feed_solids_tf	0.2		D	analogy with albumin (dsp.dryer_feed_solids_alb, C), the same platform and route: no transferrin-specific value is published. Used only in the transferrin note (scenarios/transferrin.yaml)
dsp.cex_dbc_g_L	75	50–100	B	EVERY US 2024/0083940 A1 [0700]: 'The ligand density would accommodate protein binding capacity between 50-100 g protein/L resin' (dsp.yaml iex_binding_capacity_cex_every; a design statement, not a measured dynamic capacity for lactoferrin)
dsp.cex_cv_per_cycle	16	16–25	B	EVERY US 2024/0083940 A1 Table 4: equilibration 4 + wash 2-3 + elution 3-4 + CIP 3 + regeneration 3 column volumes (midpoints; dsp.yaml iex_column_volumes_per_cycle_every). high = Intelligen's lactoferrin design, 10 + 5 + 5 + 5 bed volumes (p.9)
dsp.cex_elution_cv	3.5		B	same Table 4: elution 3-4 CV of '25 mM Citrate buffer with 1M NaCl'
dsp.cex_elution_nacl_kg_m3	58.4		B	1 M NaCl (EVERY Table 4) = 58.44 kg/m3; the dairy process elutes with 10-15% NaCl (Mao 2023), higher
dsp.cex_cip_cv	3		B	EVERY Table 4: CIP 3 CV of 1 M NaOH
dsp.cex_cip_naoh_kg_m3	40		B	1 M NaOH (EVERY Table 4) = 40.0 kg/m3
dsp.cex_regen_cv	3		B	EVERY Table 4: regeneration 3 CV of '20% EtOH and 150 mM NaCl', listed per cycle in 'a typical process' (often only for storage; per cycle is conservative)
dsp.cex_regen_etoh_kg_m3	157.8		B	20% v/v ethanol x 789 kg/m3 (EVERY Table 4)
dsp.cex_regen_nacl_kg_m3	8.77		B	150 mM NaCl (EVERY Table 4) = 8.77 kg/m3

Parameter	Value	Range	Gr.	Source
dsp.cex_flow_cv_per_h	20	5–20	B	Intelligen lactoferrin design (Feb 2026 report p.10): SP Sepharose Big Beads at 500 cm/h through a 25 cm bed = 20 column volumes per hour, loading 33 min, cycle 108 min (lactoferrin.yaml lf_cost_intelligen2026_case_b_cex; review L). low = 5, the v2.1 guess for a packed bed (E). The cycle time follows: (buffer CVs + load CVs) / flow, with load CVs = capacity / feed concentration (review BE-12). Sets only the resin volume and column hardware (capital); buffer tanks are not itemised (in-line dilution from concentrates in Intelligen's design) and fall under the missing-equipment allowance
dsp.cex_recovery	0.832	0.6–0.97	B	EVERY US 2025/0134130 Ex. 5: 'Chromatography purification recovered about 83.2% POI' (dsp.yaml iex_cex_step_recovery_every). low: Helaina's whole LF train 'recovery of the purification is >60%' (Lu 2024, C); high: lactoferrin retention 97% in Intelligen's design (p.9, B)
dsp.cex_resin_usd_L	1250	100–2000	B	Petrides 2015 p.11-33: protein IEX/HIC resins '$500 to $2,000 per liter' (midpoint); low = polymeric resins 'under $100 per liter' (both dsp.yaml). Industrial lactoferrin uses agarose SP Sepharose Big Beads (Pires 2025), a protein resin
dsp.cex_resin_life_cycles	1000	125–1000	A	Pires et al. 2025 J Chromatogr A 1760:466274 (industrial lactoferrin CEX): 'After over a thousand cycles, the column must be replaced' (a lower bound; dsp.yaml iex_resin_lifetime_industrial_lactoferrin). low = Petrides' biopharma 50-200 cycles (midpoint, B)
dsp.cex_hardware_usd_per_L_resin	1000	300–3000	E	guess: columns, skid and automation per litre of packed resin; no public cost found (lactoferrin research pass). Petrides' membrane-system curve (biopharma) is the nearest analogue. Resin first fill is added to TCI unfactored
dsp.polish_columns	0			polish columns after the cation-exchange capture ('cex' spec): 0 = capture only (lactoferrin, the dairy route); albumin 1 (products.yaml). Each repeats the capture column's cycle recipe, resin and hardware (D) at its own loading and recovery
dsp.polish_dbc_g_L	30	20–40	B	US 6,638,740 (Delta), anion exchange after the cation-exchange capture: DEAE-Sepharose FF 'loaded with 30.0 +/- 10.0 g monomeric albumin per litre of matrix' (albumin.yaml hsa_delta_aex_polish)
dsp.polish_recovery	1	0.9–1	B	US 6,638,740: over the anion-exchange step 'A yield of approximately 110% for albumin monomer has been achieved' (dimer formed in the capture step reverts to monomer); capped at 1. low = judgment (D)
dsp.cex_diafiltration_volumes	4		D	derived: 1 M NaCl in the eluate is washed out by DF to ~2% of protein at 50 g/L after 4 volumes (58.4 x e^-4 = 1.1 g/L), meeting ash <3.5% (GRN 1284); 3 volumes would leave ~6%; Intelligen's lactoferrin design uses 5 (p.10)
dsp.cex_target_frac_dry	0.9	0.77–0.94	A	lactoferrin share of the powder: GRN 1284 (All G) batches 93.6-95.2% protein x 95.7-99.4% LF of protein (~0.90-0.94); low = GRN 1219 (TurtleTree) batches 80.5-83.5% protein x >=95% (~0.77-0.79)
dsp.nacl_usd_kg	0.184		A	prices.yaml salt_food_grade_eu_import_unit_value: EUR 0.1629/kg (2025) x ECB 1.1300
dsp.naoh_usd_kg	0.585		A	prices.yaml sodium_hydroxide_eu_import_unit_value: EUR 0.5177/kg NaOH (100% basis, 2025) x ECB 1.1300
dsp.ethanol_usd_kg	0.915		A	prices.yaml ethanol_undenatured_eu_import_unit_value: EUR 0.8099/kg (2025) x ECB 1.1300

Parameter	Value	Range	Gr.	Source
dsp.isolate_uf_end_g_L	50		B	intelligen p.8: UF to 50 g/L before precipitation (their design; not a solubility limit, see dryer_feed_solids_ova)
dsp.precip_recovery	0.891		B	intelligen p.8: 90% precipitated x 99% retained by basket centrifuges
dsp.ammonium_sulfate_kg_per_m3	350		B	intelligen p.8: 350 g/L
dsp.ammonium_sulfate_usd_kg	0.18	0.18–0.33	B	intelligen Table 3: $0.180/kg; EU import UV 2025 EUR 0.19 (A); US import UV $0.33 (A, high)
dsp.ammonium_sulfate_recycled_frac	0			Intelligen sells it on as fertiliser; base assumes no recovery credit
dsp.isolate_cake_solids	0.2		B	intelligen p.8-9: 'a soft cake with a moisture content of approximately 80%', fed to the spray dryer (dsp.yaml precipitate_cake_retention_intelligen; audit R3-F4)
dsp.isolate_wash_volumes	6		B	intelligen p.9: 'The protein cake is then washed with 6 volumes of purified water' (dsp.yaml precipitate_wash_volumes_intelligen); applied to the retentate volume
dsp.acid_precip_feed_g_L	26.8		D	design choice by analogy: the permeate is ultrafiltered to the casein concentration of skim milk (26 g/kg x 1.031 kg/L = 26.8 g/L; milk_egg.yaml milk_casein_tetrapak, A as data), so the dairy recovery, acid and wash data apply to a feed like the one they were measured on (audit R9-1: a dilute permeate would not precipitate as completely). No PF casein process states a precipitation concentration
dsp.acid_h3po4_kg_per_m3	5.65	2.3–17.3	D	derived (mine; check): milk takes 0.049 mol H+/L from pH 6.6 to 4.6 (Nestle US4075196A Ex. 1: 18 kg/h HCl for 10,000 L/h skim milk; casein.yaml casein_dairy_acid_and_wash_water_per_kg, B). Applied per m3 of the UF-concentrated feed, which holds milk's casein concentration, as an analogy (D): 49 mol/m3 x 98 g/mol, one proton per H3PO4 at pH 4-4.5 (pKa1 2.15, pKa2 7.2), / 0.85 for the 85% acid priced in media.h3po4_price_usd_kg. The model's target is pH 4.5, recombinant beta-casein's solubility minimum (Chezan 2026, full text; milk's dose takes it to 4.6); Formo acidifies its alpha-s1 extract to pH 4 with H3PO4 (GRN 1312). low/high = 20/150 mol H+/m3 (guess, E). Pulling down: milk's buffering is largely calcium phosphate, phosphoserines and citrate, absent here; beta-casein itself takes up ~14-17 H+ per molecule (~16-19 mol/m3 at this concentration; audit R9-2, derived). Pulling up: medium phosphate carried into the permeate. The K. phaffii casein fermentations on file run at pH 6.8 (NewMilkBuzz; casein.yaml), near milk's 6.6. Recombinant beta-casein's zeta-potential pI is 5.5, but it is least soluble at pH 4.5 (Chezan 2026). Sulfuric or hydrochloric acid (dairy practice; NewMilkBuzz reports HCl works) would cost several times less per mol H+ and keep phosphorus out of the effluent; no price is on file
dsp.acid_precip_temperature_C	90	45–90	B	Standing Ovation US 2024/0251812 A1 Ex. 7: recombinant beta-casein at pH 4 'partial precipitation ... at 40 C and full precipitation at 90 C' for 20 min (casein.yaml casein_pf_patent_acid_precipitation_and_neutralisation). low = dairy practice for native casein, D for a recombinant one: acidify, then heat to 40-45 C (Tetra Pak ch.20; the scan's 'approx. 82 C' before that is almost certainly 32 C, audit R9-4) or cook at 50-55 C (Southward)
dsp.acid_hold_h	0.333		B	Standing Ovation Ex. 7: 20 min at 90 C (sizes the holding tank only)

Parameter	Value	Range	Gr.	Source
dsp.acid_heat_recovery	0.9	0.5–0.95	E	guess: regenerative heat exchange on the precipitation heater, set equal to the 90% heat recovery the model assumes for media HTST (plant.py block 5); no source on file for either. low = 50% for fouling and curd fusion on a 90 C slurry (audit R9-11, guess). Steam = feed x cp x (T_precip - T_ferm) x (1 - recovery)
dsp.acid_precip_recovery	0.95	0.78–0.99	D	analogy: the dairy acid-casein line recovers essentially all the casein (3 kg commercial casein per 100 kg skim milk, Southward NZIC 'Casein products', A; ~97-102% of the casein on a dry basis, casein.yaml casein_dairy_yield_from_skim_milk, derived), here on a feed at milk's casein concentration (dsp.acid_precip_feed_g_L) at pH 4.5 with the 90 C step. Supported by Chezan et al. 2026 (full text; casein.yaml casein_recombinant_beta_solubility_profile): at pH 4.5, its solubility minimum, recombinant beta-casein precipitates like native beta-casein (22.45% vs 20.67% left dissolved). low = 0.78, that assay's own value (dilute, unheated, no calcium, where native casein also leaves ~21% dissolved): a pessimistic floor. No PF casein process recovery is published
dsp.acid_wash_L_per_kg	13	11–36	C	dairy practice applied by analogy (graded as dryer_feed_solids_blg): Tetra Pak ch.20 countercurrent washing in 'two or three stages' at 45-60 C uses 'about 0.3 - 0.4 litre of water per litre of skimmilk' = 11-15 L/kg dry casein at 2.67 kg casein per 100 kg milk (casein.yaml casein_dairy_acid_and_wash_water_per_kg, derived); value = midpoint (audit R9-13). high = one concurrent wash, 1 L/L milk = 36 L/kg (Nestle US4075196A Ex. 1); decanters are sized for every stage, so the high end overstates their throughput (audit R9-8)
dsp.acid_wash_stages	3		C	dairy practice applied to PF casein by analogy (graded as dryer_feed_solids_blg; audit R9-7): Tetra Pak ch.20: 'Washing takes place in two or three stages'; each stage is a decanter pass (sizes the decanters)
dsp.acid_curd_solids	0.425	0.4–0.45	C	dairy practice applied to PF casein by analogy (graded as dryer_feed_solids_blg; audit R9-7): Tetra Pak ch.20: 'the casein is dewatered in a decanter to a DM content of 45-40%'; Southward: 'about 55% moisture'
dsp.acid_naoh_kg_per_kg	0.02	0.017–0.022	C	dairy practice applied to PF casein by analogy (graded as dryer_feed_solids_blg; audit R9-7): Tetra Pak ch.20: 'The quantity of NaOH required is generally 1.7 - 2.2% by weight of the casein solids' to pH ~6.7 (sodium caseinate); Southward ~2%
dsp.acid_dissolve_h	0.75	0.5–1	C	dairy practice applied to PF casein by analogy (graded as dryer_feed_solids_blg; audit R9-7): Tetra Pak ch.20: caseinate dissolved at 60-75 C in a 30-60 min batch (sizes the dissolving tank only)
dsp.acid_target_frac_dry	0.925	0.91–0.94	D	analogy from alpha-s1 casein (E. coli) to beta-casein (K. phaffii): FDA GRN 1312 (Formo) batches 93.2-96.5% crude protein at 0.3-1.0% moisture x >97% alpha-s1 of protein = ~0.91-0.94 of dry matter (derived); its specification minimum, >=90% protein x >97% at <=5% moisture, is ~0.92 dry (audit R9-14). casein.yaml casein_as1_formo_spec. The only public PF casein that precipitates
dsp.dryer_feed_solids_cas	0.2	0.15–0.2	C	dairy practice applied to PF casein by analogy (graded as dryer_feed_solids_blg; audit R9-7): Tetra Pak ch.20: 'The very high viscosity of sodium caseinate solutions of moderate concentration limits their solids content for spray drying to about 20%' (casein.yaml

Parameter	Value	Range	Gr.	Source
				casein_dairy_acid_process_tetrapak); low = Southward, '>15% solids ... are very viscous'. Used by both casein routes (the UF/DF variant is a mannan-rich concentrate; analogy)
dsp.dryer_MJ_per_kg_water	4.87	3.4–6	A	dsp.yaml: Baker & McKenzie survey average 4.87 GJ/t water evaporated; milk powder 3.4 (3-stage) to 4.9 (1-stage), Ramirez et al.
dsp.dryer_fan_kWh_kg_water	0.05		A	derived from Baker & McKenzie 2005 (dsp.yaml spray_dryer_sec_survey): average 4.87 GJ/t with a fuel-to-electricity ratio of about 27 -> 0.18 GJ/t = 0.05 kWh/kg
dsp.dryer_feed_solids_cas1	0.2	0.15–0.2	D	as beta-casein (dsp.dryer_feed_solids_cas): a redissolved sodium caseinate, here of alpha-s1 (analogy)
dsp.dryer_feed_solids_kap	0.2	0.15–0.2	D	as beta-casein (dsp.dryer_feed_solids_cas), analogy; kappa is only costed inside the rennet-cheese blend
dsp.dryer_recovery	0.98		D	cyclone + bag filter losses (evidence file)
dsp.spray_dryer_cost_2018usd	923650	361740–1377300	B	Yver et al. 2012 Int J Mol Sci 13:240 Table 3: spray dryer $897k (2011$, supplier-based SuperPro model) drying ~790 kg/h water, x CEPCI 603.1/585.7 to 2018$. low = BioSTEAM SprayDryer correlation (CEPCI 567, unsourced in its code) at 790 kg/h, x 603.1/567.3. high = Watson Dairy Consulting (C): mid-size milk dryer GBP 5-10M at 1,000 kg/h powder (~1,000 kg/h water) scaled to 790 kg/h at 0.68, x ECB 2025 1.3189 $/GBP, read as an installed package and divided by the model's TCI/PEC factor 4.6 and the 2018->2025 index 1.33 (audit R3-F5)
dsp.spray_dryer_ref_kg_h	790		B	Yver et al. 2012: 10% WPI feed, ~816 kg/h water less ~30 kg/h to the precipitate (derived)
dsp.spray_dryer_exponent	0.62	0.6–0.68	B	Yver et al. 2012 s.3.2.4: halving the feed concentration (~2.1x the water) needed a 'spray dryer (59% cost increase)' -> ln1.59/ln2.1 = 0.62 (derived). low = six-tenths rule; high = Watson Dairy Consulting bands GBP 5-10M (1,000 kg/h) vs 15-30M (5,000 kg/h) -> ln3/ln5 = 0.68 (C)
dsp.spray_dryer_max_kg_h	10000		C	largest single dryers offered: SiccaDania 200-10,000 kg/h powder, Tetra Pak 15,000, GEA 25,000 (Watson Dairy Consulting supplier table); 10,000 kg/h water is conservative
dsp.concentrate_target_frac_dry	0.85	0.7–0.92	D	target protein share of dry powder after UF/DF of a secreted protein; GRN 1056 minimum is 0.56 (0.70 x 0.80)
dsp.isolate_target_frac_dry	0.99		B	intelligen p.9: 'a purity of 99% on a dry basis' (audit R6-m2)
dsp.powder_moisture	0.05		B	intelligen p.9: 5% moisture; GRN 1056 max 7%
dsp.membrane_price_usd_m2	45	45–272	B	dairy spiral-wound elements: CAD 300 per 6.69 m2 module (Mercier-Bouchard 2017 J Dairy Sci Table 4 fn 3) = CAD 44.8/m2 / 1.2966 CAD per USD (ECB 2017) x CPI-U 2017->2025 1.3134 = $45/m2. high = Petrides 2015 'under $200/m2' for industrial membranes x CPI-U 2015->2025 1.3583. Flat-sheet PES quote $171/m2 (2023$; Chen 2023) lies between
dsp.membrane_life_y	0.5	0.25–2	B	dsp.yaml: Petrides 2015, industrial membranes last >2000 operating hours (lower bound)
dsp.membrane_skid_usd_m2	263	85–526	B	an INSTALLED cost: the model divides it by the convention's installation factor to get a purchased cost (review BE-8; v2 applied installation on top). Mercier-Bouchard 2017 J Dairy Sci Table 4 fn 2: 'capital investment costs (excluding membranes) of Can$321.02/m2 for SW membranes' in a 5,185 m2 dairy MF plant (the row reads 'Installation capital investment') / 1.2966 x CEPCI 603.1/567.5. low =

Parameter	Value	Range	Gr.	Source
				SuperPro help: food and water installations '$70-100/m2' of membrane (midpoint; year unknown). high = 2x, the low end of SuperPro's 2-5x multiplier for units built to biopharma specifications. Ceramic MF would be ~$3,900/m2 (same paper), not used by EVERY's process
dsp.membrane_kWh_m3_permeate	5	1–13	D	dsp.yaml: UF ~13 kWh/m3 permeate (A, upper end); typical crossflow UF lower
dsp.cip_solution_kg_per_kg	7.33		B	intelligen_ovalbumin2025 Table 3: CIP-Acid 3,430,444 + CIP-Caustic 3,901,202 kg/yr for 1,000 t/yr powder (dsp.yaml cip_chemicals_price_intelligen); counted as process water and wastewater
dsp.cip_price_usd_kg	0.023		B	intelligen_ovalbumin2025 Table 3: CIP-Acid and CIP-Caustic $0.023/kg
utilities.electricity_usd_kWh	US 0.0862; EU 0.1456	US 0.072 / EU 0.1297–US 0.1014 / EU 0.1631	A	prices.yaml: EIA industrial average 2025 US 8.62 c/kWh (Iowa 7.20 = low, Illinois 10.14 = high). EU = Eurostat nrg_pc_205 NL band IF (70-150 GWh/yr, the size of a 10 kt plant) 2025 mean EUR 0.1289/kWh excl. VAT x ECB 1.1300; EU range = band IF in BE (low, $0.1297) and DE (high, $0.1631), 2025 means
utilities.natural_gas_usd_GJ	US 4.78; EU 19.61	US 3.34 / EU 12.97–US 6.38 / EU 19.61	A	prices.yaml: EIA US industrial 2025 $5.23/Mcf / 1.036 MMBtu/Mcf (low = Henry Hub $3.52/MMBtu, high = Illinois $7.05/Mcf). EU = Eurostat nrg_pc_203 NL band I4 (0.1-1 PJ/yr) 2025 mean (S1 18.89, S2 15.81) EUR 17.35/GJ GCV excl. VAT x ECB 1.1300 (audit R5-7: NL for gas as for electricity); EU range = band I4 in BE (low, $12.97) and NL (high); DE $17.77, DK $16.81
utilities.glucose_usd_kg	US 0.4; EU 0.61	US 0.33 / EU 0.49–US 0.61 / EU 0.674	B	prices.yaml: GFI/Hawkwood 2025 Midwest DE95 $0.40/kg (B); low = Humbird contract $0.26 (2019) x CPI-U 1.2593 = $0.33; high = IMARC US dextrose Jun 2025 $611/t (C). EU (grade C, proxy; audit R5-2): EU white-sugar ex-factory average 2025 EUR 537/t x 1.13 = $0.61/kg (EC agri-food portal, A as data; sugar used as the EU sweetener price reference, not as feedstock: K. phaffii does not use sucrose). EU low = Comext glucose-syrup export UV 2025 EUR 0.4305/kg as traded x 1.13 (a lower bound on the dry basis); EU high = IMARC Germany dextrose Jun 2025 $674/t (C)
utilities.ammonia_usd_kg	0.61	0.38–0.86	B	prices.yaml: US anhydrous NH3 import unit value 2025 $0.611/kg (UN Comtrade, A proxy; EU import UV EUR 0.53 = $0.60); low = Humbird SI $0.30 (2019) x CPI-U 1.2593; high = USDA Iowa retail 2025 $784/short ton. Priced per kg NH3 contained (fed as 30% w/w)
utilities.water_usd_m3	US 1.5; EU 1.62	US 0.33 / EU 1.62–US 3 / EU 1.62	B	prices.yaml: GFI/Hawkwood 2025 Midwest $1.50/m3 (B); low = Humbird well water $0.26 (2019) x CPI-U 1.2593; high = Intelligen $3/t; EU = VEMW 2025 NL large-user average EUR 1.43/m3 (C; no EU range on file)
utilities.wastewater_usd_m3	1	0.4–4	B	dsp.yaml: $0.2-0.5/m3 below 1 g/L BOD (Petrides); Chicago $0.42/kg BOD implies ~$4/m3 at 10 g/L (A). EU priced as US: the NL levy is per kg COD and effluent COD is not modelled (audit R5-8, a limitation)
utilities.steam_gas_GJ_per_t	3.29		D	physics: LP steam ~2,740 kJ/kg from feed water at 25 C (105 kJ/kg), steam tables, at an 80% boiler efficiency (judgment) = 3.29 GJ of gas per t steam, no condensate return (audit R5-6). Steam price = gas price x this
utilities.steam_usd_t	—			override only (calibration anchors); otherwise derived from the gas price
utilities.biomass_disposal_usd_t_dry	US 458; EU 211	US 0 / EU 0–US 458 / EU 686	C	spent GM biomass is today mostly sterilised and landfilled (Garnica-Garcia 2026 New Biotechnol, abstract). Fees per wet tonne are converted at 15% dry

Parameter	Value	Range	Gr.	Source
				solids, the model's own centrifuge paste (30% dry wet cells x 50% extracellular liquid; review BE-9: v2 used Humbird's 20% cream yeast, i.e. $343 and $158, while no dewatering is costed). US = EREF 2024 national average MSW landfill tipping fee $62.28/short ton (Waste Advantage, C) = $458/t DM, excl. transport. EU = WRAP 2024/25 UK anaerobic-digestion gate fee median GBP 24/t wet (C) x ECB 2025 1.3189 = $211/t DM (the Kalundborg route: Novonesis/Novo Nordisk biomass to biogas). low = 0: a free outlet as feed (21 CFR 573.587 allows dried K. pastoris in broiler feed) or fertiliser; credits are possible but unquantified. high: EU WRAP range top GBP 78/t wet = $686; US: no higher source
utilities.rt_refrig	0.2629248030187212		B	Humbird RawUtil!BY4 (1.24 hp/ton, Couper)
utilities.cooling_water_dT_C	8		B	Humbird A.1!C158
utilities.building_W_m2	50		B	Humbird Equip Table9: Sanitary 50 W/m2
utilities.sterilizer_peak_factor	2		E	guess: HTST sterilizer sized for twice the average media + feed flow (sizing only; v1 hard-coded value, audit R3-F13)
utilities.boiler_peak_factor	3		E	guess: boiler sized for three times the average steam demand, SIP being intermittent (sizing only; v1 hard-coded value, audit R3-F13)
labour.operator_salary_usd	US 60430; EU 59150	US 58320 / EU 55250–US 79970 / EU 80870	A	prices.yaml: BLS OEWS May 2025 Iowa 51-9011 chemical equipment operators $60,430 (US mean $60,710; Humbird's basis was $50k in 2018); low = Iowa 51-8091 $58,320, high = US 51-8091 chemical plant operators $79,970. EU (grade B, derived): Eurostat SES 2022 NL ISCO OC8 gross earnings EUR 43,822 x NL manufacturing (C) hourly labour cost 2022->2024 (47.3/42.0) x NL industry (B-E) 2024->2025 (50.7/47.8; C not yet published) x ECB 2025 1.1300 = $59,150 (audit R5-9). EU range = the same chain on DE (EUR 40,934) and DK (EUR 59,910) OC8 earnings
labour.burden	1		B	Humbird A.1!C246: 100% labour burden
labour.shifts_per_position	4.5		B	Humbird A.1!C225
labour.employees_per_supervisor	10		B	Humbird A.1!E242
labour.supervisor_factor	1.4		B	Humbird RawUtil!BG24: supervisors paid 1.4x the operator salary
labour.attention_production	0.1		B	Humbird A.1!C231: 10% of online+offline time
labour.attention_seed	0.1		B	Humbird A.1!C230 (large seed)
labour.worker_efficiency	0.8		B	Humbird A.1!C224
labour.continuous_fte_per_shift	(table in file)		D	Humbird A.1 continuous-area FTE rules (0.25-1 per area) extended to the DSP areas; isolate_extra only for the precipitation routes (isolate; acid, by analogy), chromatography only for the cex route
labour.fermenters_per_continuous_crew	12		E	guess: one crew for the continuous areas per ~12 production fermenters (v1 hard-coded value, audit R3-F13); Humbird's rules are per plant of 6-8 fermenters
capital.cost_index_2018_to_2025	1.33	1.229–1.524	B	prices.yaml: CEPCI 2018->2023 x1.323 (B) and 2025 = 1.016 x 2024 (publisher) -> ~x1.33-1.35; BLS PPI machinery & equipment WPU11 x1.327 (A). low = BLS PPI chemical manufacturing PCU325325 x1.229 (A), the index Humbird's Equip sheet uses (290.2 in 2018); high = PPI pumps & compressors WPU1141 x1.524 (A). Equipment-specific PPIs run higher still: heavy-gauge tanks x1.92, heat exchangers x1.64
capital.location_factor	US 1; EU 1.09	US 1 / EU 1.075–US 1 / EU 1.09	B	IEAGHG 2018/04 (Amec Foster Wheeler), Chapter E Tables 5-6: the same plants costed in the Netherlands and the US Midwest at 3Q2016 prices and 1.13 $/EUR: TPC ratio NL/US 1.087-1.090 (coal cases; value, high) and 1.075-1.082 (gas-turbine cases, whose turbine is world-priced; low). Applied to TCI. A PF plant (imported stainless equipment) may sit at the low end

Parameter	Value	Range	Gr.	Source
capital.basis	humbird			humbird ('price-anchored capital (Humbird method)', base: Humbird's facility-class factors, 9%/yr fixed costs and capital charge; with his factors and fixed costs his baker's-yeast workbook lands inside the market-implied cost band, which the lighter NREL stack falls below) | nrel ('design-report capital (NREL-style)': NREL 2011 installation + Table 27 indirects, 3% ISBL maintenance + 0.7% FCI insurance): structural sensitivity
capital.fermenter_quote	vendor			vendor (base since the Phase F review: capital.vendor_fermenter, an industrial enzyme-plant fermenter, at the vessel-price level of the yeast anchor, whose A.1 bubble column is $0.78M at 199 m3 in 2018$, and of Intelligen's food-PF design) | acce_aseptic (Humbird 2021 STRFERM: ACCE bare vessel + agitator to the ASME BPE-2016 aseptic cell-culture specification, 316L and full vacuum, text p.37; 2.9x the vendor quote at 300 m3; structural sensitivity). Review BE-1/HM-2/SC-3
capital.seed_fermenter_quote	—			null = seed vessels priced like the production fermenters (base) | acce_aseptic: Humbird's STRFERM quotes, which cover 1-20 m3, instead of the 300 m3 vendor quote scaled down (sensitivity)
capital.vendor_fermenter	(table in file)		B	humbird2011 App. A: F-400 $3,628,673 + A-400 $5,255,007 for 9 units (2007$) = $987,076 per 300 m3 fermenter with agitator (F-400: '80,000 gal, 1 atm, 28 C, Internal coil, SS316'), x CEPCI 603.1/525.4 to 2018$ (audit R4-17/R5-5); exponent 0.7 (Area 300 fermentors). Used for production and seed fermenters in both conventions
capital.nrel_basis	(table in file)		B	humbird2011: install F-400 2.0 and A-400 1.5, cost-weighted 1.70; other equipment 1.8 (Table 25 skidded equipment). Table 27: site development 9% + piping 4.5% + other 4% of ISBL (the 4% warehouse is inside this, so no warehouse building on this basis); prorateable 10% + field 10% + home office 20% + contingency 10% + other 10% of TDC. Table 31: maintenance 3% of ISBL installed, insurance 0.7% of FCI; maintenance_frac_fci is for other design reports (Voutilainen 1.5% of FCI)
capital.fermenter_exponent_above_table	0.69	0.6–0.998	B	used only with fermenter_quote acce_aseptic. Humbird SI Equip!BM74:BO74: power fit of the STRFERM bare-vessel quotes, exponent 0.69, used beyond the largest quote (196 m3). low = six-tenths rule (D); high = the last quoted segment 100 -> 196 m3 (0.998), the v1 extrapolation (audit R3-F3)
capital.facility_class	commodity_aerobic			commodity_aerobic (food-grade, indoor sanitary) | commodity_anaerobic (outdoor, lysine-type)
capital.facility_classes	(table in file)		B	Humbird Equip Table14 (Petrides classes, as tabulated by Humbird): ISBL non-bioreactor DC x PEC, OSBL DC x PEC, E&C x TDC, fees & contingency x TPC, fermentation-area building class. Humbird tabulates the commodity-aerobic column but none of his sheets uses it (his 4FB/4P sheets use cGMP biopharm; A.1 uses a flat Lang factor of 4.5). The model applies the ISBL factor to the fermenters too, where Humbird enters bioreactors at their installed ACCE cost (Table14 bioreactor install factor 1): with the aseptic quote and the 20% allowance this gives $7.5M per 200 m3 installed (2018$), against his own installed aseptic STR at $6.8M ($2.4M bare; text Table 3.3). Review HM-5

Parameter	Value	Range	Gr.	Source
capital.pioneer_factor	1		A	1.0 = nth-of-a-kind plant (the base; SCOPE). pioneer = 1/0.78: across 106 estimates for 44 pioneer process plants the mean ratio of estimated to actual capital cost, excluding external factors such as inflation, was 0.78 (SD 0.194) (Merrow, Phillips & Myers 1981, RAND R-2569-DOE, Table 4.8, p.58; engineering.yaml rand_pioneer_cost_growth). Applied to all capital as a separate sensitivity (scenarios.PIONEER), never in the base. RAND's regression (Table 4.9) would predict less growth for a plant of commercially proven steps; the mean is used as the plain pioneer case
capital.missing_frac	0.2	0.1–0.2	B	Humbird A.1 (baker's yeast): 20% missing capital on an abbreviated equipment list (TCI = Lang 4.5 x PEC / 0.8), the case closest to this model's list, which omits CIP and feed tanks per fermenter, sterile filter housings, glucose storage, biomass inactivation, powder packing and effluent pretreatment (review BE-4). low = Humbird 4FB sheets: 10% on a fuller list (the v2 value). As a fraction of the total: allowance = m/(1-m) x listed PEC
capital.packing_factor	0.15		B	Humbird 4FB-020!D232: fraction of area occupied by fermenters
capital.building_usd_m2_level_2018	(table in file)		B	Humbird Equip Table9 (ACCE 2016$ escalated to 2018)
capital.shop_area_frac	0.25		B	Humbird SI 4FB-020 row 237: 'Shop | % of ferm area | 0.25'
capital.warehouse_area_frac	0.25		B	Humbird SI 4FB-020 row 238: 'Warehouse | % of ferm area | 0.25'
capital.dsp_area_factor	0.5	0.3–1	E	guess: downstream + drying building area as a fraction of fermentation area; no source found (placeholders item 2). Check: Petrides Table 11.3 puts buildings at 0.45 x equipment purchase cost for enzyme plants (range 0.1-3.0); design.building_to_pec reports the model's ratio
capital.office_m2_per_employee	15		B	Humbird 4FB-020!D235
capital.lab_m2	400	150–600	D	analogy: Humbird 2021 SI 4FB-020 rows 233-234, a 6.8 kt/yr cultured-meat plant has 441 m2 of labs (cell lab 294 + QC lab 147). low = the QC lab alone; high = judgment
capital.compressor_building_m2_each	20		E	guess: floor area per air compressor (v1 hard-coded value, audit R3-F13)
finance.ccf_rate	0.075		B	humbird2021 p.39: i = 7.5%, n = 10 y -> CCF ~15% TCI/y
finance.ccf_life_y	10		B	humbird2021 p.39
finance.maintenance_frac_tci	0.04		B	humbird2021 p.46: 'Annual maintenance is here taken to be 4% TCI/y, which is assumed to include CIP chemicals in addition to the repair and preventative maintenance of rotating equipment, etc.'
finance.insurance_tax_frac_tci	0.05		B	humbird2021 p.46: 'Annual insurance is taken to be 5% TCI/y.' Humbird's whole fixed-cost ('overhead') rule is these two; the model has no separate plant-overhead, QC or local-tax line, so 9% stands for all of them (typical insurance alone is ~0.4-1%; review BE-3)
finance.fixed_cost_rule	—			null = follow capital.basis (humbird: 4% + 5% of TCI; nrel: 3% of installed ISBL + 0.7% of FCI); set to show the two rules on one plant (report, capital section)
finance.maintenance_includes_cip_chemicals	yes		B	humbird2021 p.46 (quote under maintenance_frac_tci): with the price-anchored convention the CIP chemicals are not charged again (review HM-8)
finance.dcf_irr	0.1		B	humbird2011 Table 33
finance.dcf_life_y	30		B	humbird2011 Table 33 (construction 8/60/32% and 7-y MACRS are fixed in finance.py with the same source)

Parameter	Value	Range	Gr.	Source
finance.tax_rate	0.21		A	US federal corporate rate since 2018 (NREL 2011 used 35%)
finance.working_capital_frac_fci	0.05		B	humbird2011 Table 33

B Evidence base

Evidence file	A	B	C	D	E	Total
data/evidence/hosts.yaml	66	19	3	4	8	100
data/evidence/dsp.yaml	7	72	5	9	0	93
data/evidence/engineering.yaml	40	38	8	1	0	87
data/evidence/prices.yaml	66	23	15	0	0	104
data/evidence/placeholders.yaml	13	30	13	0	1	57
data/evidence/milk_egg.yaml	46	0	0	0	0	46
data/evidence/lactoferrin.yaml	22	7	6	0	0	35
data/evidence/casein.yaml	40	19	5	1	0	65
data/evidence/ovomucoid.yaml	27	6	3	0	0	36
data/evidence/albumin.yaml	27	10	15	1	3	56
data/evidence/alac.yaml	26	5	12	1	1	45
data/evidence/heme.yaml	24	20	6	3	1	54
data/evidence/feed.yaml	16	16	7	21	0	60
data/evidence/kinetic.yaml	27	0	0	12	0	39
data/evidence/transferrin.yaml	14	5	9	2	0	30
data/evidence/binder.yaml	6	6	8	0	0	20

Table 22: Evidence entries by grade. Each entry carries a source, a locator and, where a document was read, a verbatim quote.